

TRAINING WORKSHOP

FOR

HIV - AIDS HEALTH WORKERS
&
MINISTRY OF HEALTH & WELLNESS (MOHW) EMPLOYEES

ON

THE USE OF

**STATISTICAL PACKAGE FOR SOCIAL SCIENCE (SPSS) FOR DATA
ANALYSIS**

APRIL 24th – MAY 30th, 2024

WORKSHOP REFERENCE MANUAL

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1.0 Introduction

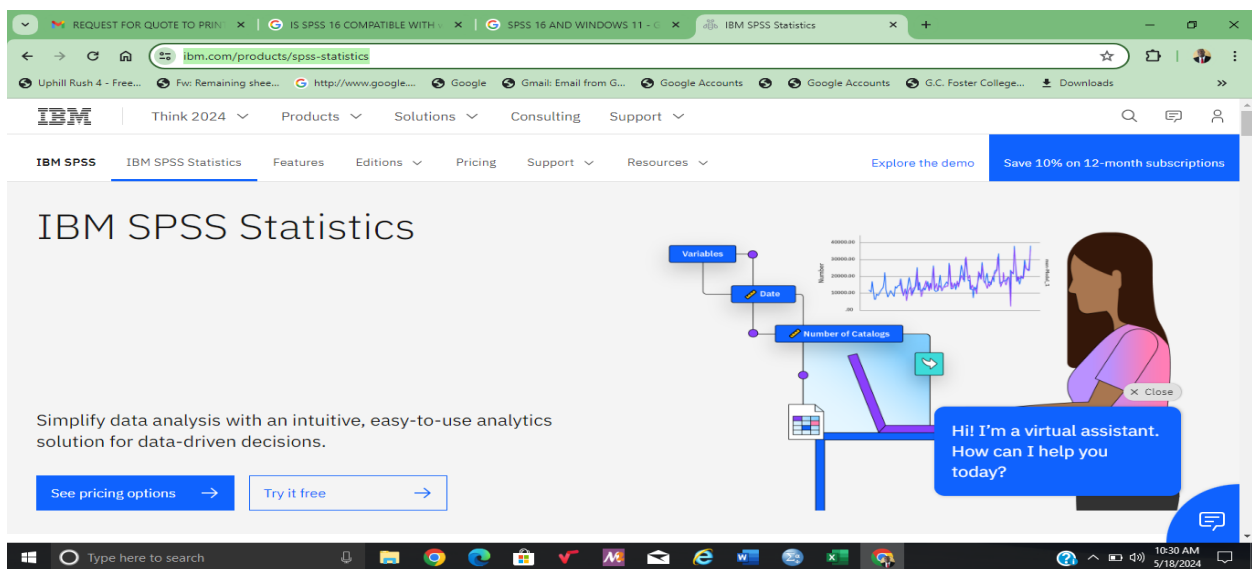
The focus of this Workshop is on the application of Statistical Package for Social Science (SPSS) for Windows in the analysis of data from patient records and other sources. It is noteworthy to remind all participants that there are several versions of SPSS and for this Training the applications are exclusively based on SPSS versions 16, 20 and 25. This manual provides a step-by-step guide on selected statistical procedures for quantitative research on health-related topics including HIV-AIDS. At the end of the Workshop, participants should be able to manipulate datasets for ready import into SPSS 16, 20 and 25, use these SPSS versions to conduct descriptive statistics, inferential statistics using selective non-parametric and parametric statistical tests. In addition, participants should also be able to formulate research questions, null and alternative hypotheses, use appropriate p-value to either accept or reject null hypothesis and interpret Outputs from SPSS.

Upon completion of the Workshop, participants will be assessed on all the aspects of the Training via an online multiple-choice test in which every participant will have three attempts to obtained a score of eighty – percent (80%) or more to be awarded certificate of competency while participants who obtained seventy-nine percent (79%) or below will be awarded certificate of participation.

Participants will be placed into groups according to Regions and Departments to collaborate on any research topic of interest for a period of four -weeks as post continuing education activity after the Workshop. Each group will be required to make a presentation at a seminar to be organized by the Ministry of Health and Wellness.

SPSS and Windows Operating System

To officially obtain the software please visit <https://www.ibm.com/products/spss-statistics> on Google Search Engine or any browser on your personal computer. The link will take you to the screen below:



Click on the “Try it free” to kick start your experience with the latest version of the software.

2.0 Day 1: Data Manipulation Techniques & Descriptive Statistics

2.1 Learning Objectives

At the end of the session, participants should be able to:

- Open SPSS Version 16 or Higher Versions on Personal Computers
- Differentiate Data View from Variable View
- Explain Variable Level of Measurements & Importance
- Select Correct Level of Measurement in Variable View
- Create a Template in Variable for Data Collection Form
- Import /Export Data from Microsoft Excel to SPSS
- Merge Cases and Variables in SPSS
- Use logical Operators to Select Cases
- Recode Variables into Groups
- Explain Descriptive Statistics
- Use Descriptive Statistics in SPSS
- Explain Descriptive Statistics Output from SPSS
- Create Frequency Tables and Charts in SPSS
- Explain Inferential Statistics
- Define p-value and Confidence Levels
- Differentiate Parametric & Non-Parametric Statistical Tests in SPSS
- Complete Day 1 Assessment Items

2.2 Introduction to SPSS Views (Data View & Variable View)

The SPSS can be found in the list of programs or apps on a personal computer. Figure 1.0 illustrate a similar icon from a personal computer and users of the software are expected to click of the SPSS version “IBM SPSS Statistics 25, 20, or SPSS16.0” to lunch the software.

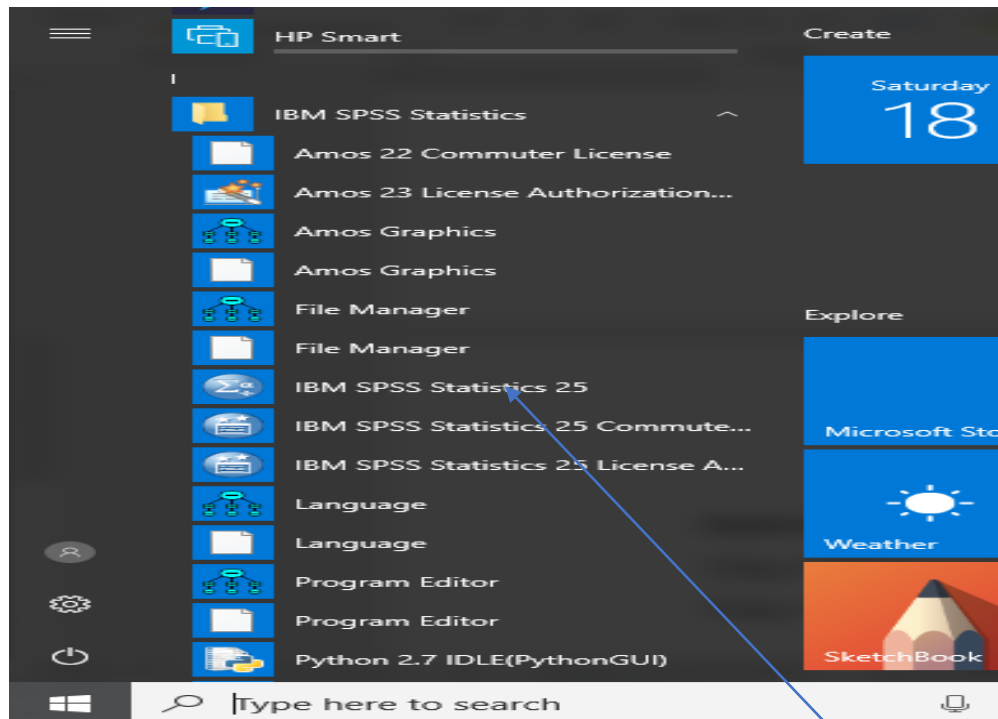


Figure 1.0

The SPSS has two views: -

- Data View
- Variable View

The views are located at the bottom of the screen once the software is lunched as illustrated below

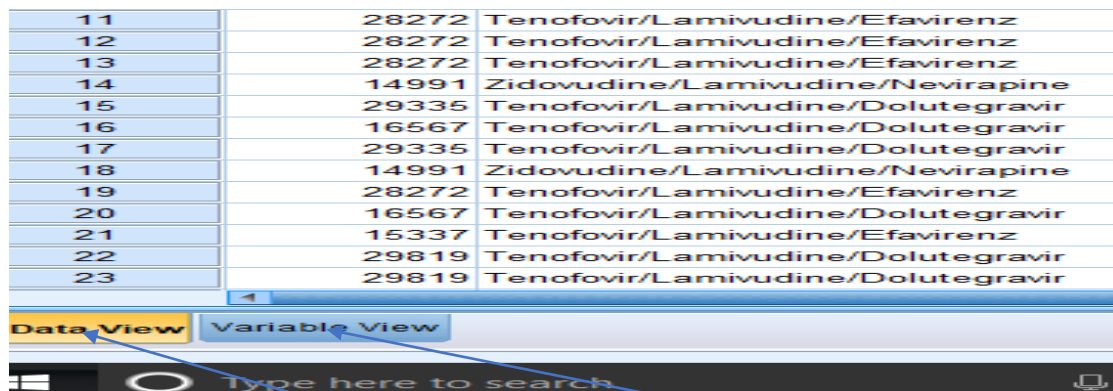


Figure 2.0

Data View highlighted

Variable View not highlighted

Variable View: The Variable is where all the variables on a data collection form are coded into the SPSS with the appropriate **Name, Type, Width, Decimals, Label, Values, Missing, Columns, Align, Measures and Role**. See Figure 3.0. The **Name** of the variable may abbreviations such as “Q1” to represent Question 1 for example. The **Name** must not have any space in between the characters. Click on the Type or format of the Variable. For open- ended Variables please select “**String**” and “**Numeric**” for any Variable that requires the use of values to represent the responses (Male =1, Female = 2, Yes =1, No = 2) and any other format that suitable for the Variable (See Figure 4.0). The **Width** is the require space to contain the response for the Variable. As indicated earlier, open -ended responses will require higher space, hence 250 or more is highly recommended. For responses with few characters, a Width of 10 or less may be appropriate. Generally, **Decimals** are not necessary for all responses except for precise measurements. Therefore, it can be set to zero where it is not necessary. The **Label** is the **description** of the Variable. For example, “Q2, what is your gender?”. The space under the **Label** expands as the user enters the Variable’s description.

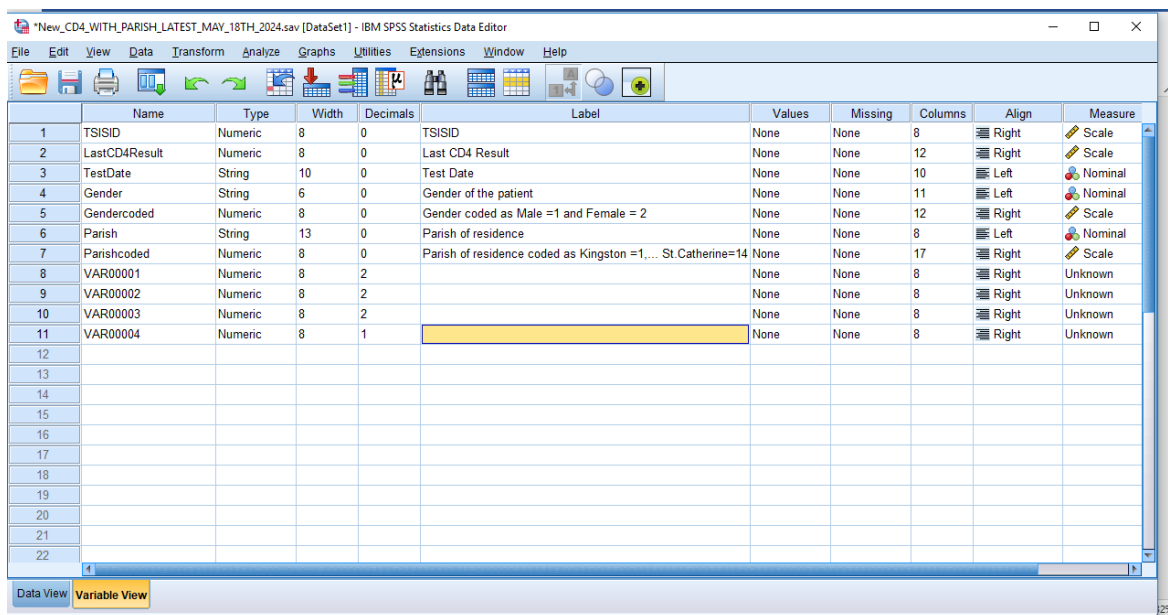


Figure 3.0

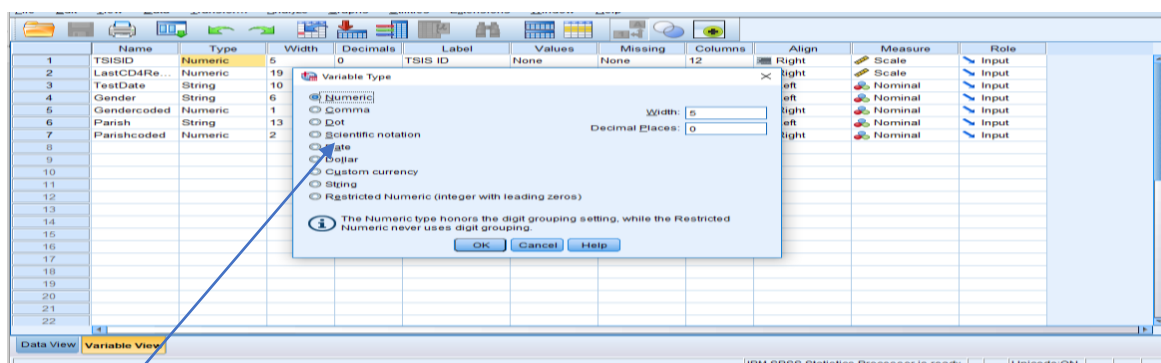


Figure 4.0

Values are necessary for response in form of a category. This illustrated in Figures 5.0 & 6.0 for a gender variable, click on the corner of the cell under the Values enter “1” for value and “Male” for Label, then “add”. Repeat the process for “2” and “Female” for Label. See Figures 5.0 and 6.0

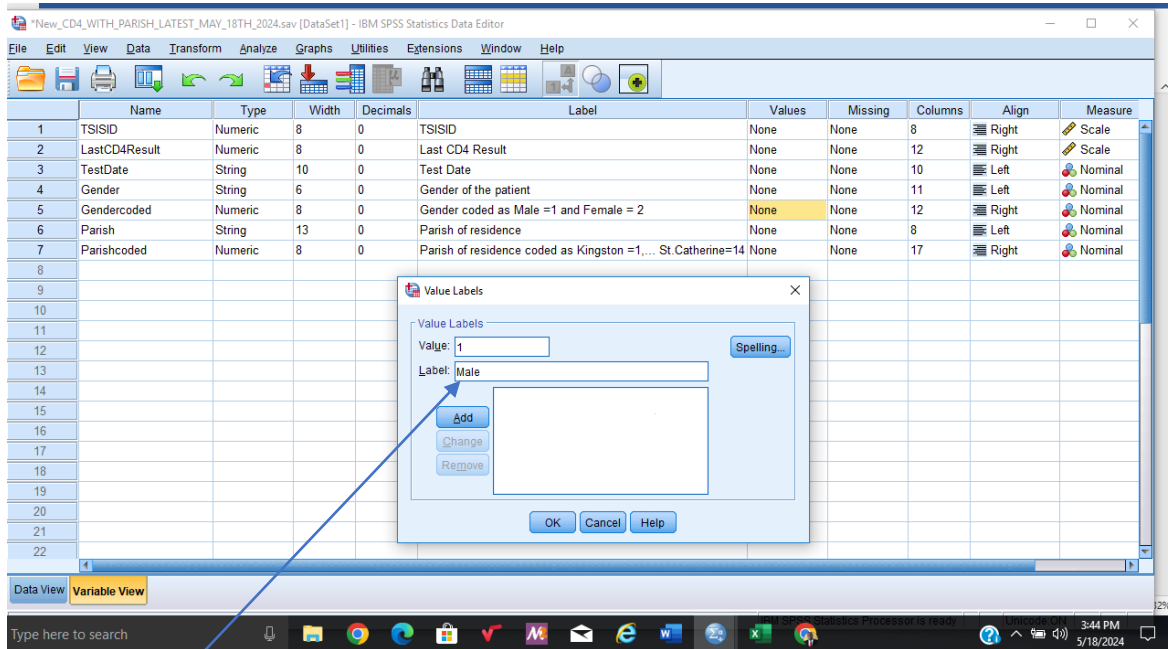


Figure 5.0

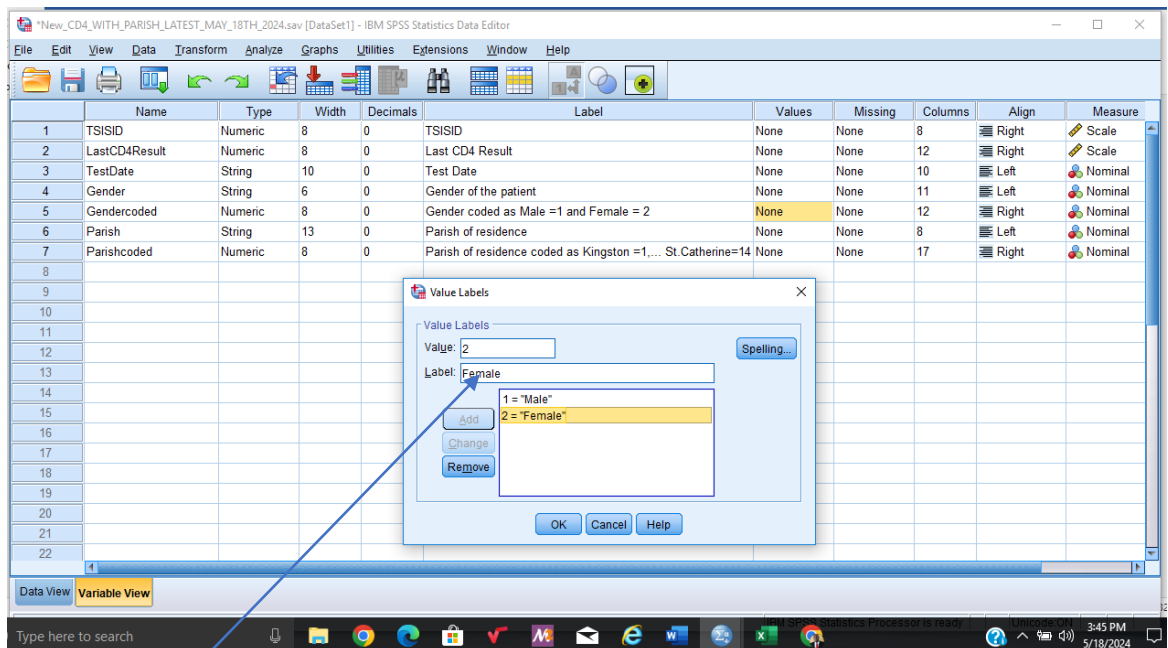


Figure 6.0

Variable Levels of Measurement & the Importance

- Ratio:- Variables that are measured on precise estimate. For example, height, weight, salary etc
- Interval: - Variables that are measured on reference or baseline and also precise. For example, temperature, pressure etc
- Ordinal: - Variables that are measured on an evaluative response or ranking (Excellent, Very Good, Good, Fair, Poor). Likert Scale items such as strongly agree, agree, undecided, disagree, strongly disagree etc.
- Nominal: - Variables that are measured on categories of responses, such as the marital status, race, parish, countries etc
-

Levels of Measurement in SPSS

- SCALE :- This is a combination of Ratio and interval
- NOMINAL :- As defined earlier
- ORDINAL :- As defined earlier

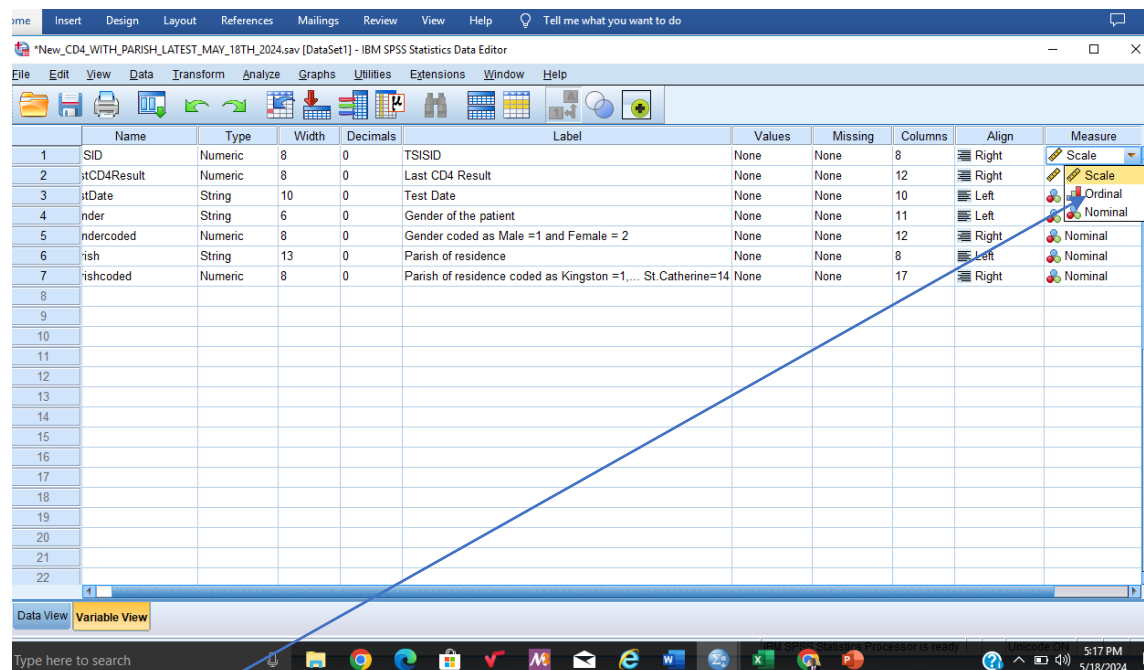


Figure 7.0

Importance

These levels of measurement help the data analyst to indicate to the software the appropriate measurement to facilitate the correct analysis. For example, the continuous variables level of measurement will be “SCALE” and finding averages in the descriptive statistics is suitable for

these variables compare with the variables measured at the “NOMINAL” level of measurement which will be more suitable for counting to generate a Frequency Table. In summary:

- It facilitates the application of the correct statistical test
- It also facilitates the relevant statistical analysis

2.3 SPSS Template /Data Entry /Merging Cases & Variables /Logical Operators/ Recoding

SPSS Template for Data Entry

To create a template for a data entry of survey instrument or other data collection instruments, **go to the SPSS Variable View**. See example below:

Assuming that you are going to create a data entry template for the following items :
Questionnaire ID (QID), Age, Gender (Male =1, Female =2), Marital Status (Married =1 , Single = 2, Widow = 3, Common Law = 4)
Gender & Marital Status are Nominal level of Measurement Variables
QID & Age are Scale level of Measurement Variables (See Figures 8.0 & 9.0)

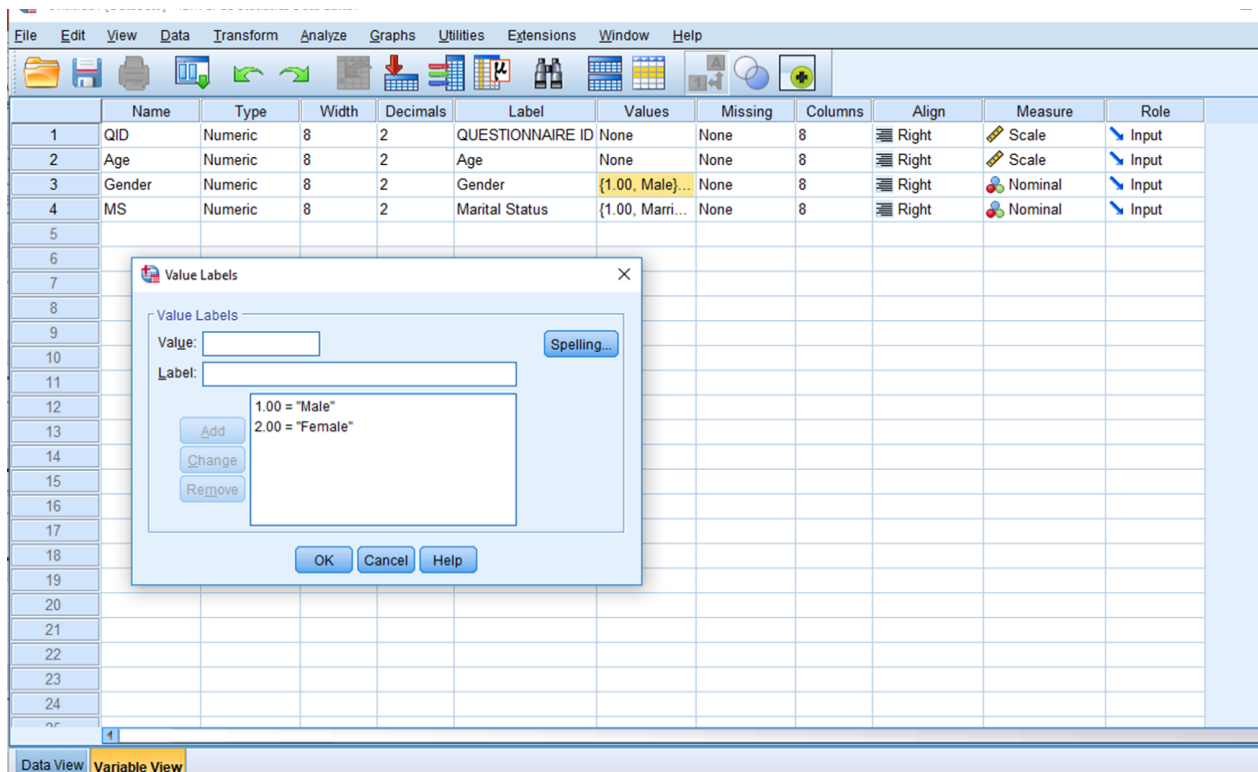


Figure 8.0

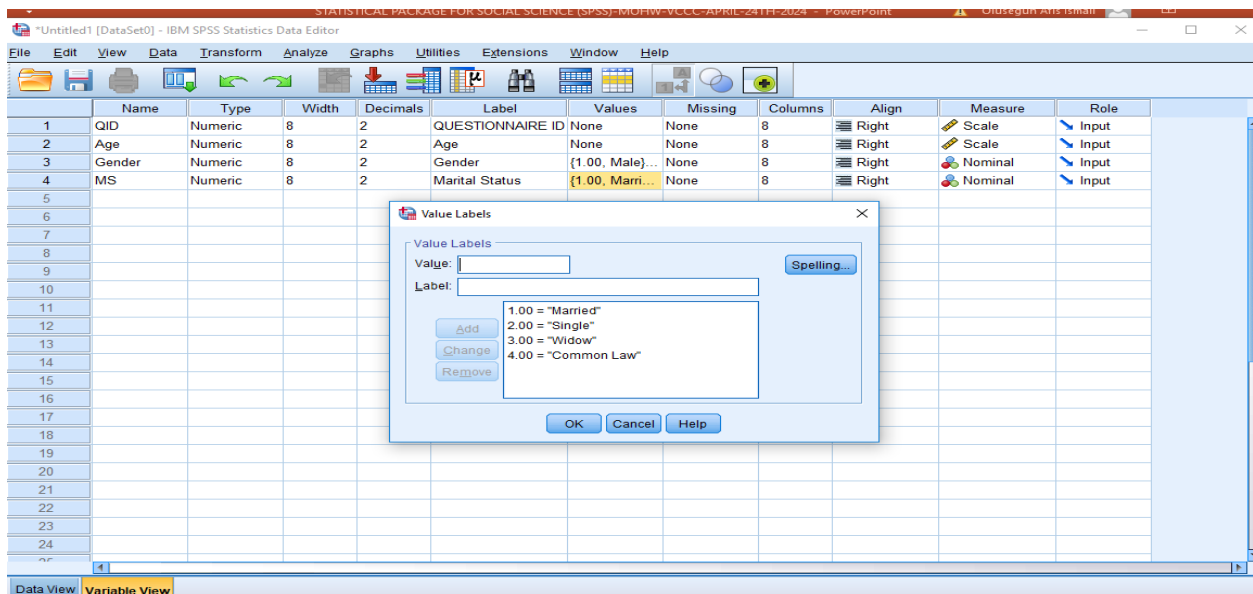


Figure 9.0

Importing Data from Microsoft Excel to SPSS & Vice Versa

Importing from Microsoft Excel to SPSS will require the user to go to **File** then **Import Data** then select the file appropriate file format as illustrated below (See Figures 10.0,11,12 &13)

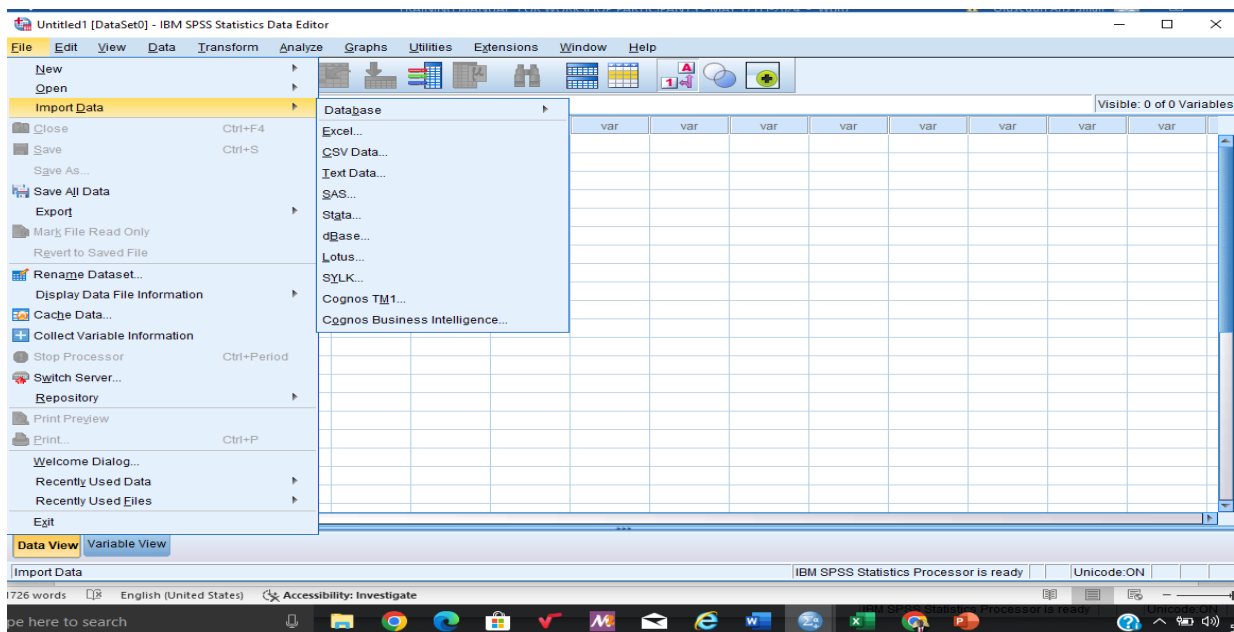


Figure 10.0

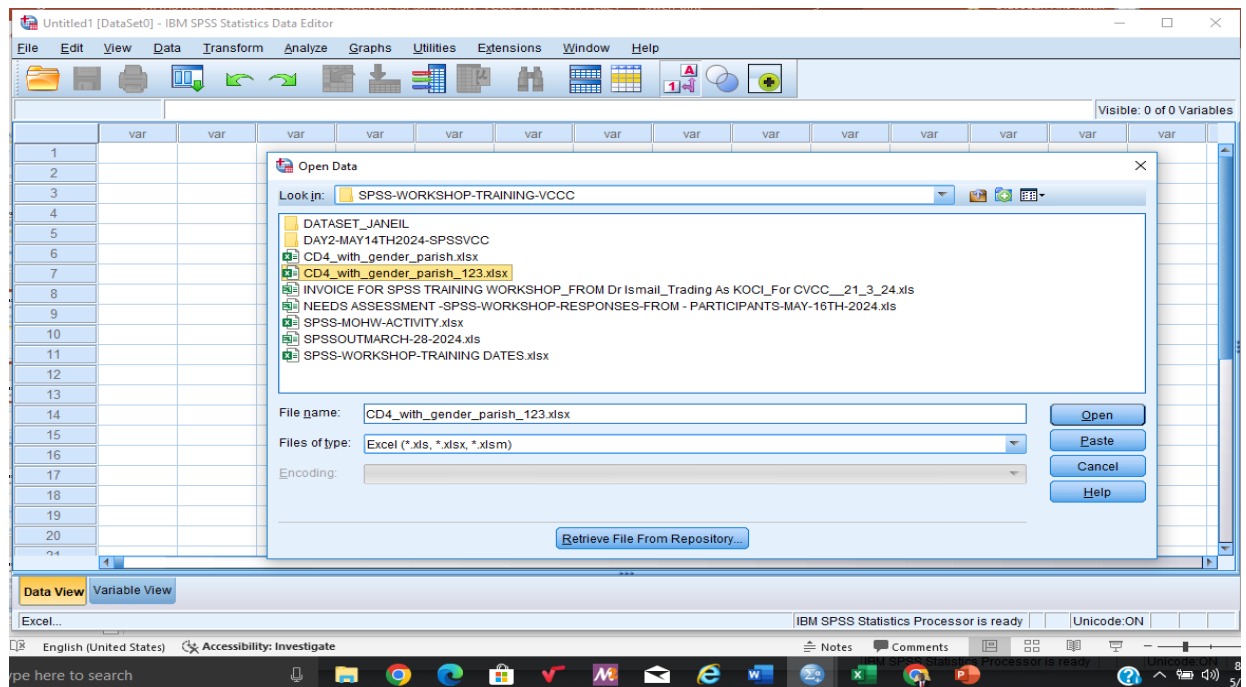


Figure 11.0

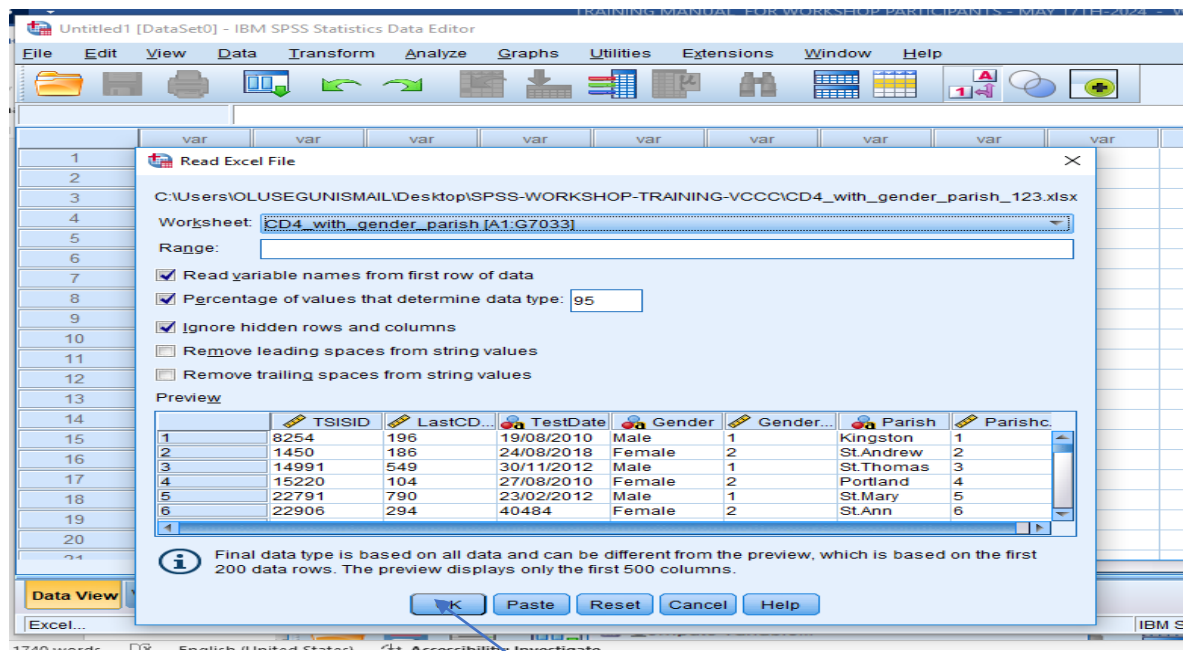


Figure 12.0

Click OK

Importing CSV file format into SPSS

To import CSV, start with by clicking on File then **CSV Data** as illustrated in **Figure 13.0** follow by the steps in **Figures 14.0, 15.0 & 16.0** respectively.

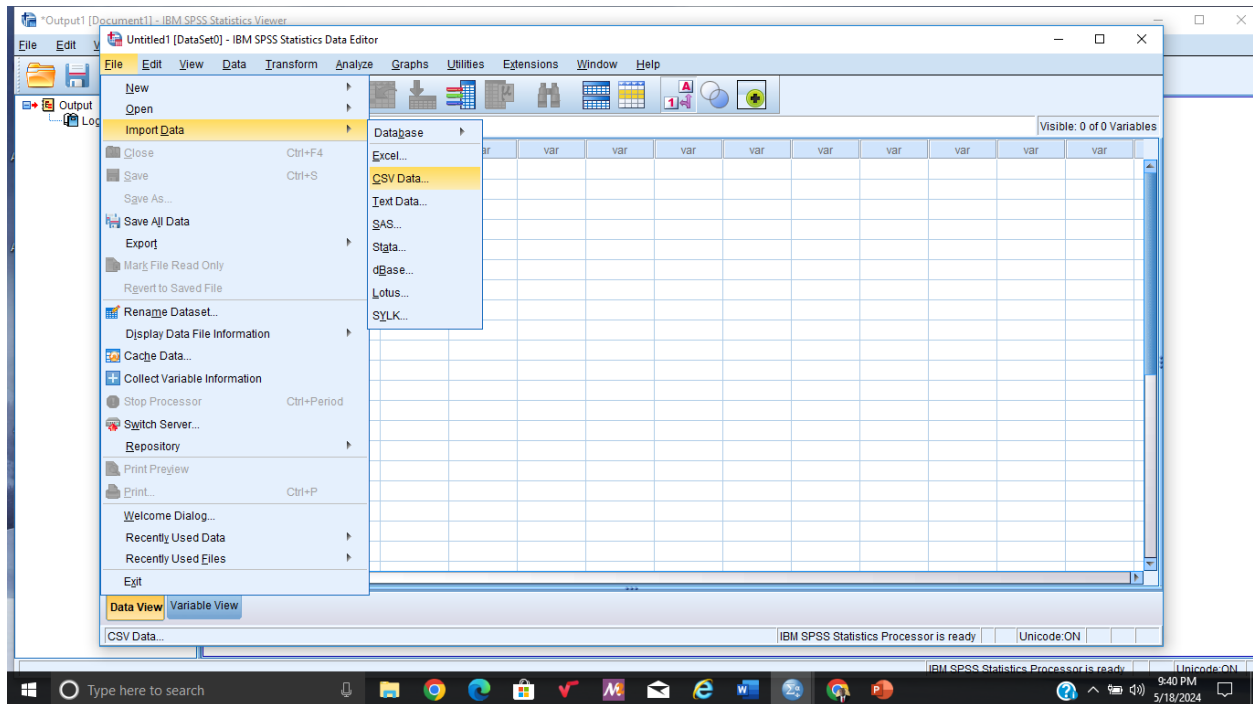


Figure 13.0

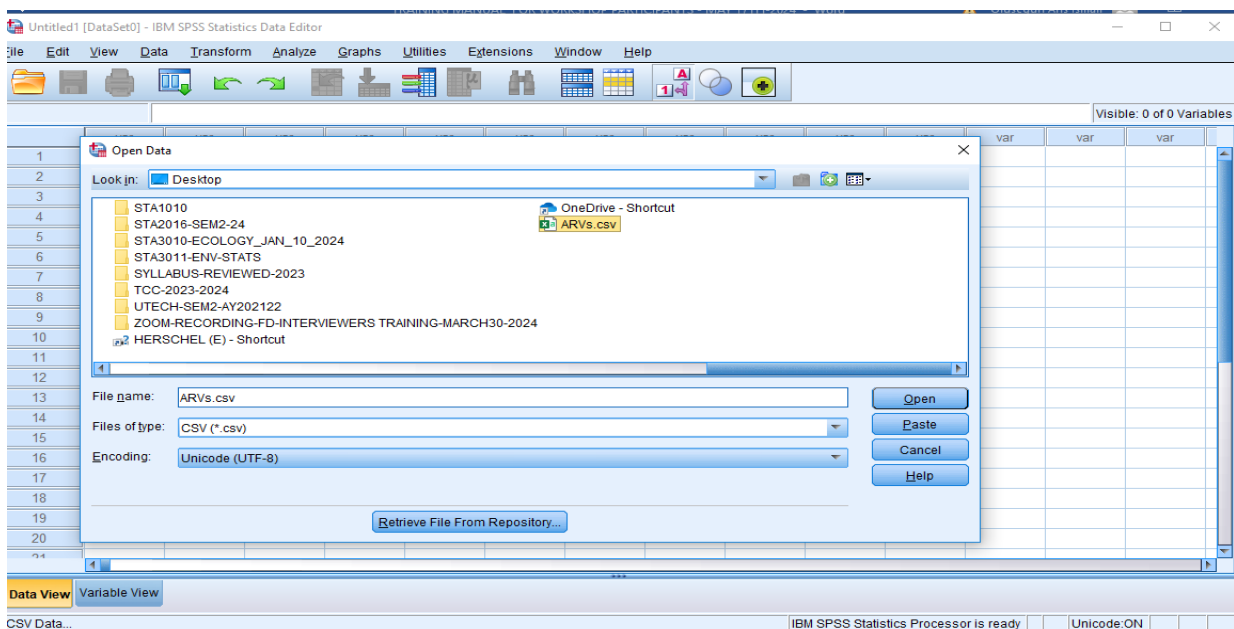


Figure 14.0

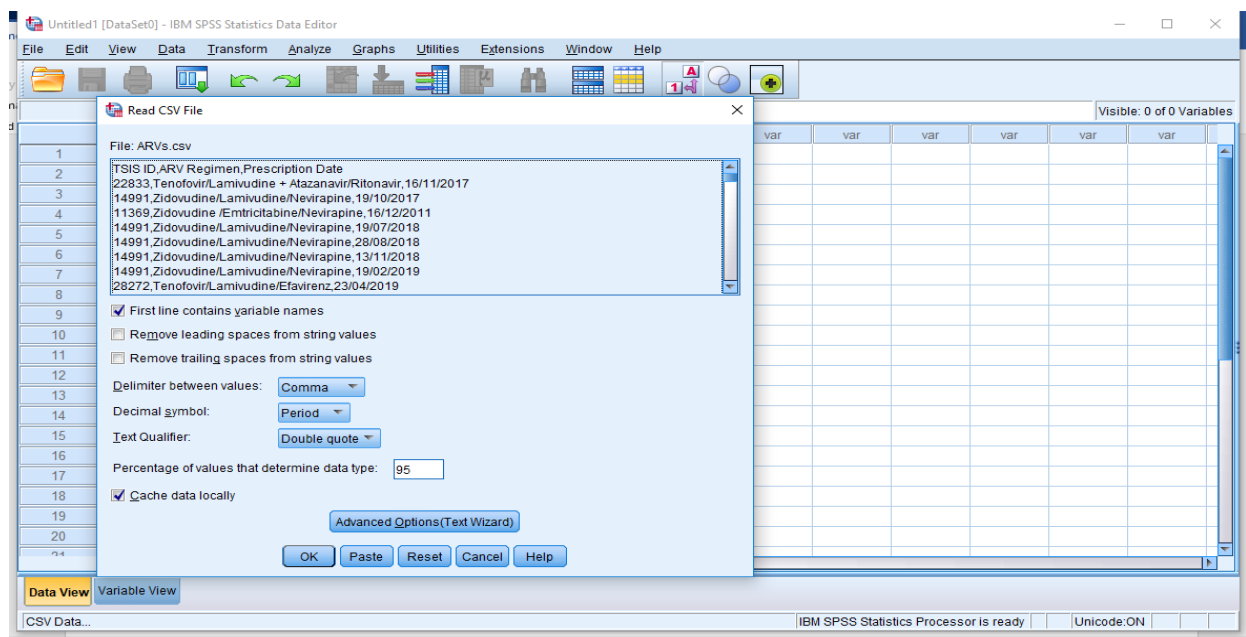


Figure 15.0

	TSISID	ARVRegimen	Prescription Date	var	var	var	var	var	var	var	var	var	var
1	22833	Tenofovir/Lamivudine + Atazanavir/Rito...	16/11/2017										
2	14991	Zidovudine/Lamivudine/Nevirapine	19/10/2017										
3	11369	Zidovudine /Emtricitabine/Nevirapine	16/12/2011										
4	14991	Zidovudine/Lamivudine/Nevirapine	19/07/2018										
5	14991	Zidovudine/Lamivudine/Nevirapine	28/08/2018										
6	14991	Zidovudine/Lamivudine/Nevirapine	13/11/2018										
7	14991	Zidovudine/Lamivudine/Nevirapine	19/02/2019										
8	28272	Tenofovir/Lamivudine/Efavirenz	23/04/2019										
9	28272	Tenofovir/Lamivudine/Efavirenz	27/06/2019										
10	28272	Tenofovir/Lamivudine/Efavirenz	20/11/2018										
11	28272	Tenofovir/Lamivudine/Efavirenz	20/09/2018										
12	28272	Tenofovir/Lamivudine/Efavirenz	9/8/2018										
13	28272	Tenofovir/Lamivudine/Efavirenz	31/07/2018										
14	14991	Zidovudine/Lamivudine/Nevirapine	20/08/2019										
15	29335	Tenofovir/Lamivudine/Dolutegravir	14/01/2020										
16	16567	Tenofovir/Lamivudine/Dolutegravir	30/01/2020										
17	29335	Tenofovir/Lamivudine/Dolutegravir	11/2/2020										
18	14991	Zidovudine/Lamivudine/Nevirapine	18/02/2020										
19	28272	Tenofovir/Lamivudine/Efavirenz	27/02/2020										
20	16567	Tenofovir/Lamivudine/Dolutegravir	5/3/2020										

Figure 16.0

Transform: Computing & Combining & Recoding Variables in SPSS

Transforming a variable from one form to another is done under the “Transform” menu in the SPSS. This menu offers several boxes for data manipulation such as computing of new variable, merging of cases, combining of variables and recoding of a variable. **Figure 17.0** illustrate the Transform menu option in SPSS.

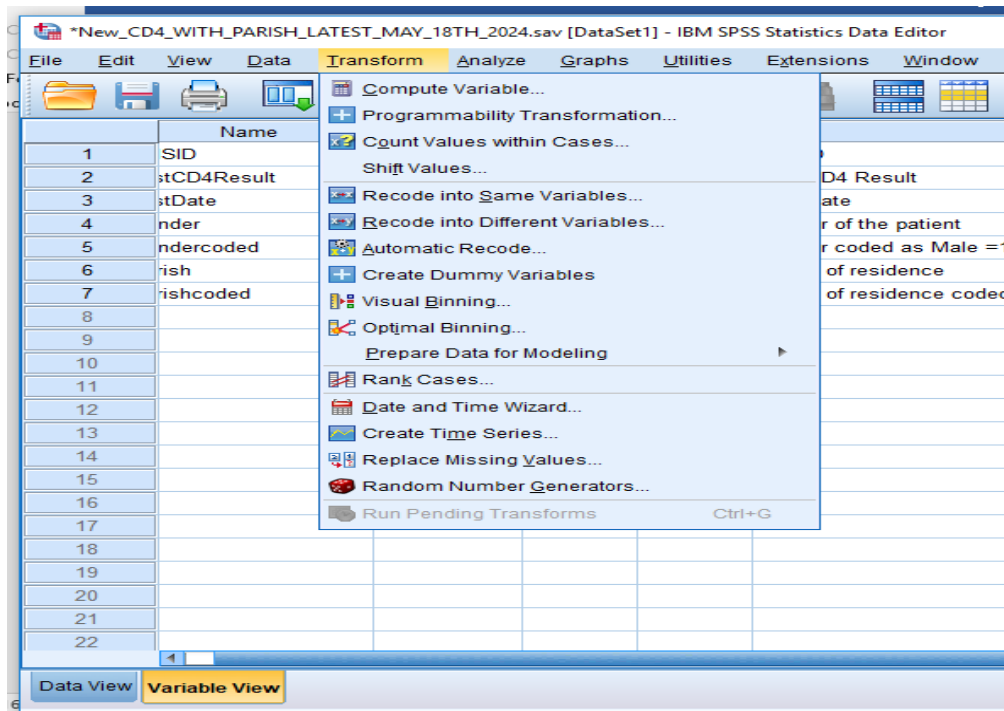


Figure 17.0

Computing a Variable & Use of Logical Operators

The compute menu facilitates the use of the logical operators to create a new variable. The logical operators are simply arithmetic signs to communicate the intention of the data analyst. To create a new variable, click on the **Transform** menu (**Figure 18.0**) follow by the **Compute Variable** (**Figure 18.0**). An example is the creation of the variable “HIGHERTSID” which will contain all the cases with **TSISD >130** (See **Figure 19.0**).

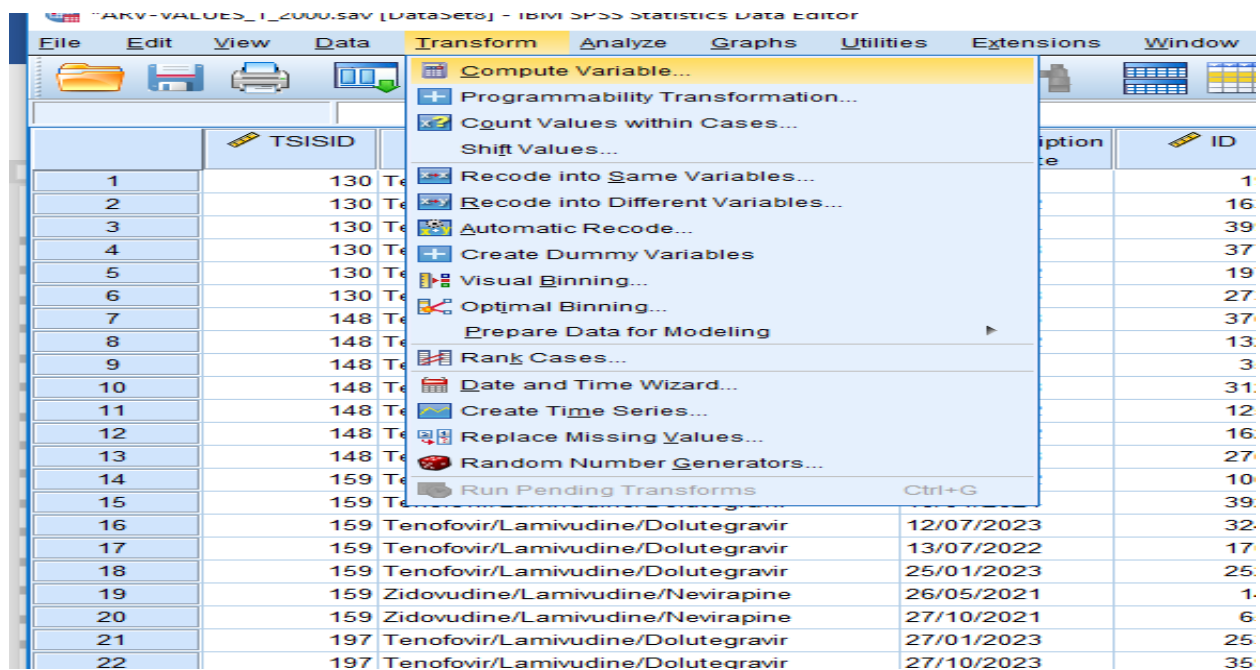


Figure 18.0

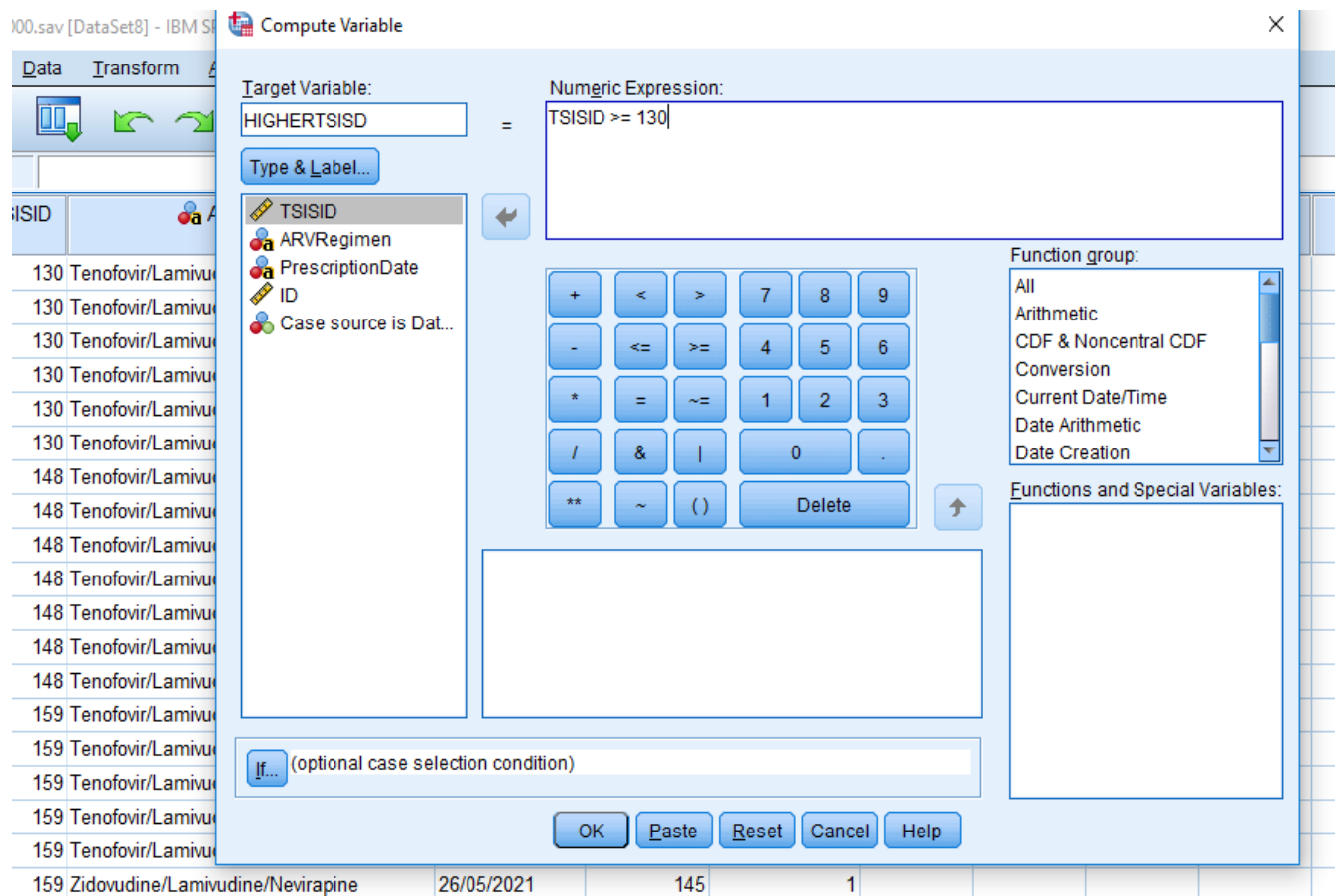


Figure 19.0

Recoding Variables in SPSS

The **Recode** menu can be used to recode a **ratio** or **interval level of measurement** into **Same** variable or **Different** variable (Figure 20.0). The **Same** variable menu is used to collapse cells or groups.

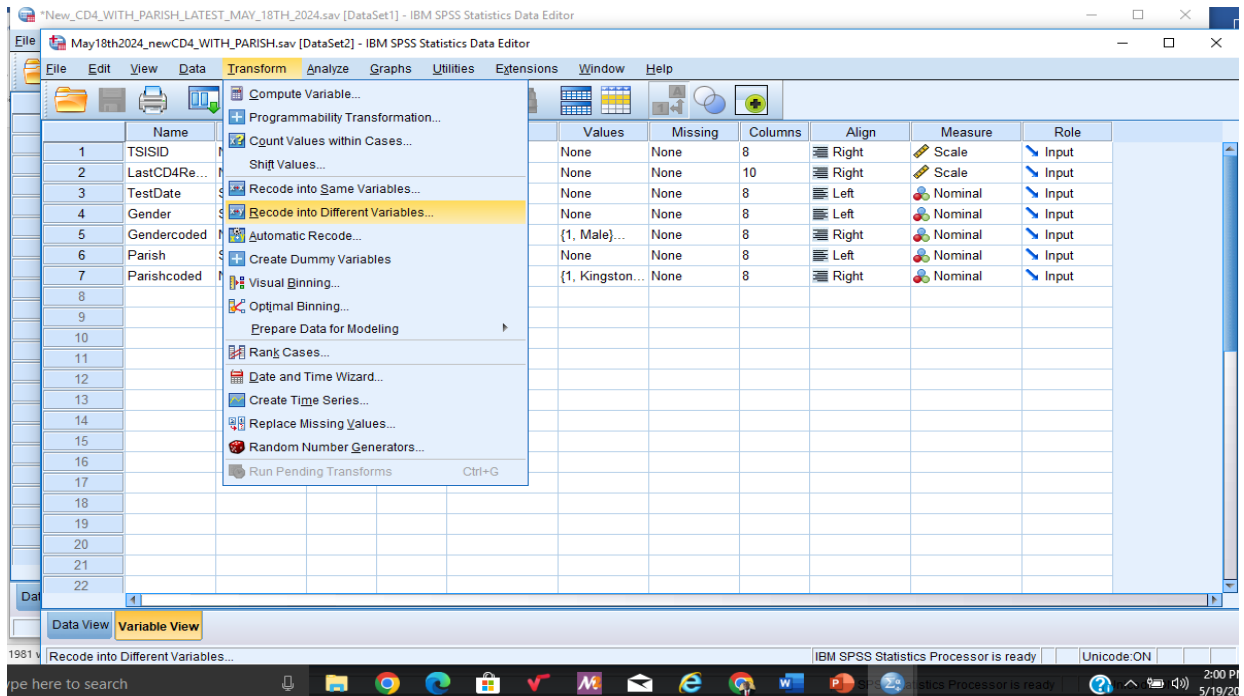


Figure 20.0

Recode into Different Variable. This is the menu used to create a new variable from the existing variable. An example is the recoding of the TSISID data into a new variable that will take the format of a category. The new variable will take the **Name** "NewTSISD", this name will be entered under **New** and the **Label**. Click on **Change** to complete the process. (See Figure 21.0)

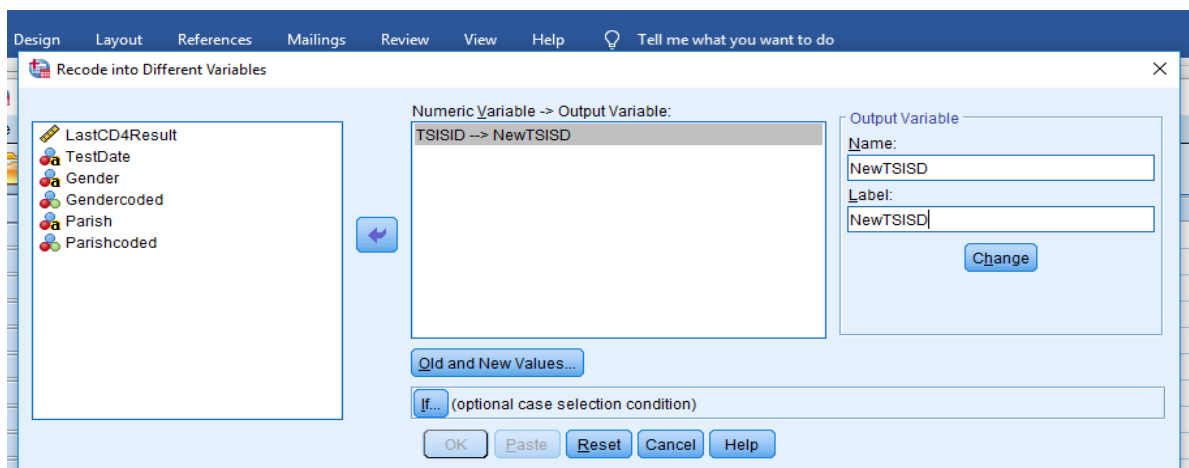


Figure 21.0

In the recoding steps, values from 1 to 400 is coded as 1; then the user should click on “Add”. This process is repeated for 401 through 600 coded as 2. (See Figures 22.0 & 23.0)

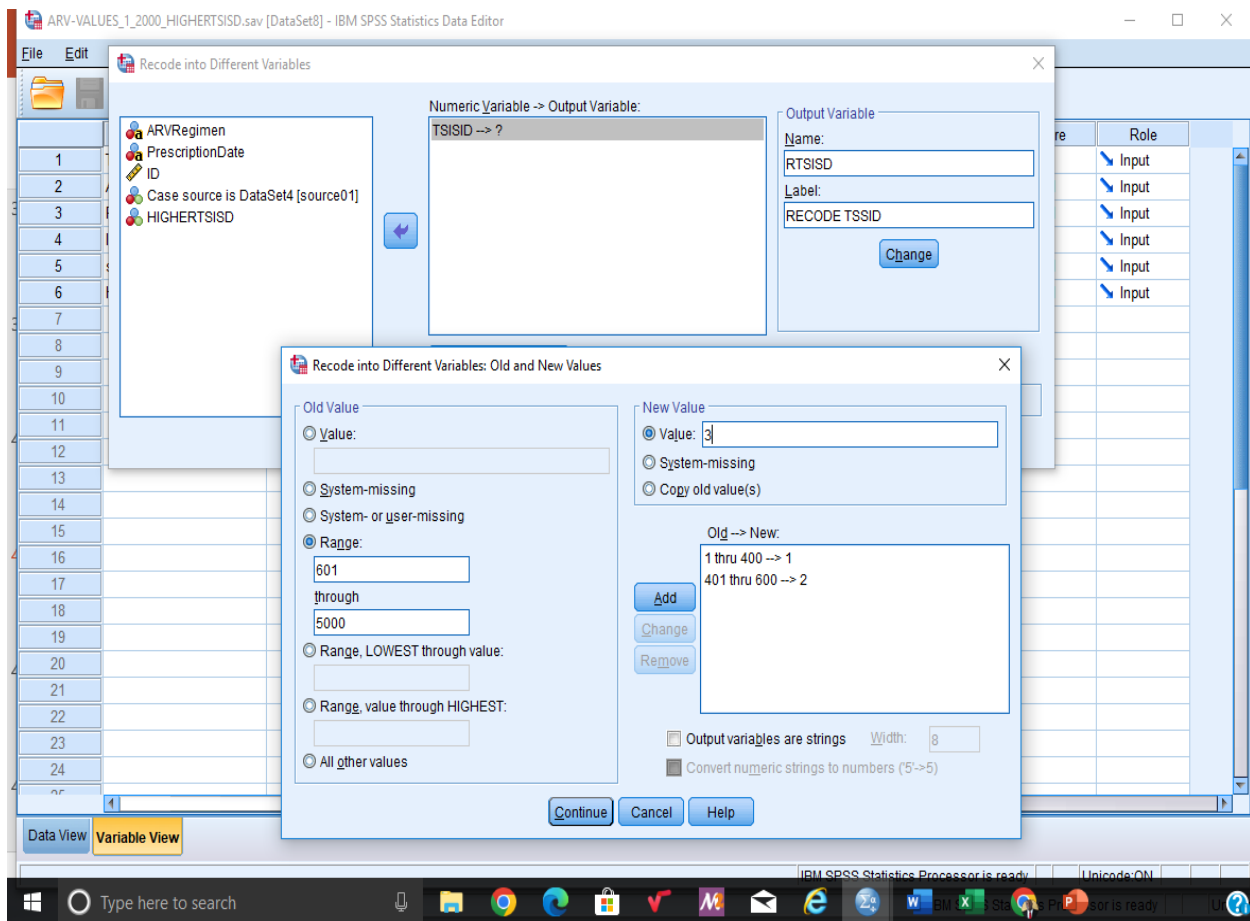


Figure 22.0

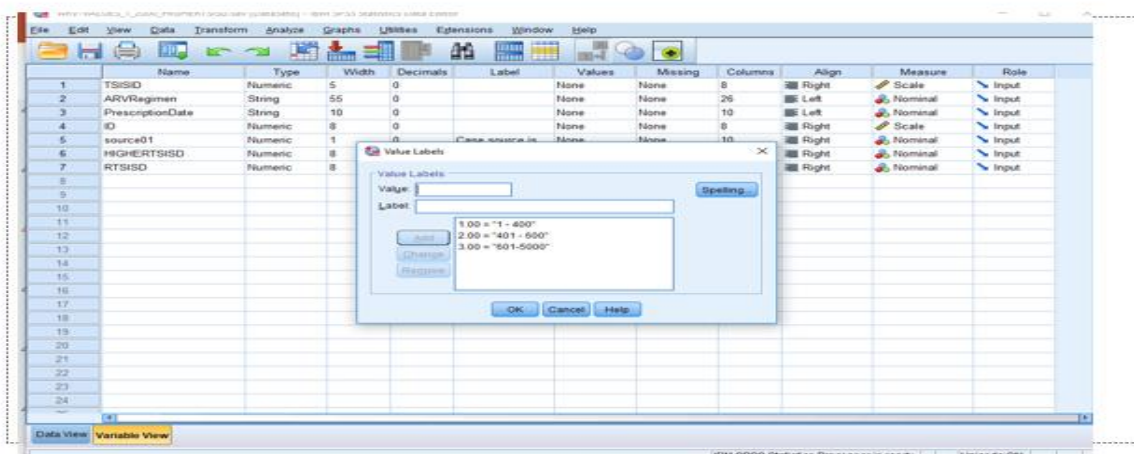


Figure 23.0

Data Menu Option - Merging Cases and Variables in SPSS

The Data menu provides several analytic options including the merging of cases and process of combining variables. To merge cases, it is important to have a unique identifier. **Figure 24.0** illustrates the Data menu followed by **Add Cases** or **Add Variables**.

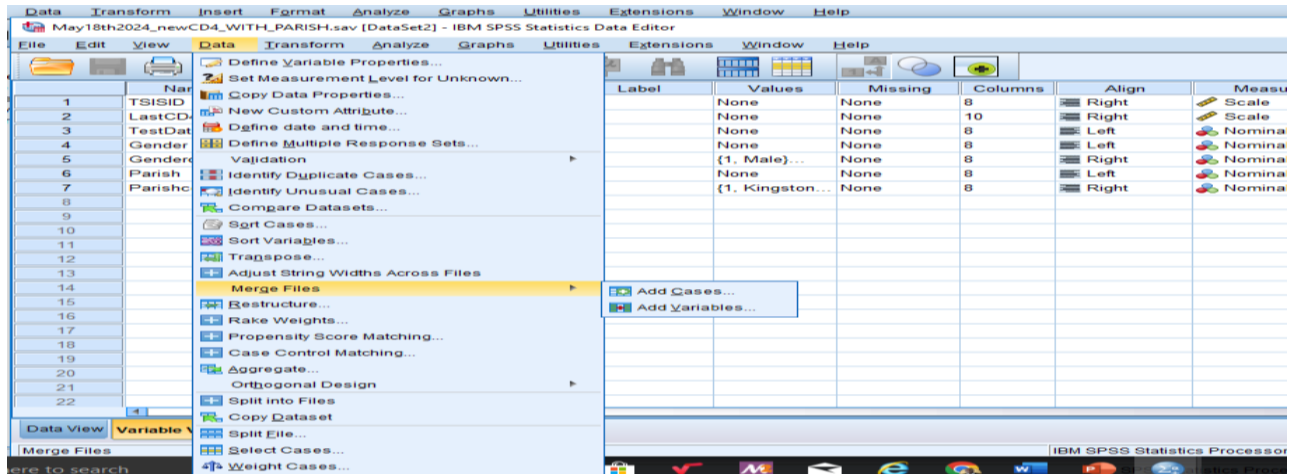


Figure 24.0

Click on **Add Cases** to merge data, it is important to select the source (open or external), See **Figure 25.0**

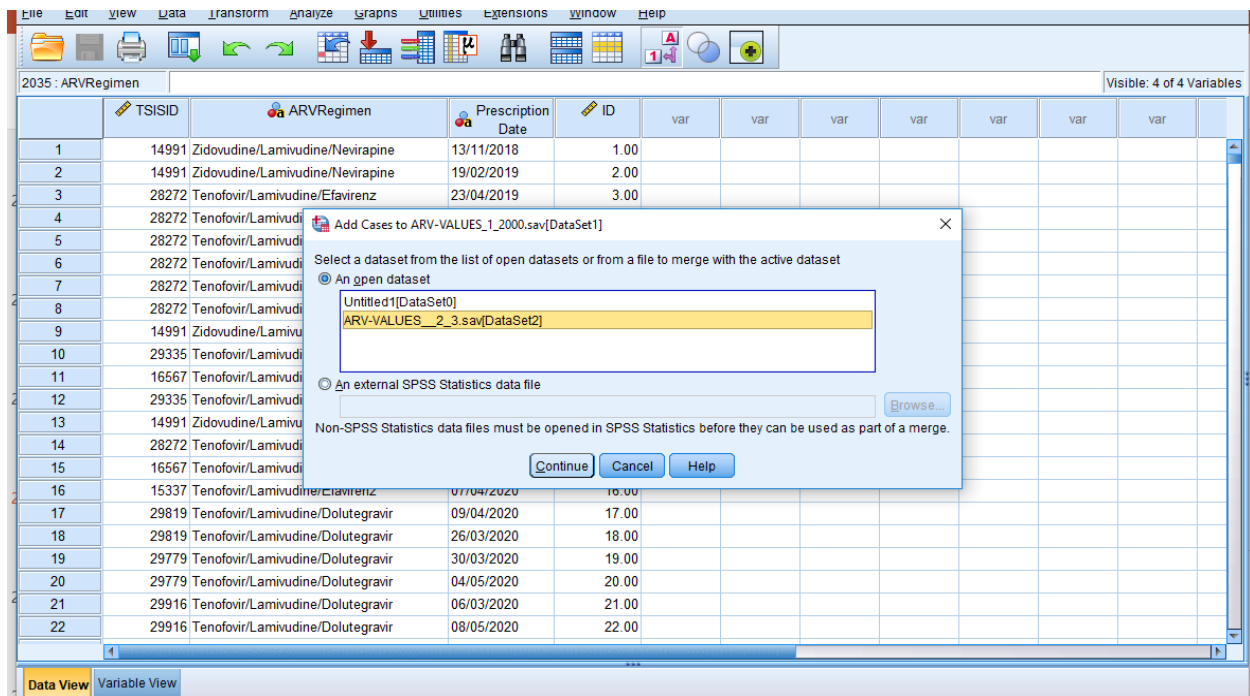


Figure 25.0

Descriptive Statistics

The descriptive statistics consist of measures of central tendency, measures of dispersion and graphical illustrations. The measures of central tendency include the mean, median and mode. These measures are used to represent averages of continuous variables. The **mean** is average of all the values in the dataset. The **median** is the middle number after the values are arranged in ascending order while the **mode** is the most occurring number in a data set. The graphical illustration includes pie charts, bar charts, histogram, box-plot and stem – leaf diagram.

The measures of dispersion include range, standard deviation, variance, mean deviation, quartile, coefficient of variation and skewness. Dispersion implies spread of the data. The variance is a measure of the variation within the data when compared with the average. The square root of the variance is the standard deviation. In some research findings, the mean and the standard deviation are reported for the descriptive statistics regarding continuous variables.

Measures of Dispersion

Coefficient of Variation : The ratio of the standard deviation to the average in percentage terms.

Skewness : - A measure of the data symmetry. A dataset may be positively skewed (Mean > Median > Mode) . May be negatively skewed (Mean < Median < Mode)

Normal :- A data set may follows the normal distribution if it is symmetrical (Mean approximately equal to the Median & Mode)

Figure 26.0 illustrate the descriptive statistics regarding the variable LastCD4Results

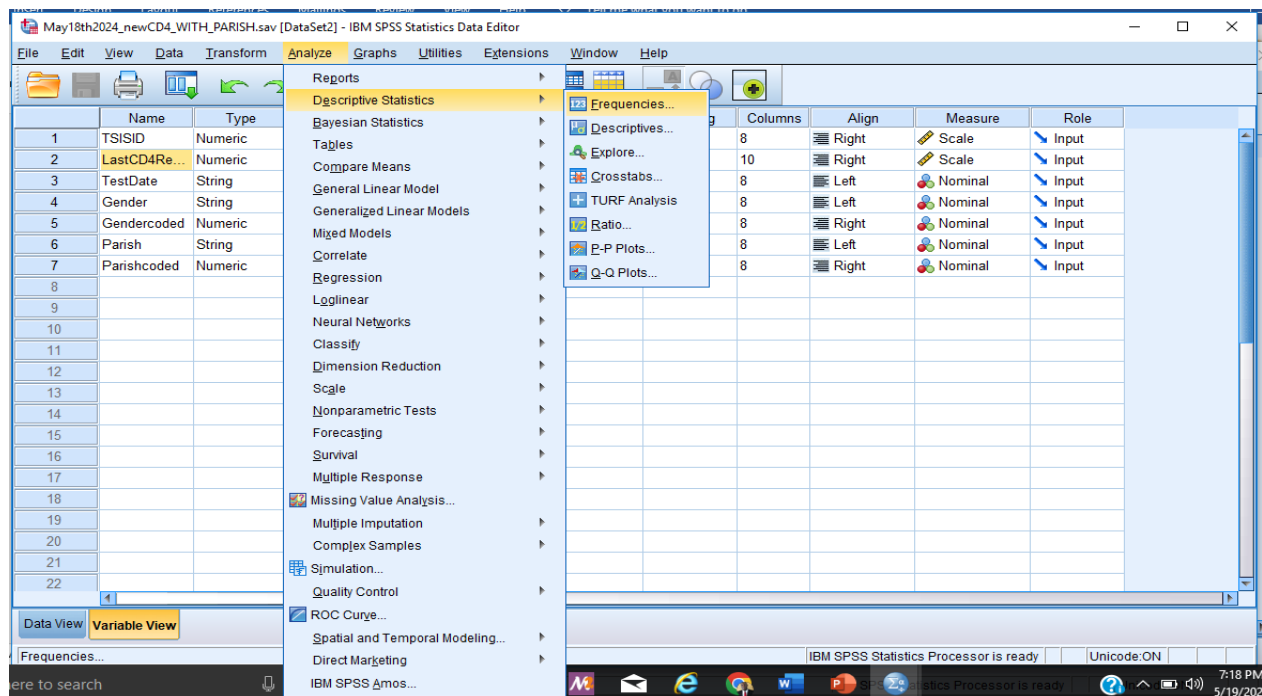


Figure 26.0

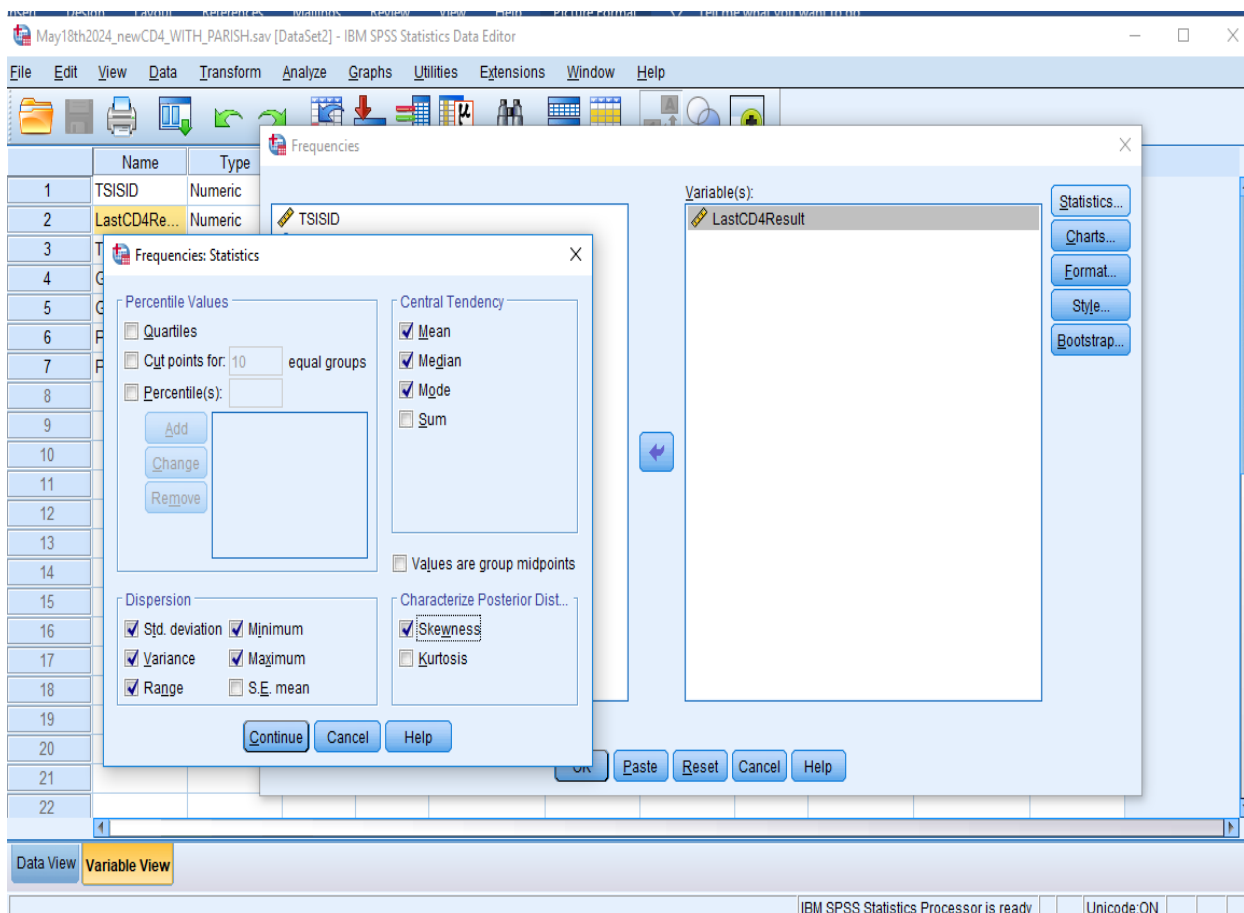


Figure 27.0

SPSS Output- Descriptive Statistics (Measures of Central Tendency & Dispersion)

<i>Statistics</i>		
LastCD4Result		
N	Valid	7030
	Missing	2
Mean		556.86
Median		439.00
Mode		68
Std. Deviation		2430.522
Variance		5907439.359
Skewness		40.282
Std. Error of Skewness		.029
Range		129833
Minimum		0
Maximum		129833

Graphs in SPSS (See Figures 28.0 & 29.0)

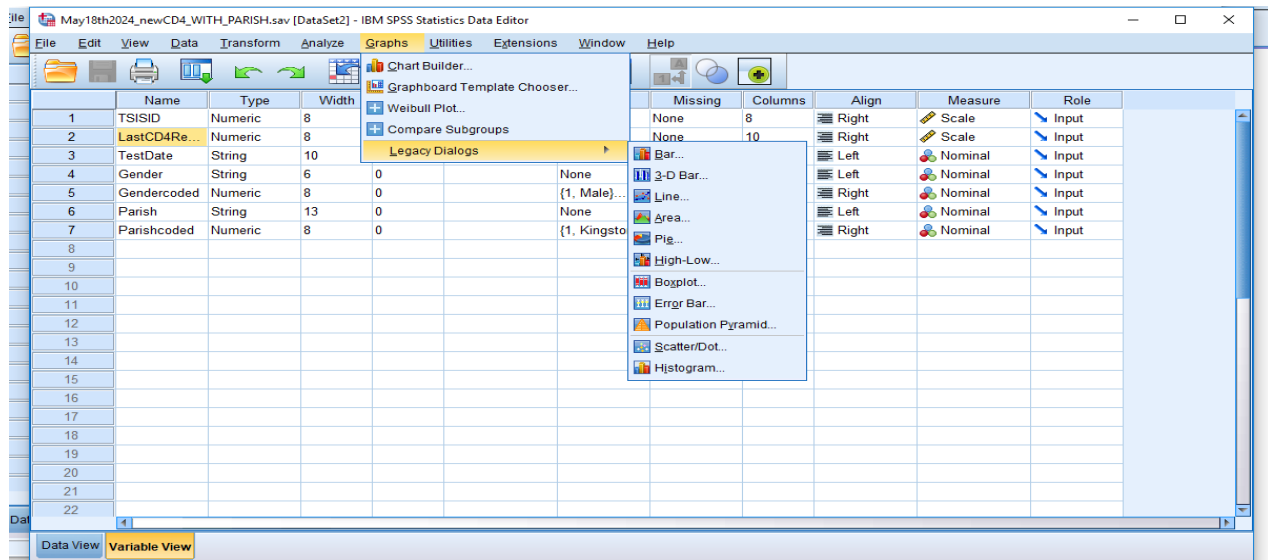


Figure 28.0

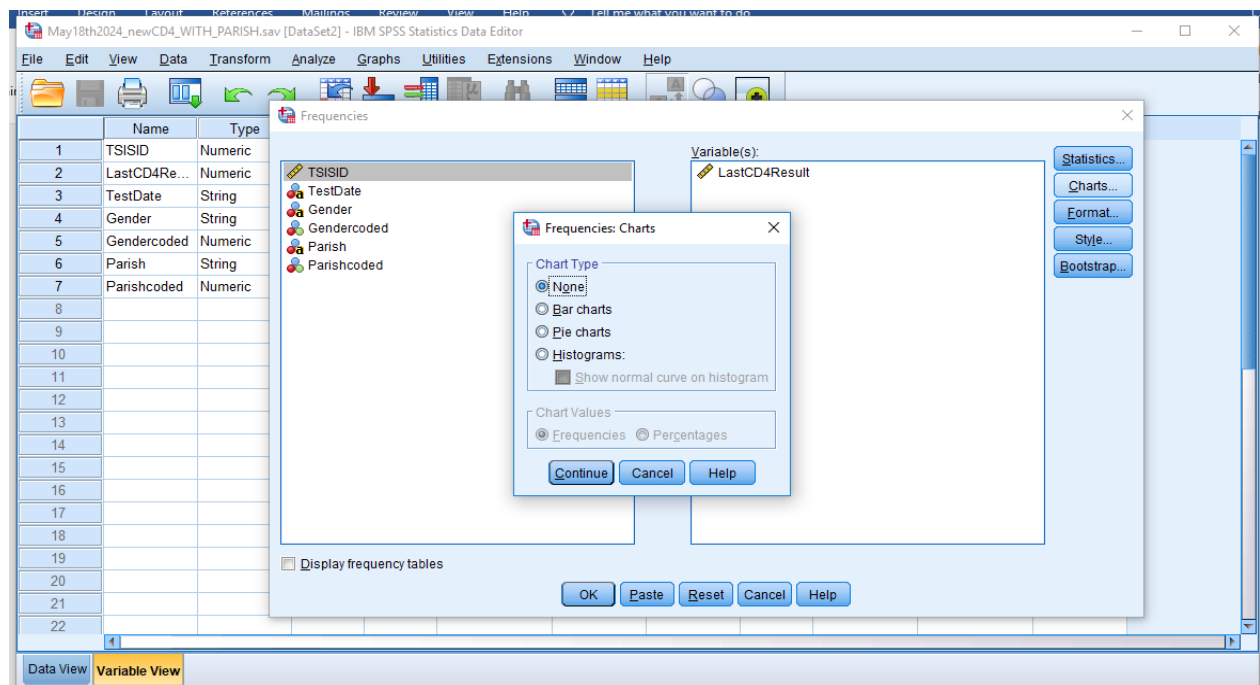


Figure 29.0

Variables that are measured at the nominal levels of measurement are best illustrated with Figures (bar chart, pie chart). Variables that are measured at the scale levels of measurement are best illustrated using histogram, box –plot, stem and leaf diagram to see the spread of the data

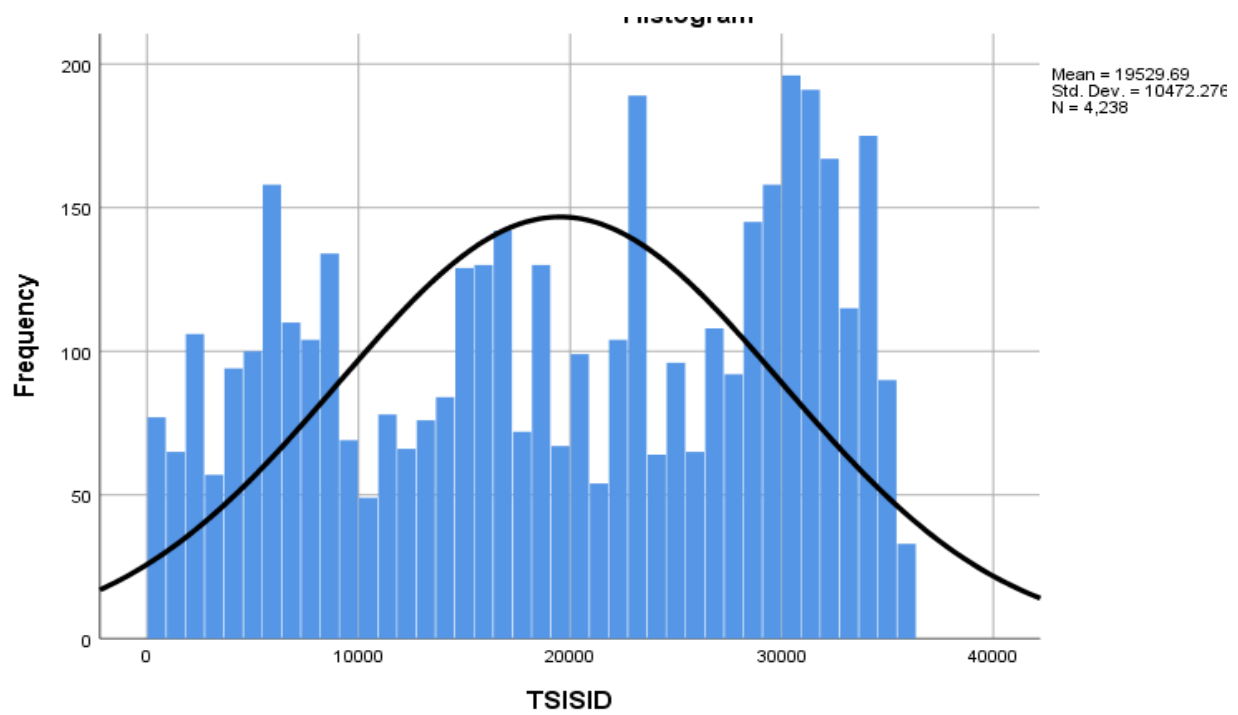


Figure 30.0

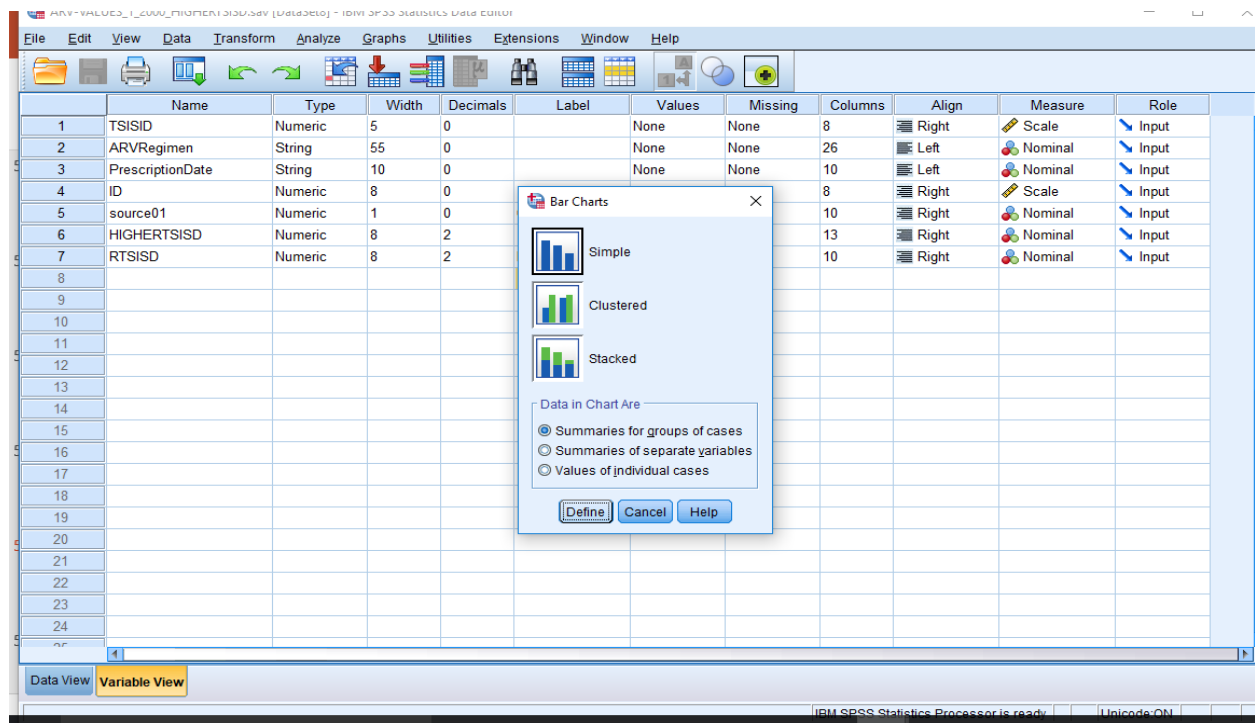


Figure 31.0

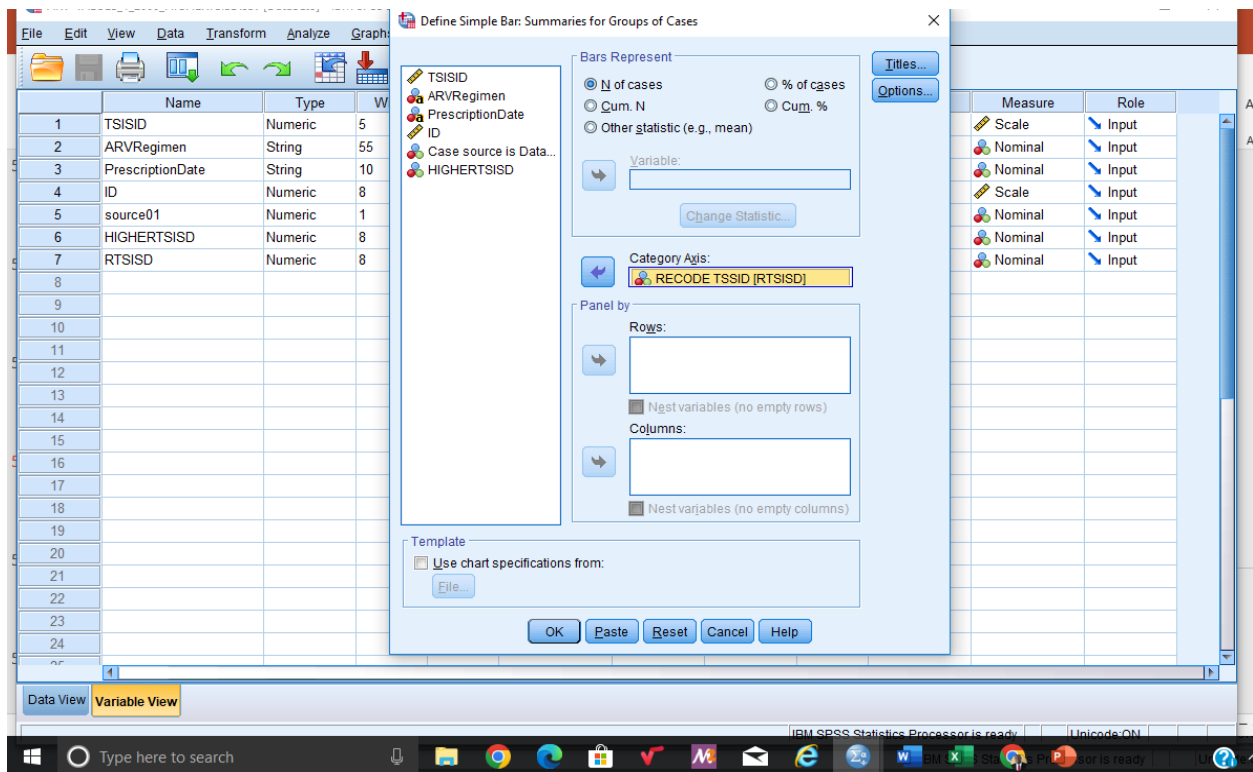


Figure 32.0

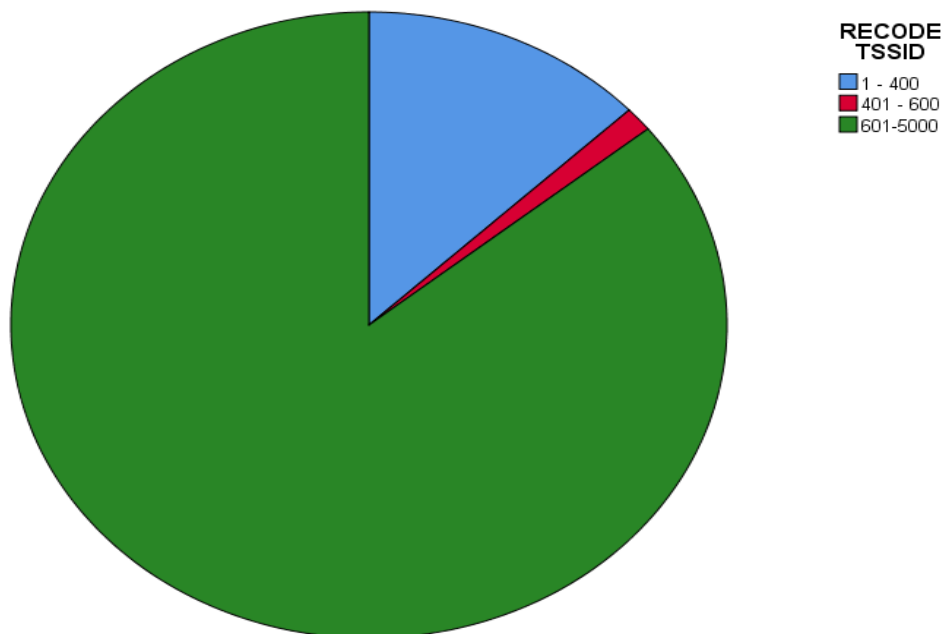


Figure 33.0

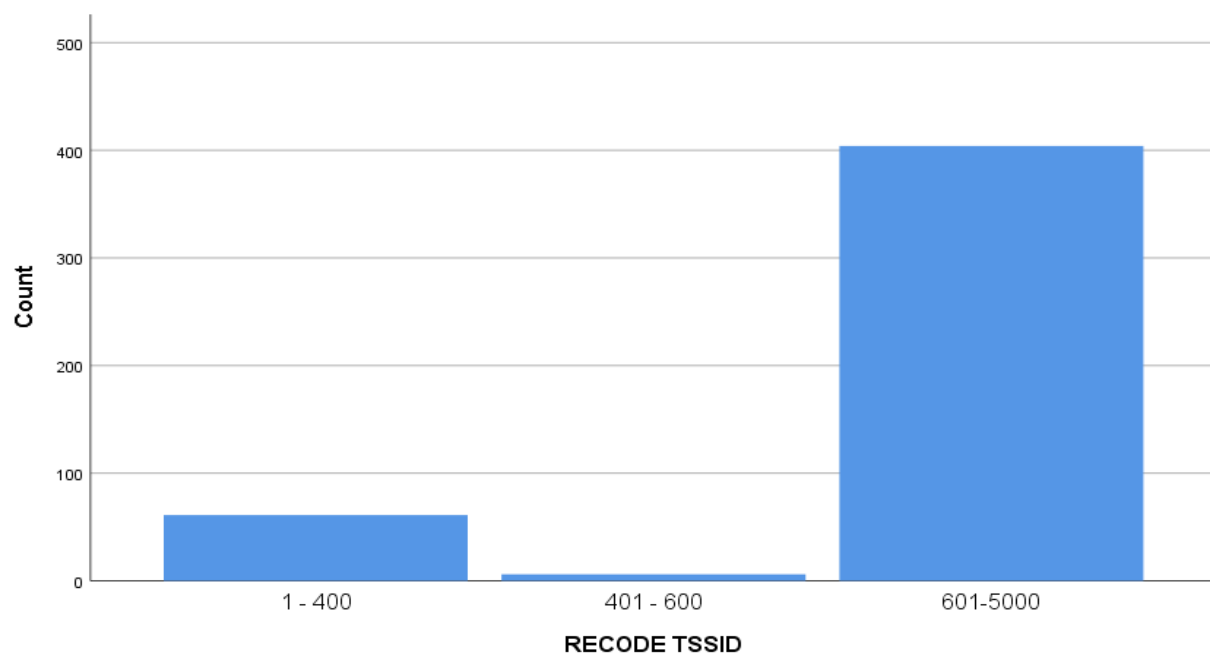


Figure 34.0

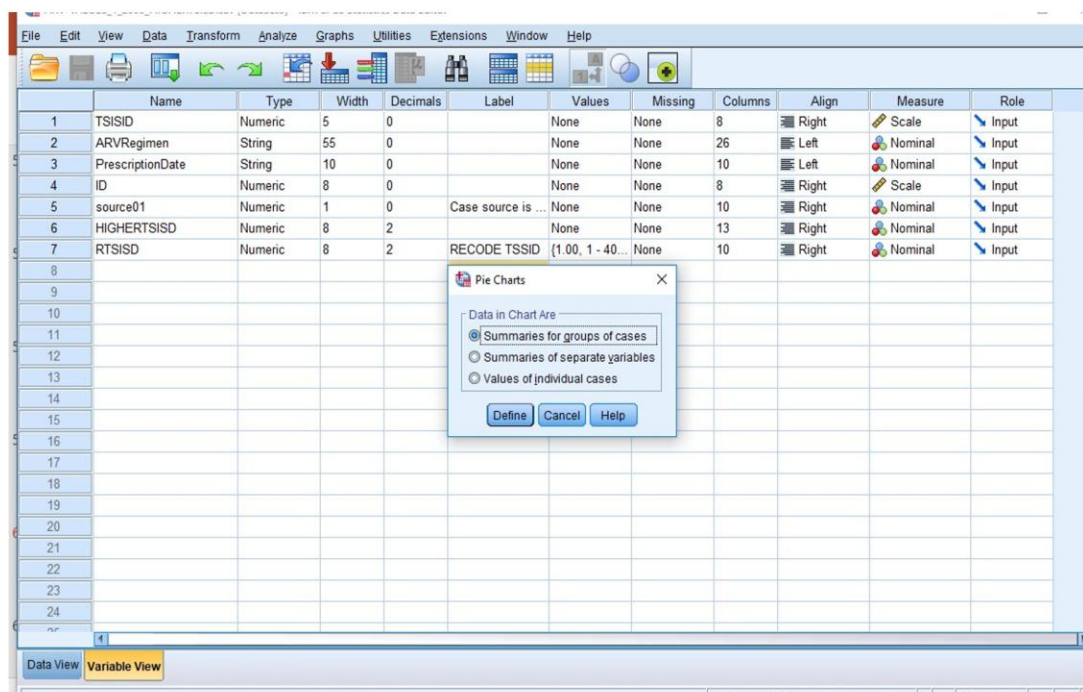


Figure 35.0

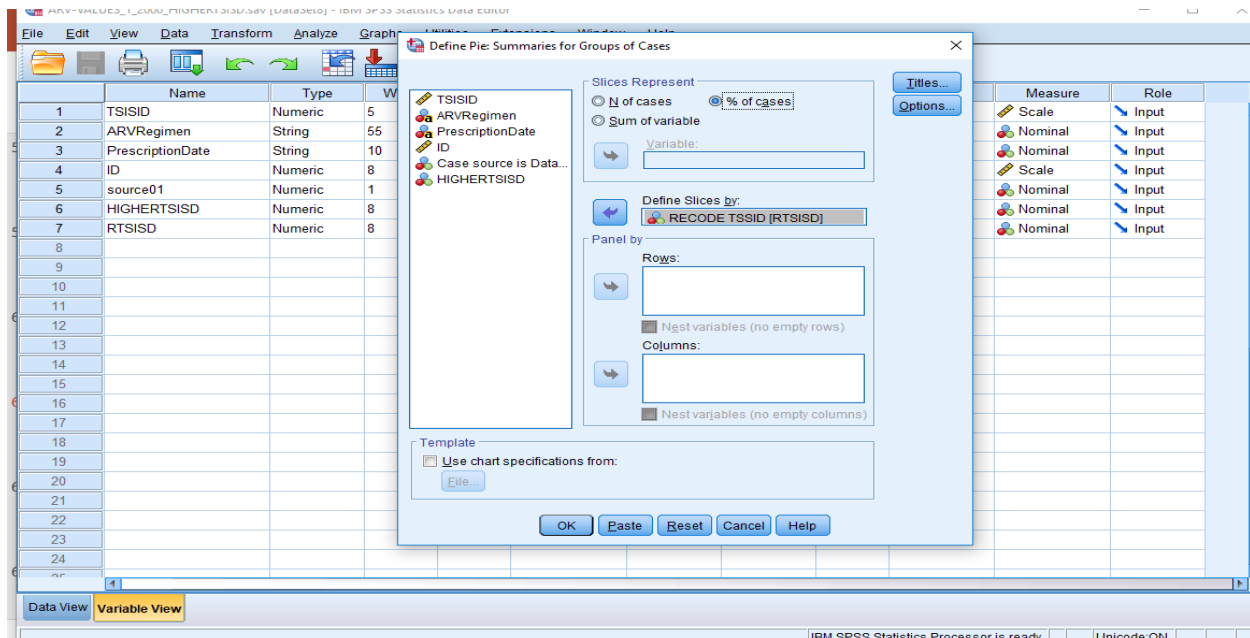


Figure 36.0

Sample Frequency Table

<u>RECODE TSSID</u>	<u>Frequency</u>	<u>Percent</u>	<u>Valid Percent</u>	<u>Cumulative Percent</u>
1 - 400	61	1.4	13	13
401 - 600	6	0.1	1.3	14.2
601-5000	404	9.5	85.8	100
Total	471	11.1	100	
Missing	3767	88.9		
<u>Total</u>	<u>4238</u>	<u>100</u>		

Pasting in Microsoft Excel for graphical illustrations-Bar Chart

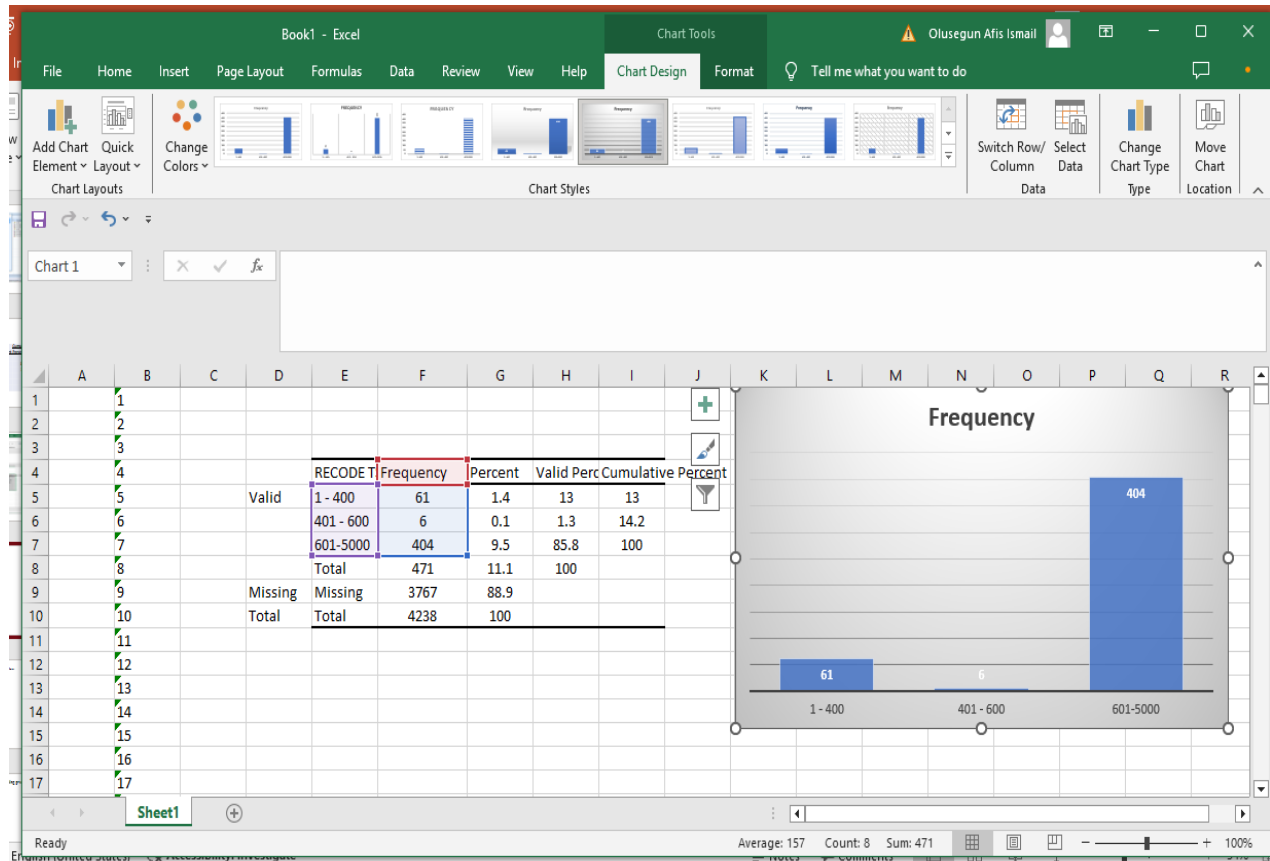


Figure 37.0

Inferential Statistics & p-value & Confidence levels

Inferential Statistics involves hypothesis testing. This is carried out by stating the null and alternative hypotheses or by verbally assuming the hypotheses.

Null Hypothesis (H₀) focuses on the hypothetical statement without evidence. For example: An hypothetical on there is no relationship between the sexual behavior of an adult and HIV Status

Alternative Hypothesis (H₁) focuses on the alternative to the null hypothesis. For example: the hypothetical statement on “there is a relationship between the sexual behavior of an adult and HIV Status”.

Level of Confidence of Estimates: - In statistical analysis, estimates are derived. Some are point estimates but some are based on the level of confidence from 0 no confidence to 100. In most quantitative studies, confidence can range from 90% to 99%. In the Literature, 95% is popular level of confidence.

p-value: - This is the probability of rejecting the null hypothesis and can vary from 1% to 10%. In the literature, 5% is very popular. p-values are **complement** of the levels of confidence.

Introduction to Statistical Tests available in SPSS - Parametric & Non-Parametric Statistical Tests in SPSS

The inferential statistics requires test statistic, a formula to judge the hypothetical calculations against the sample information. To do this, we need to understand the which Statistical Tests

Parametric Statistical Test: - Parametric statistics is a **branch of statistics which leverages models based on a fixed (finite) set of parameters.**

Parametric tests assume a normal distribution of values, or a “bell-shaped curve.” For example, height is roughly a normal distribution in that if you were to graph height from a group of people, one would see a typical bell-shaped curve. This distribution is also called a Gaussian distribution. Parametric tests are in general more powerful (require a smaller sample size) than nonparametric tests.

<https://www.sciencedirect.com/topics/medicine-and-dentistry/parametric-test>

Parametric Statistical Test: - Parametric statistics is a branch of statistics which did not leverages models based on a fixed (finite) set of parameters.

Nonparametric tests are used in cases where parametric tests are not appropriate. Most nonparametric tests use some way of ranking the measurements and testing for weirdness of the distribution.

<https://www.sciencedirect.com/topics/medicine-and-dentistry/parametric-test>

Parametric Tests	Nonparametric Tests
Independent-Samples T Test	Mann-Whitney Test
Paired-Samples T Test	Wilcoxon Signed-Rank Test
One-way ANOVA	Kruskal-Wallis Test
One-way Repeated Measures ANOVA	Friedman's ANOVA

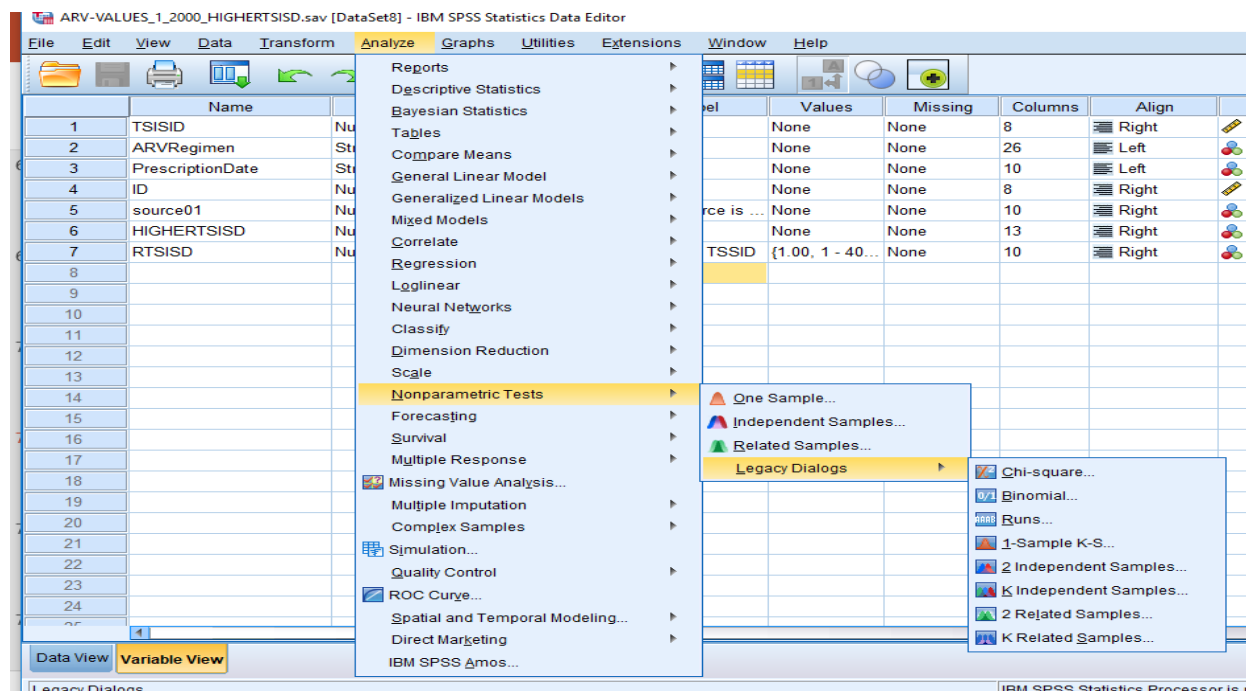


Figure 38.0

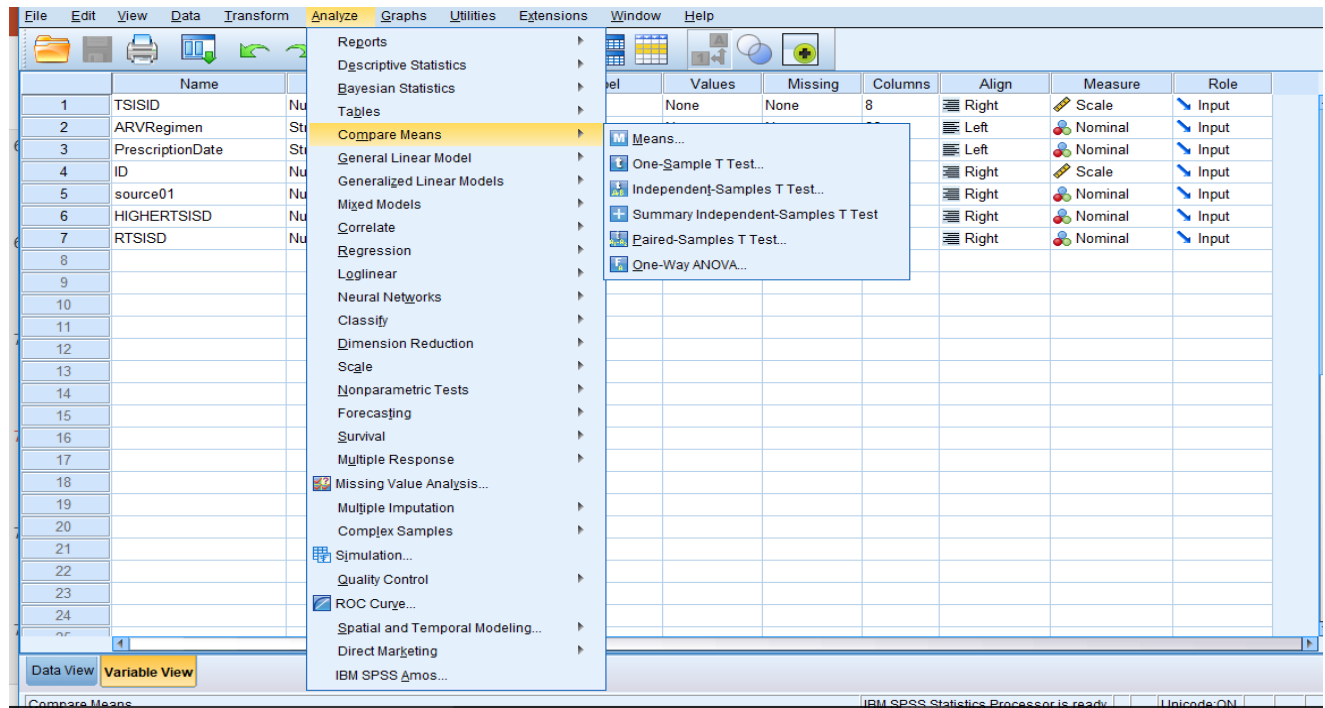


Figure 39.0

Group Case Study Assignment Discussion

Form a group of Five (5)

Each group should import the CD4 data into SPSS, Recode the CD4 Dataset into four groups (Stage 1, 2,3 &4).

Present a frequency Tables to illustrate the stages

Select each stage calculate use the descriptive statistics to describe the data

Present your findings in a power point and submit to the facilitator via email

Day 2

Learning Objectives

At the end of the session, participants should be able to:

- Define Non-Parametric Statistics
- State Advantages / Disadvantages of Non-Parametric Statistics
- Find Non-Parametric Statistical Tests in SPSS
- List Non-Parametric Statistical Test in SPSS
- Use Mann-Whitney Test / Mann- Withney U Test / Wilcoxon Rank -Sum Tests in SPSS
- Interpret SPSS Output of Mann-Whitney Test
/ Mann- Withney U Test / Wilcoxon Rank -Sum Tests

Introduction to Non-Parametric Statistics/Advantages/Disadvantages

The non-parametric test is also known as distribution free test. The non-parametric statistical test is useful when conducting hypothesis testing (Inferential Statistics) that is not based on the distribution of the variable (s) involved in the test.

Reasons for Using Non-parametric Test

- When the distribution of the variable is skewed, a non-parametric test is used. For skewed distributions, the mean is not the best measure of central tendency, hence, parametric tests are not used.
- If the sample size of the data is too small then validating the distribution of the data becomes difficult. Thus, in such cases, a non-parametric test is used to analyze the data.
- If the data is nominal or ordinal, a non-parametric test is used. This is because a parametric test can only be used for continuous data.

Advantages

- The advantages of a non-parametric test are listed as follows:
- Knowledge of the population distribution is not required.
- The calculations involved in such a test are shorter.
- A non-parametric test is easy to understand.
- These tests are applicable to all data types.

Disadvantages

- The disadvantages of a non-parametric test are given below:
- They are not as efficient as their parametric counterparts.
- As these are distribution-free tests the level of accuracy is reduced.

Types of Non-Parametric Statistical Test

- Mann – Whitney Test or Mann – Whitney U Test
- Median Test
- Wilcoxon Rank –Sum Test
- Wilcoxon Signed Rank Test / Sign Rank Test
- Sign Test
- Kruskal Wallis Test
- Chi-Square Test

Mann-Whitney Test / Mann- Withney U Test / Wilcoxon Rank -Sum Tests in SPSS

The Test is suitable under the following assumptions regarding two independent samples:

1. All the observations from both samples /groups are independent of each other,
2. The data or responses are at least ordinal
3. Under the null hypothesis H_0 , the distributions of both populations are identical (Alternatively, the medians are the same)
4. The alternative hypothesis H_1 is that the distributions are not identical (Alternatively, the medians are not the same)

Non-Parametric Tests for 2 Independent Samples

- Median Test is based on median for each group or sample
- Mann – Whitney U –Test is based on ranking of the data set
- Wilcoxon Rank Sum Test is based on ranking of the data set
- All the three can be generated in SPSS same time

Mann – Whitney Test or Mann – Whitney U Test / Wilcoxon Rank –Sum Test - Application

Mann-Whitney U test is used for every field, but is frequently used in psychology, healthcare, nursing, business, and many other disciplines. For example, in HIV study, it is used to compare CD4 counts of males versus females or urban areas dwellers versus rural areas dwellers. In medicine, it is used to know the effect of two medicines and whether they are equal or not.

How to use SPSS –Mann-Whitney Test

- Scenario: A researcher wants to investigate the possibility of significant difference between the CD4 counts of males and females in a given record provided by Monitoring Officers. The records provided contains the following variables:
- TSISD
- Last CD4 Results
- Test Date
- Gender
- Gender Recoded
- Parish
- Parish Recoded

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	TSISID	Numeric	5	0	TSIS ID	None	None	12	Right	Scale	Input
2	LastCD4Result	Numeric	19	15	Last CD4 Result	None	None	12	Right	Scale	Input
3	TestDate	String	10	0	Test Date	None	None	10	Left	Nominal	Input
4	Gender	String	6	0		None	None	6	Left	Nominal	Input
5	Gendercoded	Numeric	1	0		{1, Male}...	None	12	Right	Nominal	Input
6	Parish	String	13	0		None	None	13	Left	Nominal	Input
7	Parishcoded	Numeric	2	0		{1, Kingston...	None	12	Right	Nominal	Input
8											

Figure 40.0

- Dependent Variable: Last CD4 Results
- Independent Variable: Gender or
Gender Recoded (Male = 1 Female =2)

Target samples (Male / Female) = Independent Samples

- The Test can be found on SPSS as follows:
- Go to Analyze
- then Nonparametric Tests

then Independent Samples

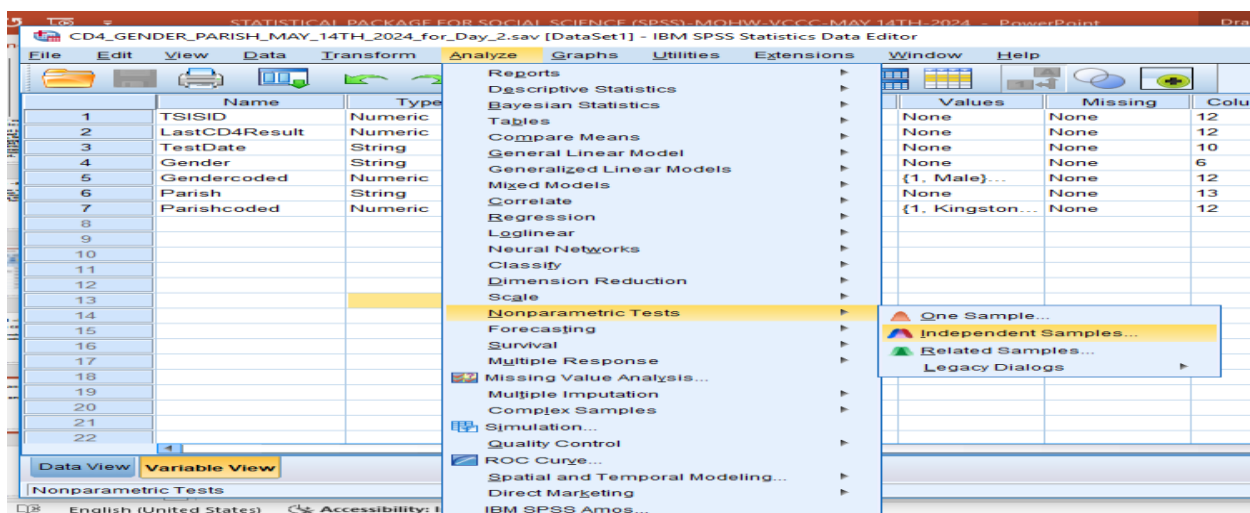


Figure 41.0

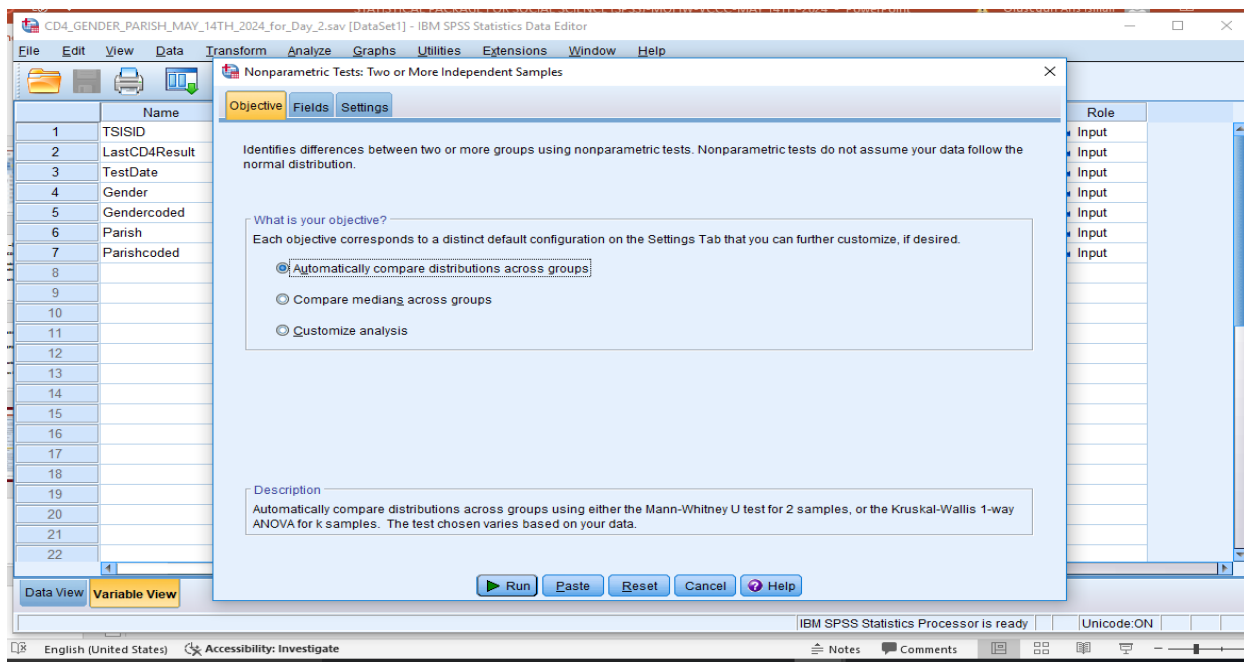


Figure 42.0

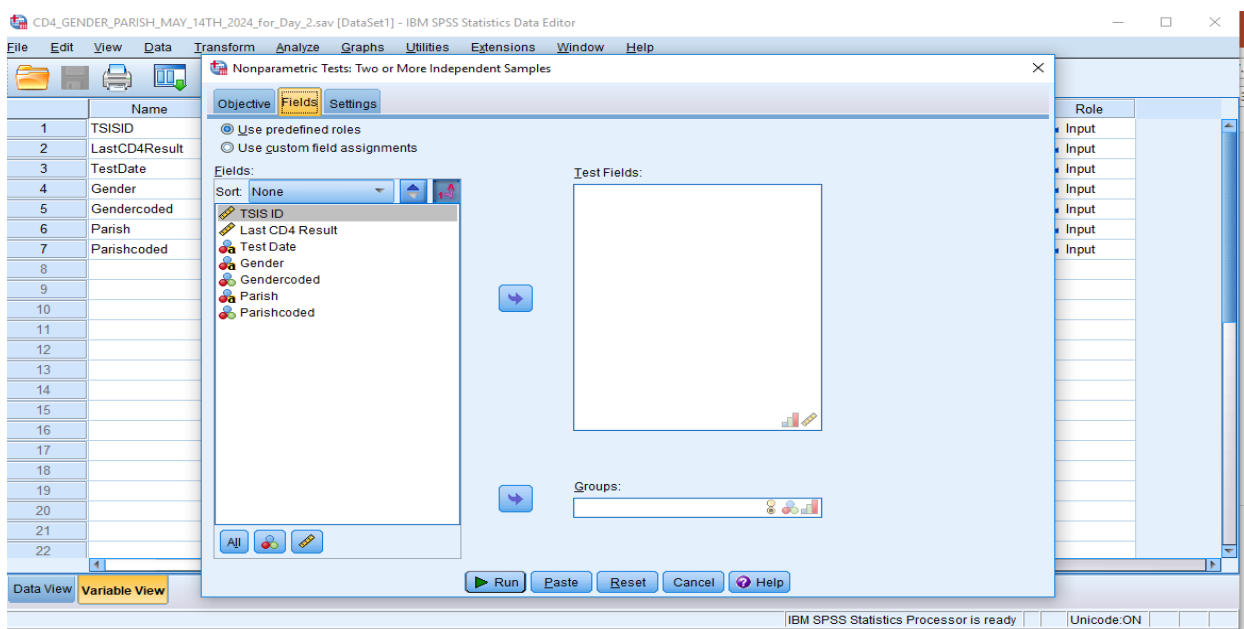


Figure 43.0

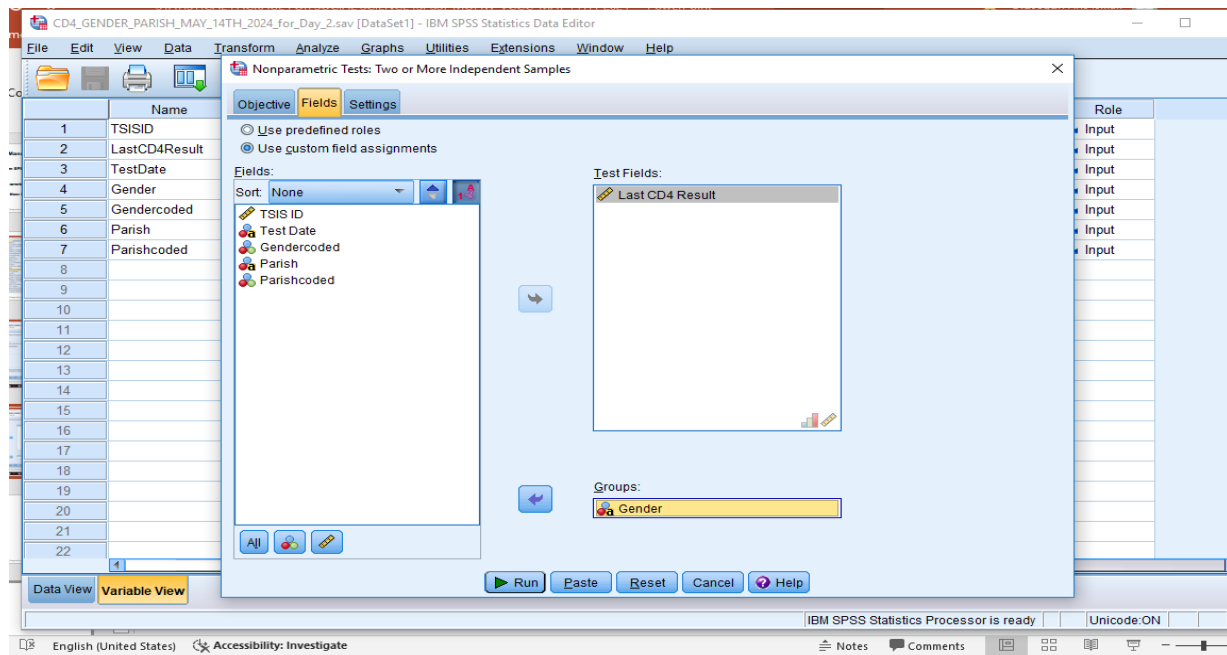


Figure 44.0

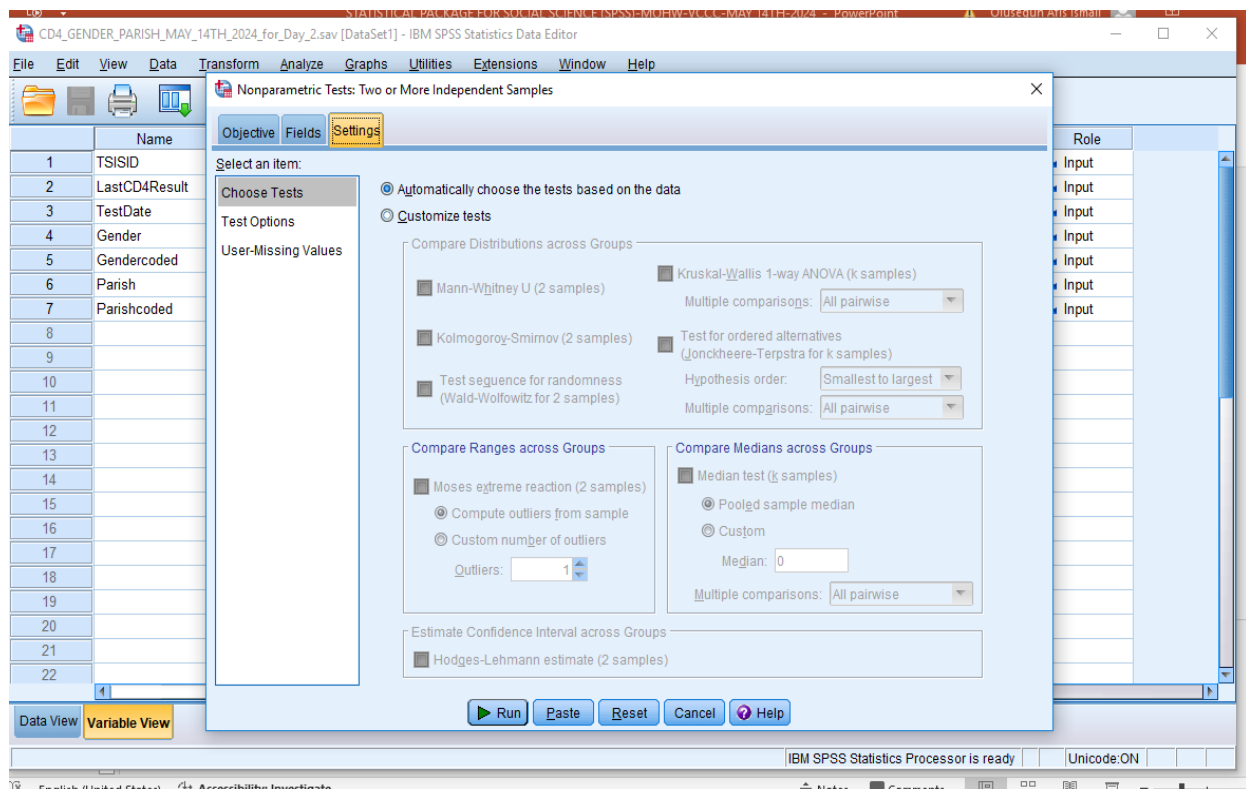


Figure 45.0

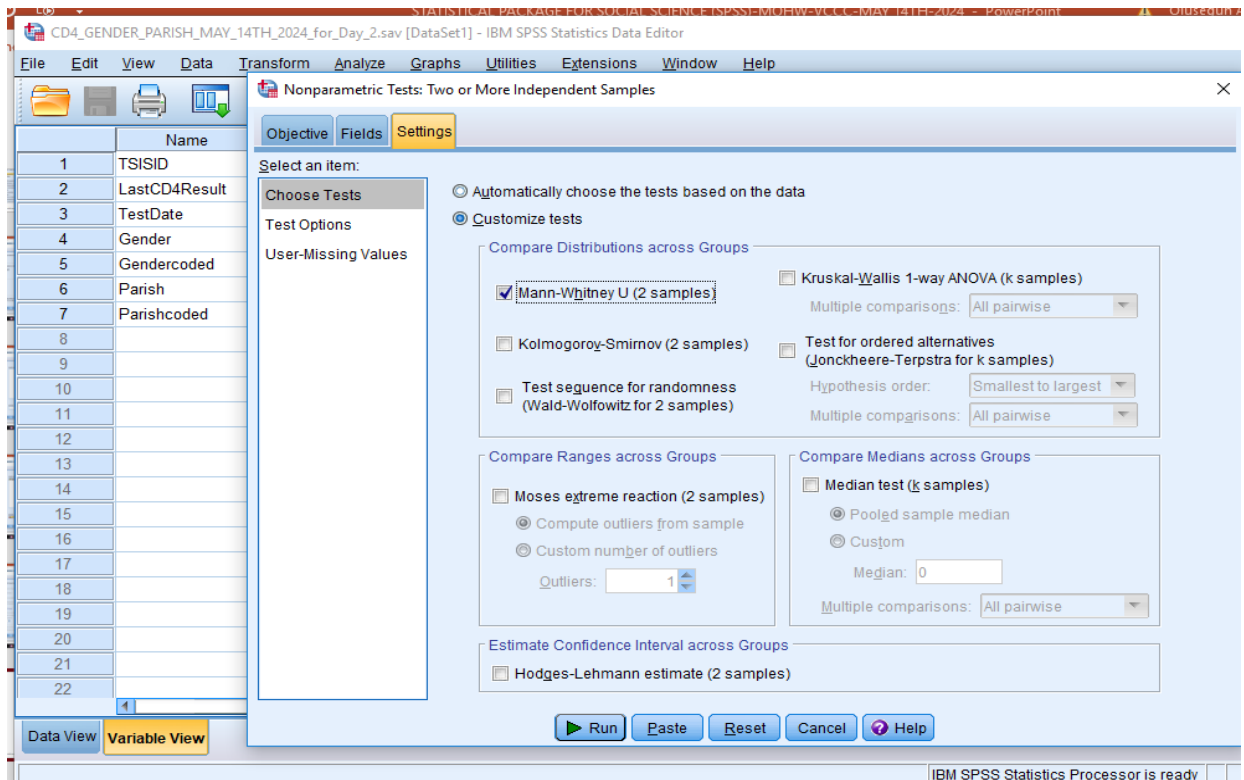


Figure 46.0

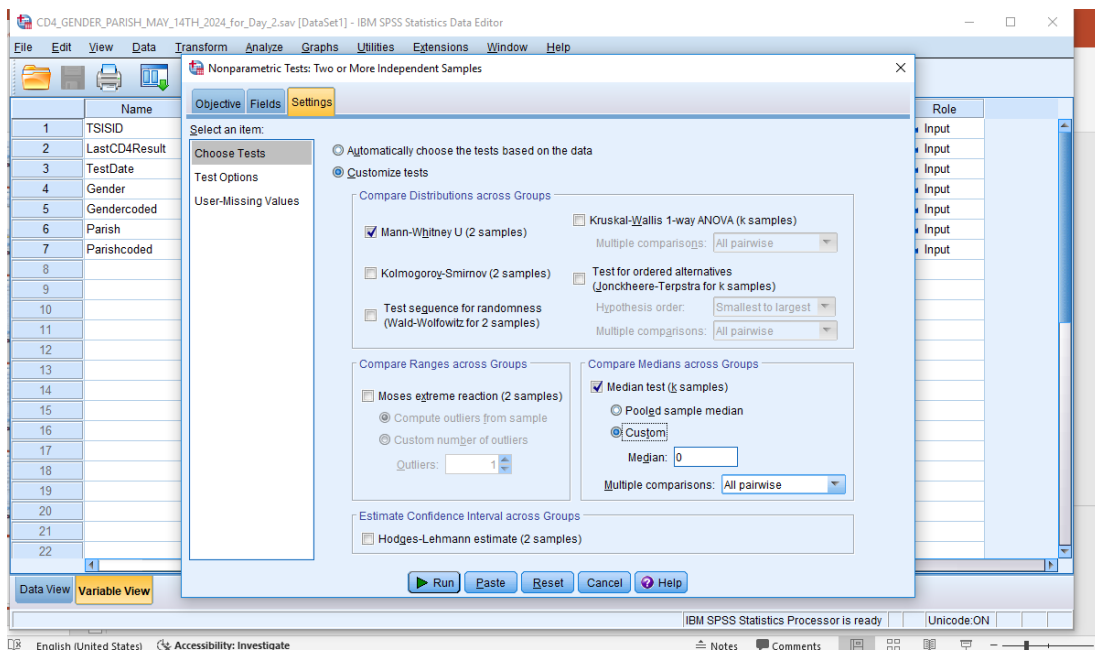


Figure 47.0

Hypothesis Test Summary

	Null Hypothesis	Test	Sig.	Decision
1	The medians of Last CD4 Result are the same across the specified categories of Gender.	Independent-Samples Median Test	.682	Retain the null hypothesis.
2	The distribution of Last CD4 Result is the same across categories of Gender.	Independent-Samples Mann-Whitney U Test	.051	Retain the null hypothesis.

Asymptotic significances are displayed. The significance level is .05.

Figure 48.0

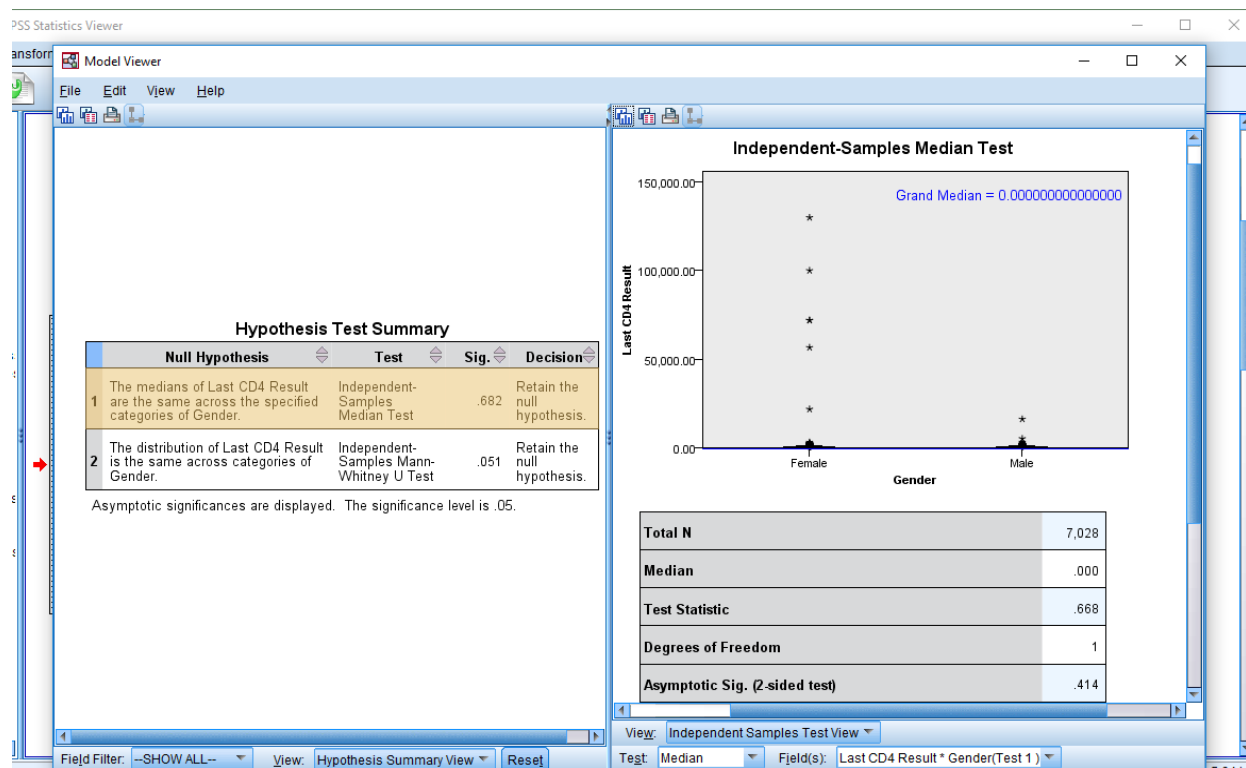


Figure 49.0

Interpretation of Results

- Research Question: Is there a difference between CD4 Counts for the male and female groups?
- From Output 1: (2) The null hypothesis is accepted that the distribution are the same for the CD4 count for both male and female $p=0.051 > 0.05$. Same result for Median Test , $p=0.682$

Non-parametric Test for 2 Independent Samples

This is useful when the data analyst wants to identify any significant difference between two independent samples

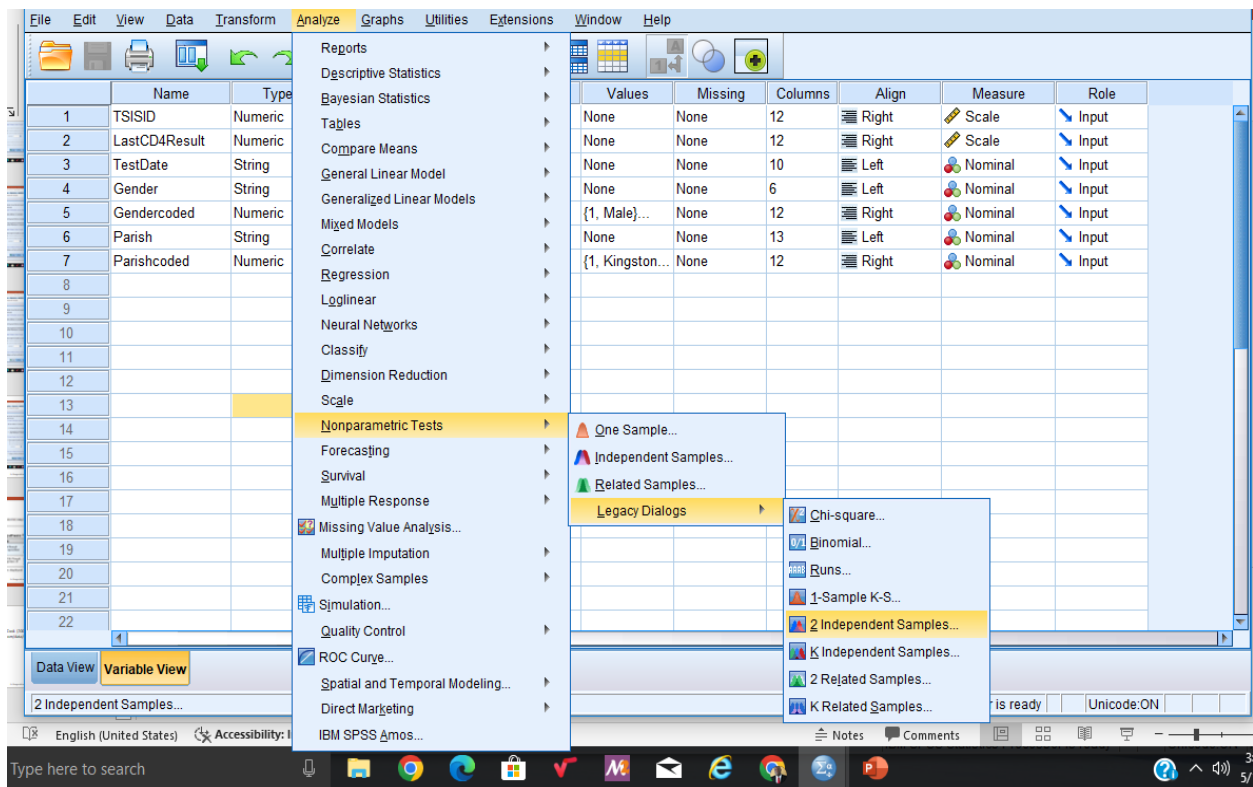
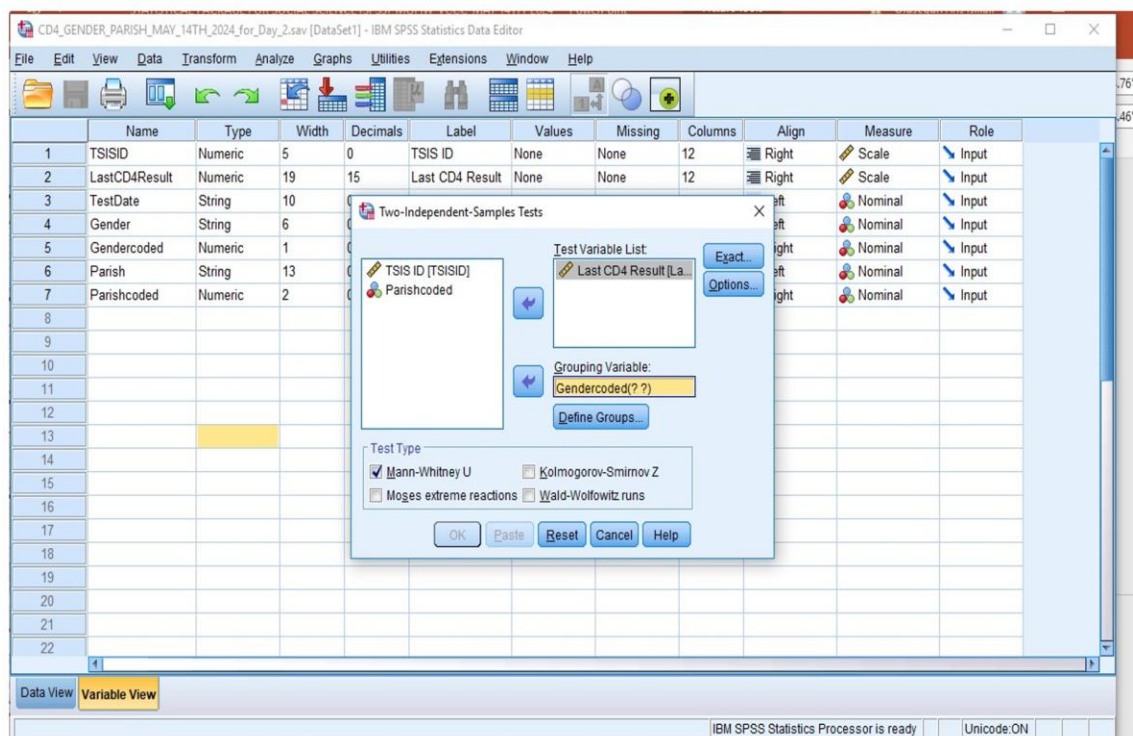


Figure 50.0

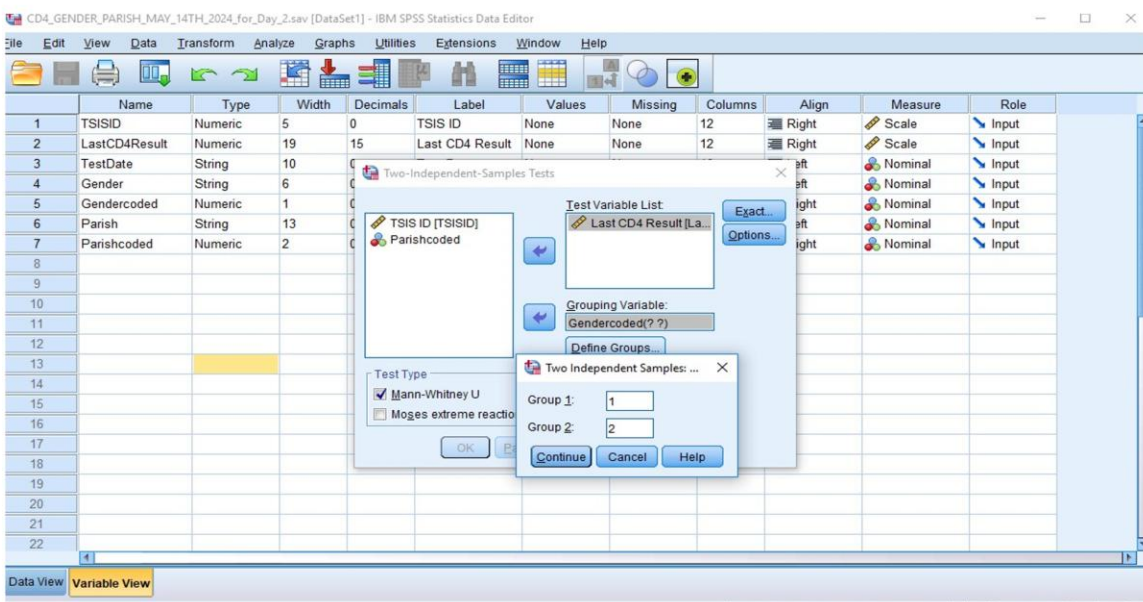
The example in **Figure 51.0** seeks to illustrate how we can compare the LastCD4Results of the male population with the female population. The Dependent Variable is the LastCD4Results while the Grouping Variable or Independent Variable is Gender.



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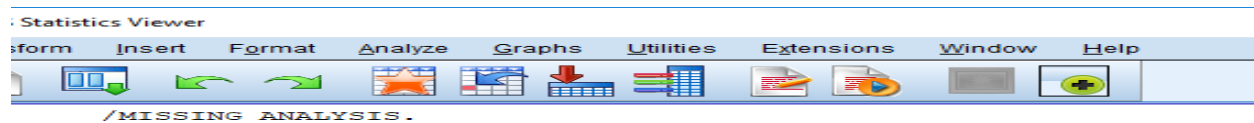
Figure 51.0



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Figure 52.0



→ NPar Tests

Mann-Whitney Test

<i>Ranks</i>				
	Gendercoded	N	Mean Rank	Sum of Ranks
Last CD4 Result	Male	3515	3467.31	12187581
	Female	3513	3561.72	12512325
	Total	7028		

<i>Test Statistics^a</i>	
	Last CD4 Result
Mann-Whitney U	6008211.0
Wilcoxon W	12187581
Z	-1.951
Asymp. Sig. (2-tailed)	.051

a. Grouping Variable:
Gendercoded

Figure 53.0

Interpretation 2

- From this data, it can be concluded that CD4 Count in the male group was not statistically significantly higher than CD4 Count in Female group ($U = 6008211.0$, $p = .051$). Similar explanation for the Wilcoxon $W = 12187561$, $p = 0.051$
- Conclusion: The gender of the patients is not related or influence the patient CD4 Count. The spread of the CD4 Count is almost the same for both gender.
- Graham Upton and Ian Cook (2001).Oxford Dictionary of Statistics
- <https://www.cuemath.com/data/non-parametric-test/>

Day 3

Learning Objectives for Day 3

At the end of the session, participants should be able to:

- Use Non – Parametric Test - Sign -Rank Test in SPSS
- Use Non -Parametric Test – Kruskal – Wallis Test in SPSS
- Use Non-Parametric Test - Chi Square Test in SPSS
- Define Parametric Tests
- Know when to use Parametric Test
- Use Parametric Test – One sample Test / Independent Samples Test
/ Paired Sample Test in SPSS
- Use Parametric Test - One-Way Analysis of Variance

Non – Parametric Test Sign -Rank Test

The Sign-Rank Test or Sign Test is used to determine whether there is a median difference between paired or matched data. A matched or paired data is a record obtained from the same patient on two occasions. For example, is trying to identify any difference in weight of patients before taking recommended drugs for HIV-AIDS and after taken the recommended drugs. For this example, we will use the May20th dataset to conduct a Sign Test in SPSS. Using **Figure 54.0**, go to Analyze , then Legacy Dialogue, then 2 Related Samples.

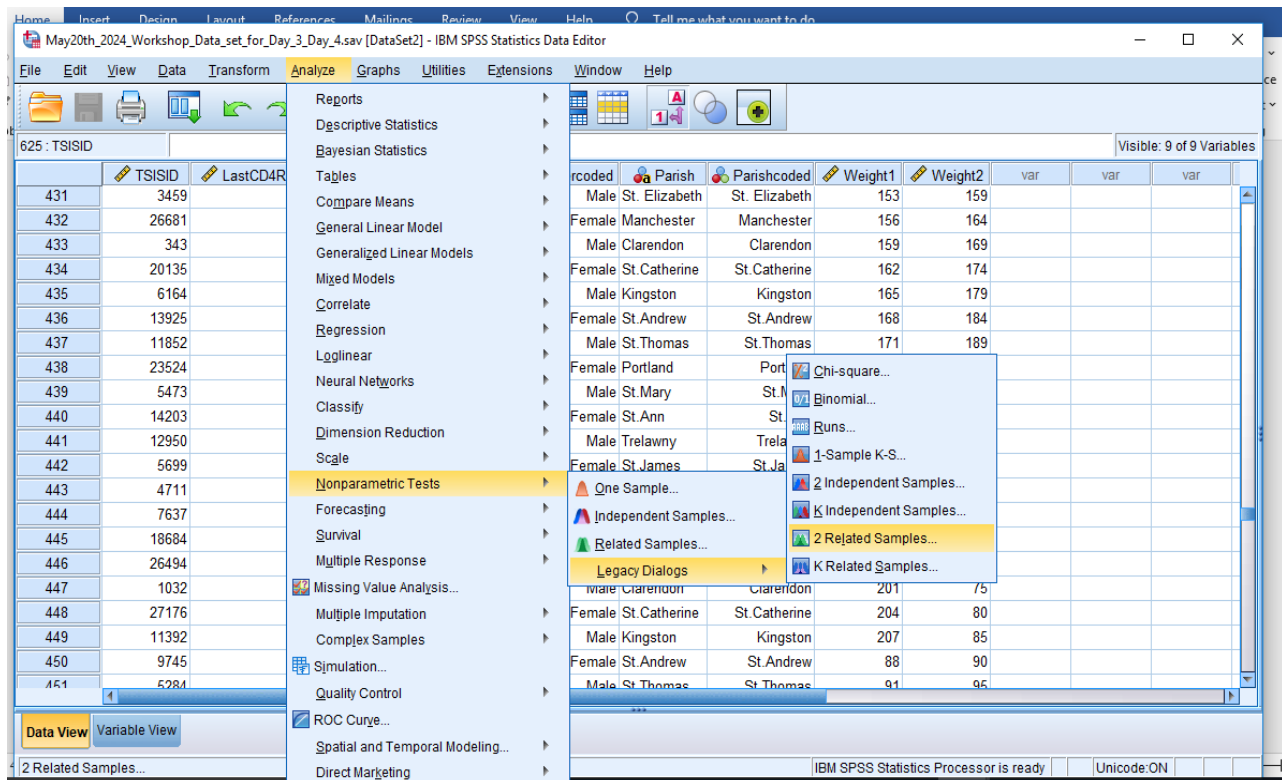


Figure 54.0

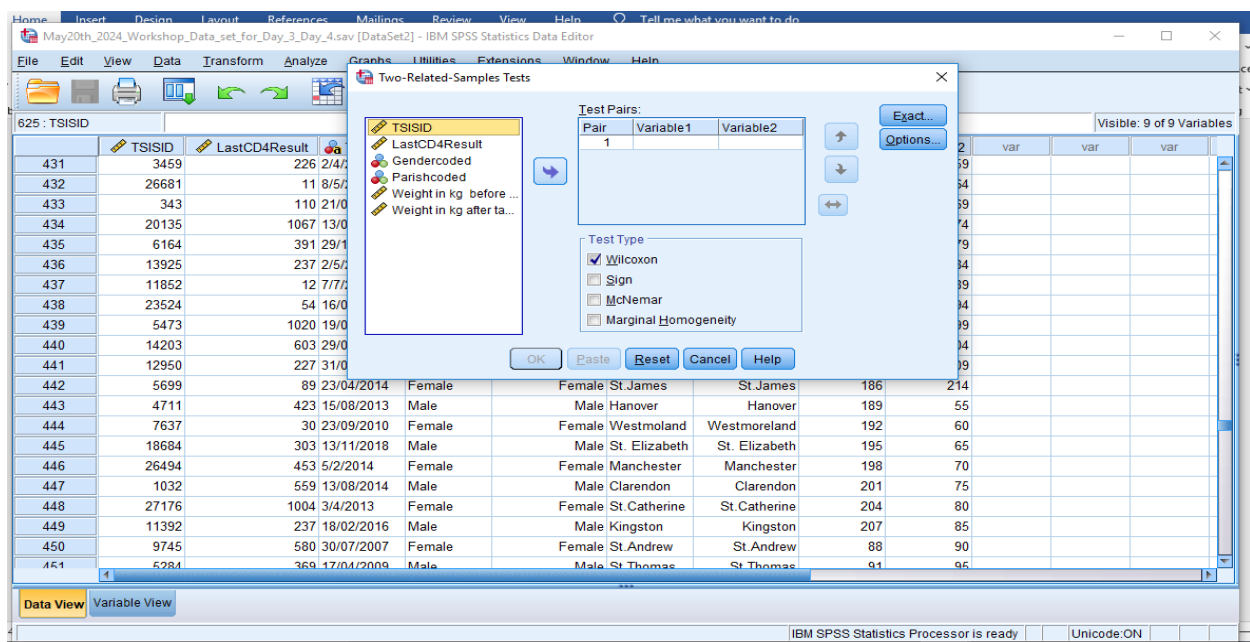


Figure 55.0

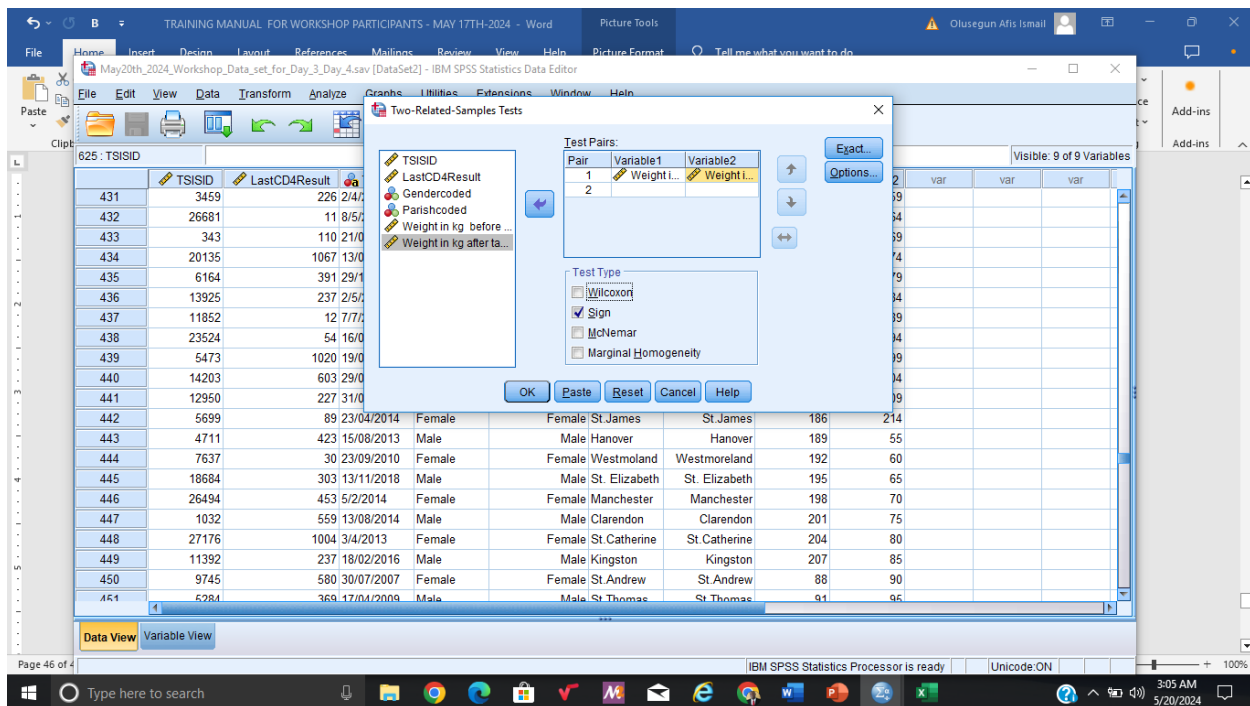


Figure 56.0

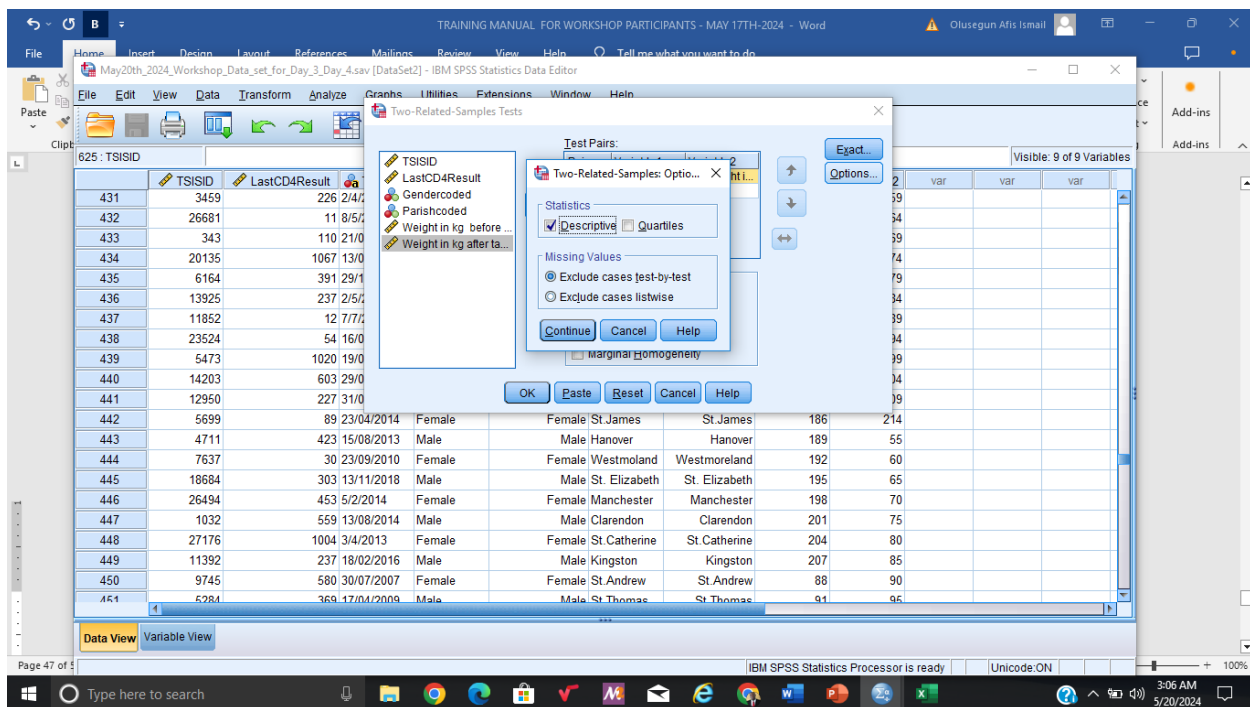


Figure 57.0

SPSS Output

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum
Weight in kg before taking the recommended drugs	624	149.26	40.216	56	229
Weight in kg after taken the recommended drugs	624	121.92	40.178	46	214

<i>Frequencies</i>		
		N
Weight in kg after	Negative Differences ^a	418
taken the recommended	Positive Differences ^b	203
drugs - Weight in kg	Ties ^c	3
before taking the	Total	624
recommend drugs		
a. Weight in kg after taken the recommended drugs < Weight in kg before taking the recommend drugs		
b. Weight in kg after taken the recommended drugs > Weight in kg before taking the recommend drugs		
c. Weight in kg after taken the recommended drugs = Weight in kg before taking the recommend drugs		

Test Statistics^a

Weight in kg after taken the recommended drugs - Weight in kg before taking the recommend drugs	
Z	-8.588
Asymp. Sig. (2-tailed)	.000

a. Sign Test

Null Hypothesis: There is no difference in the weight of patients before and after taken the recommended drugs

Alternative Hypothesis: There is a difference in weight of the patients

The findings revealed that a statistically significant median difference in the weight of the patient before and after taking the recommended drugs, $p=0.00$ and $p<0.05$.

Non -Parametric Test – Kruskal – Wallis Test

The Kruskal – Wallis Test is used to identify any significant differences across groups. It is based on ranking of the dataset from each group and then compare the mean rank. An example is the comparison of the LastCD4Results across the fourteen parishes to identify differences.

Figure 58.0 illustrates the steps in SPSS to conduct the Test

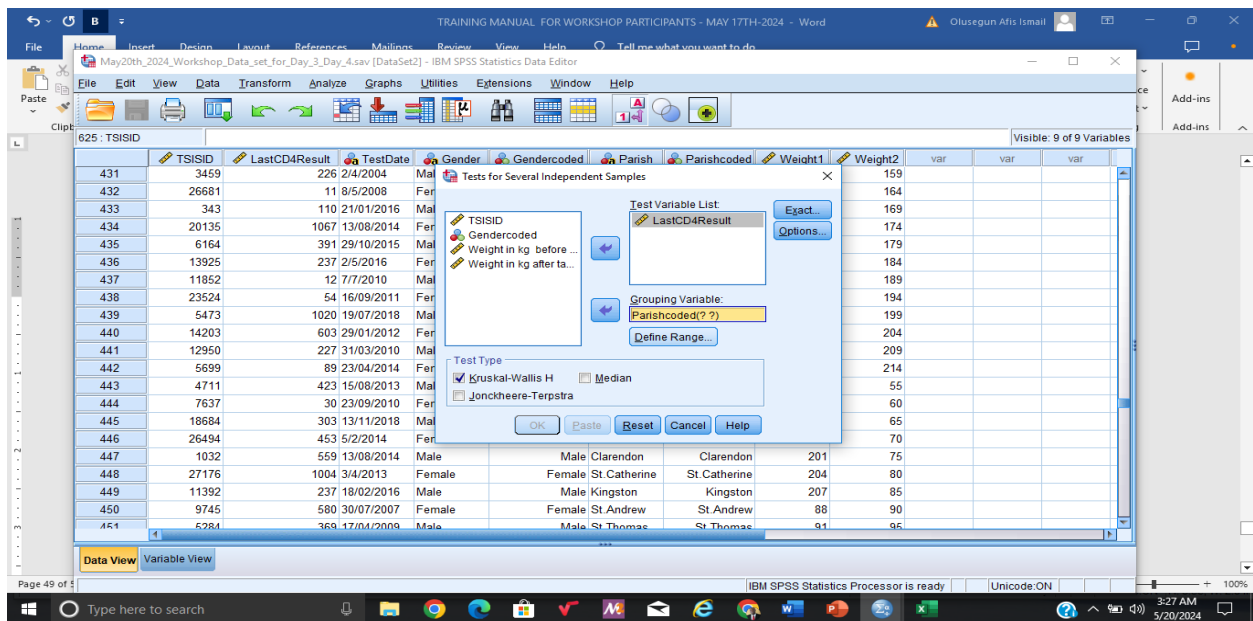


Figure 58.0

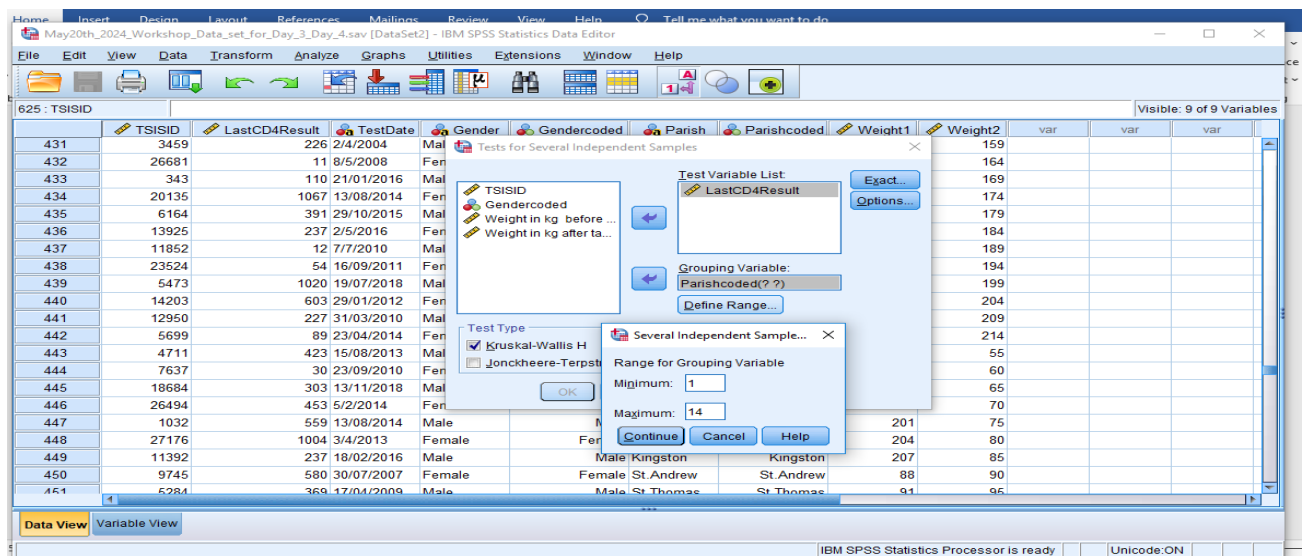


Figure 59.0

Ranks

	Parishcoded	N	Mean Rank
LastCD4Result	Kingston	45	324.72
	St.Andrew	45	293.83
	St.Thomas	45	325.94
	Portland	45	298.21
	St.Mary	45	335.07
	St.Ann	45	359.78
	Trelawny	45	273.88
	St.James	45	297.61
	Hanover	44	291.09
	Westmoreland	44	302.67
	St. Elizabeth	44	339.83
	Manchester	44	287.89
	Clarendon	44	337.23
	St.Catherine	44	307.05
	Total	624	

Test Statistics^{a,b}

	LastCD4Result
Kruskal-Wallis H	10.845
df	13
Asymp. Sig.	.624

a. Kruskal Wallis Test

b. Grouping Variable: Parishcoded

Interpretation

A Kruskal -Wallis H Test revealed that there is no statistically significant difference between the parish of residence of the patients LastCD4 Results Chi-Square value = 10.845 (df =13), $p=0.624 > 0.05$.

Non-Parametric Test -Chi Square Test

The Chi-Square Test of Independence is used to identify any significant relationship between two categorical variables. For example, trying to identify any relationship between the gender of the patients and level of education. **Figure 60.0 to 63.0** illustrate the steps.

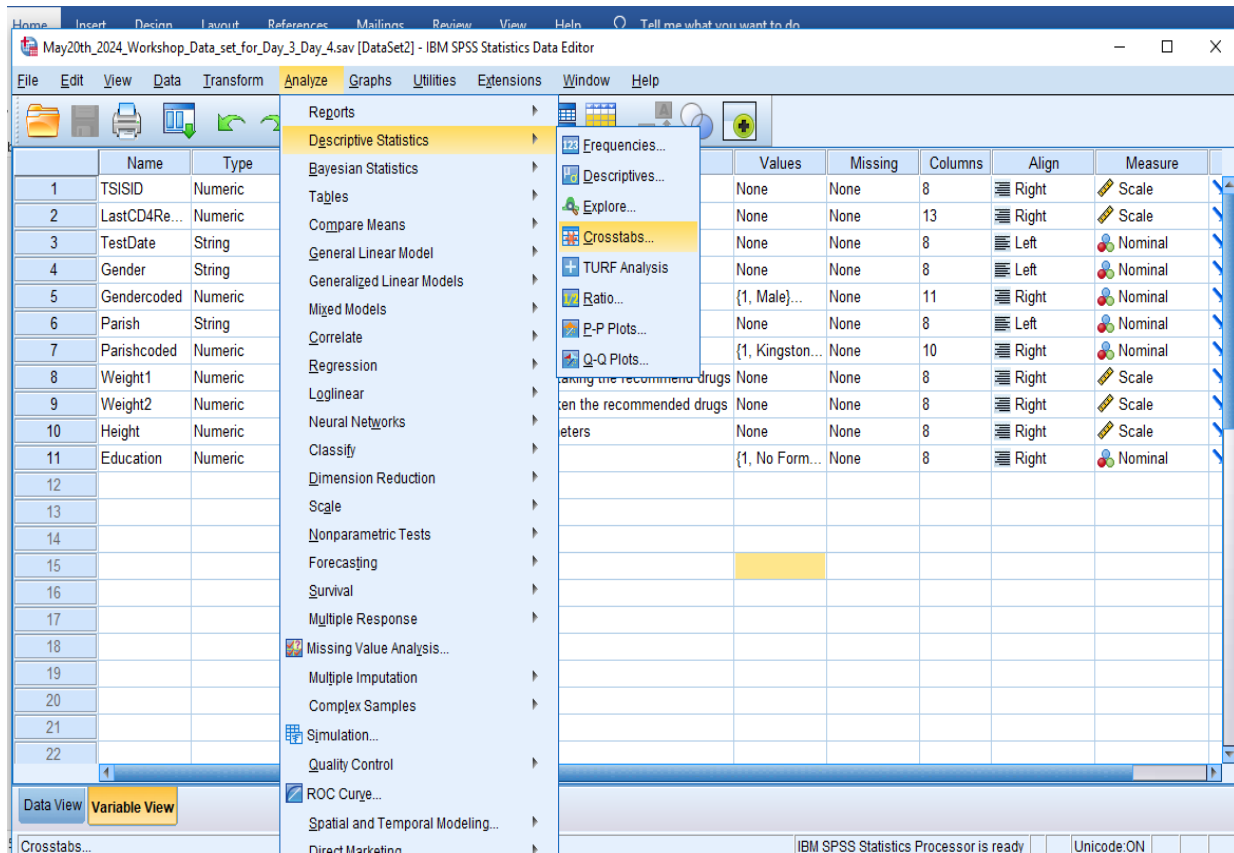


Figure 60.0

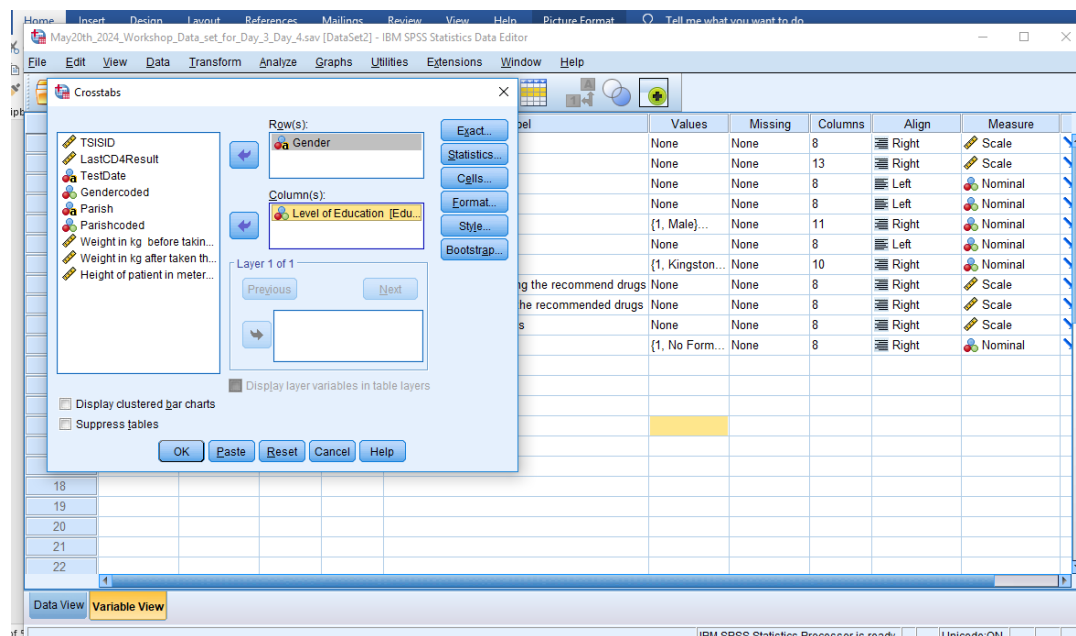


Figure 61.0

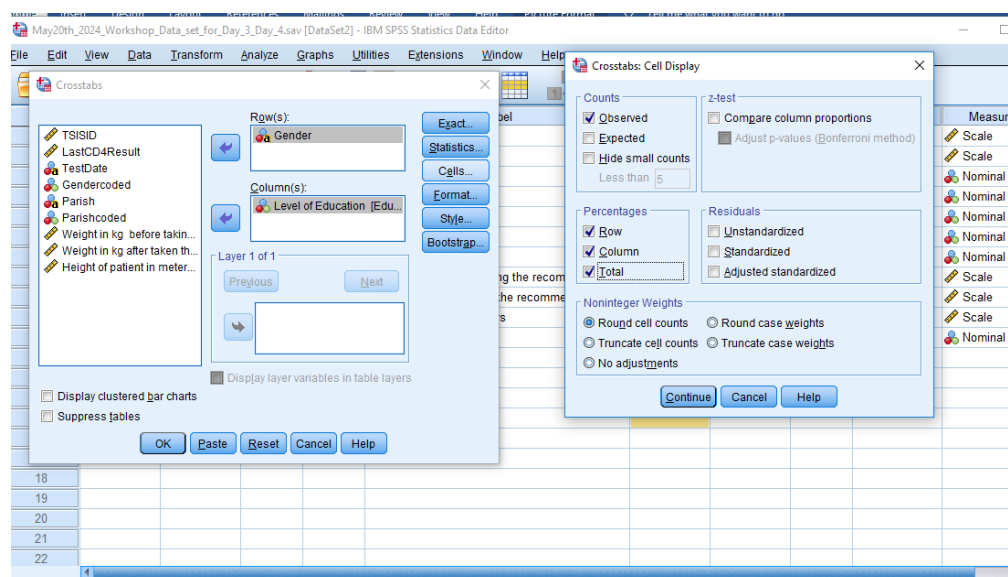


Figure 62.0

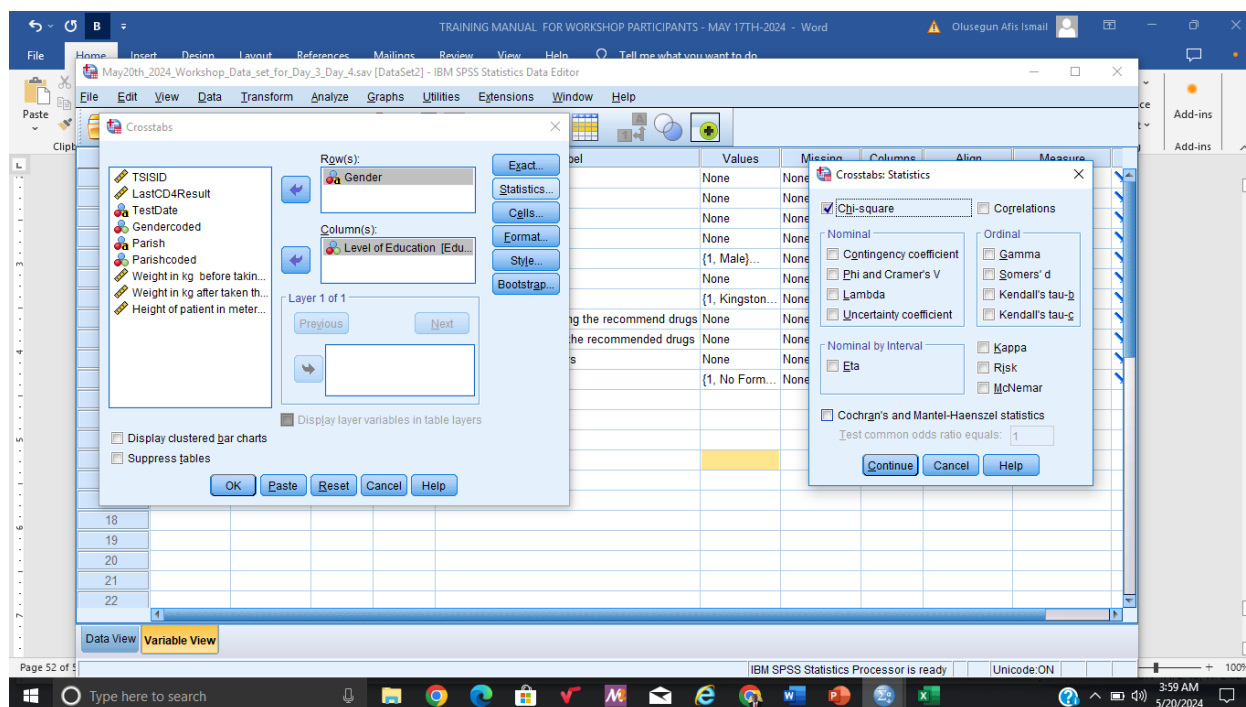


Figure 63.0

Gender * Level of Education Crosstabulation

			Level of Education				
			No Formal Education	Primary School	High School	Tertiary	Total
Gender	Female	Count	155	1	2	154	312
		% within Gender	49.7%	0.3%	0.6%	49.4%	100.0%
		% within Level of Education	99.4%	0.6%	1.3%	98.7%	50.0%
		% of Total	24.8%	0.2%	0.3%	24.7%	50.0%
	Male	Count	1	154	155	2	312
		% within Gender	0.3%	49.4%	49.7%	0.6%	100.0%
		% within Level of Education	0.6%	99.4%	98.7%	1.3%	50.0%
		% of Total	0.2%	24.7%	24.8%	0.3%	50.0%
Total	Count		156	155	157	156	624
	% within Gender		25.0%	24.8%	25.2%	25.0%	100.0%
	% within Level of Education		100.0%	100.0%	100.0%	100.0%	100.0%
	% of Total		25.0%	24.8%	25.2%	25.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	600.256 ^a	3	.000
Likelihood Ratio	798.046	3	.000
N of Valid Cases	624		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 77.50.

Interpretation

There is a statistical significant relationship between the gender and the level of education of the patients, $p = 0.00 < 0.05$

Parametric Test – One sample T - Test / Independent Samples Test / Paired Sample Test

Parametric Test – One sample Test

The One sample T- Test is used to compare hypothetical population mean with a sample mean. If we choose to compare the average height of the patient with 1.5m.

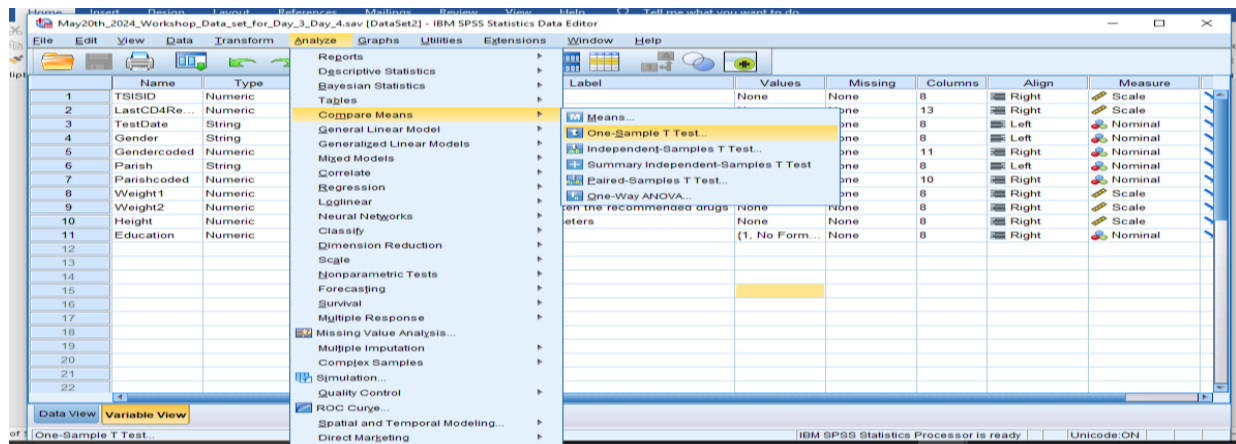


Figure 64.0

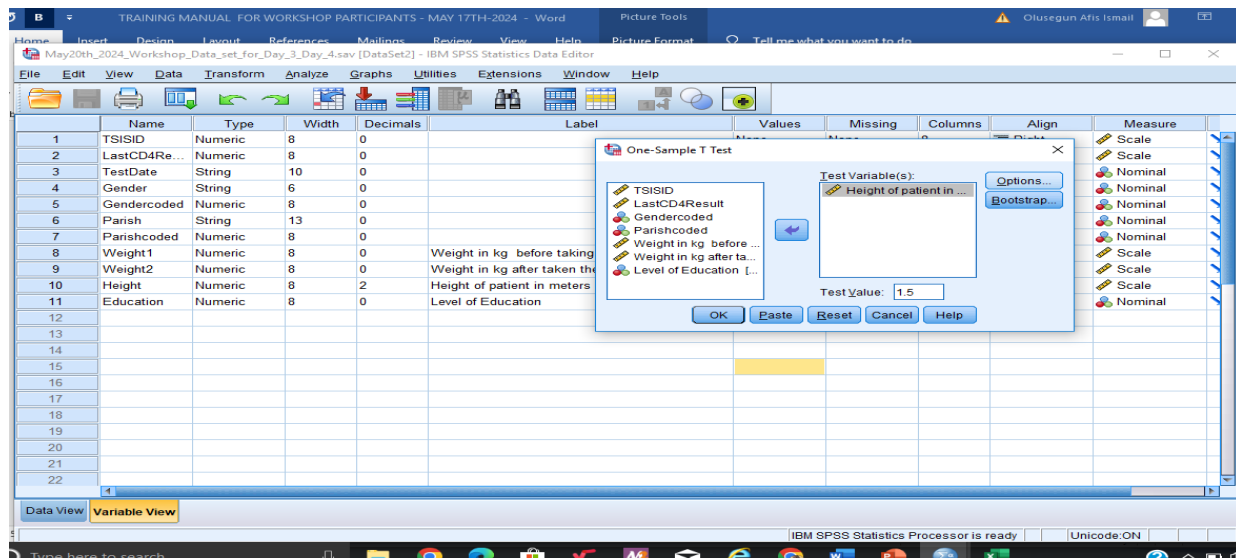


Figure 65.0

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
Height of patient in meters	624	1.4108	.15632	.00626

One-Sample Test

Test Value = 1.5						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Height of patient in meters	-14.246	623	.000	-.08915	-.1014	-.0769

Independent Samples Test

This is used to compare the average of two independent samples to identify any statistical significant difference

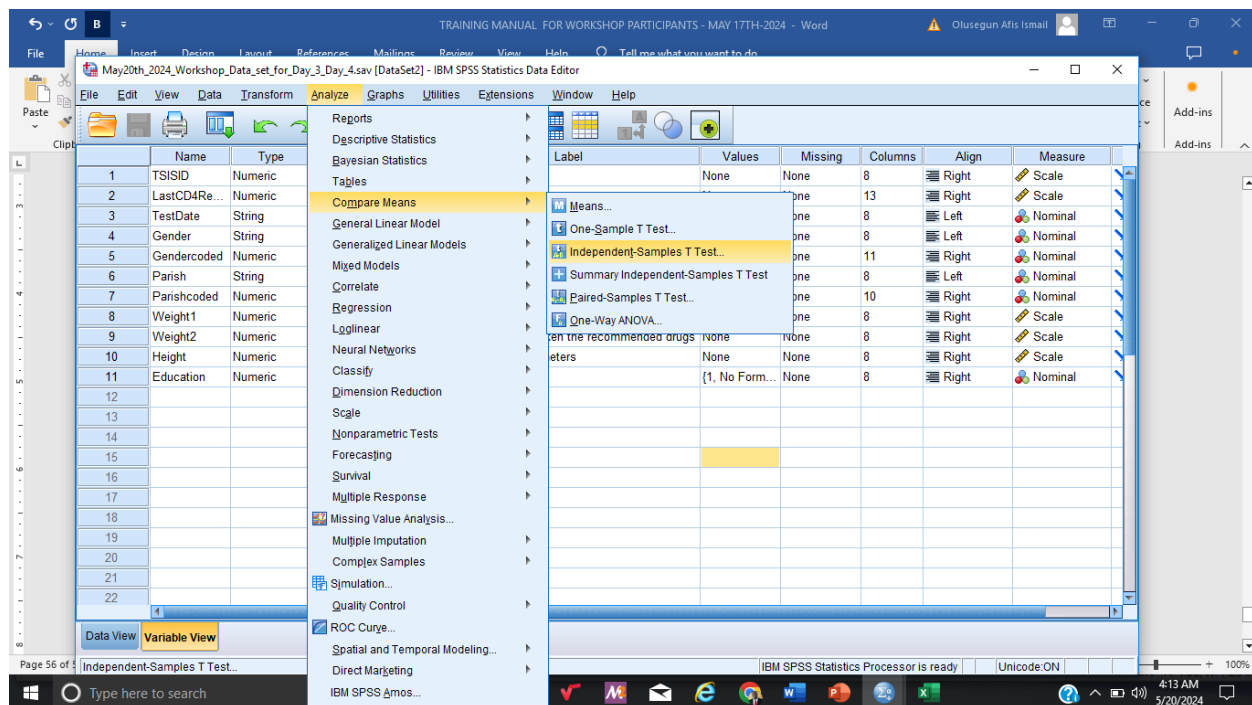


Figure 66.0

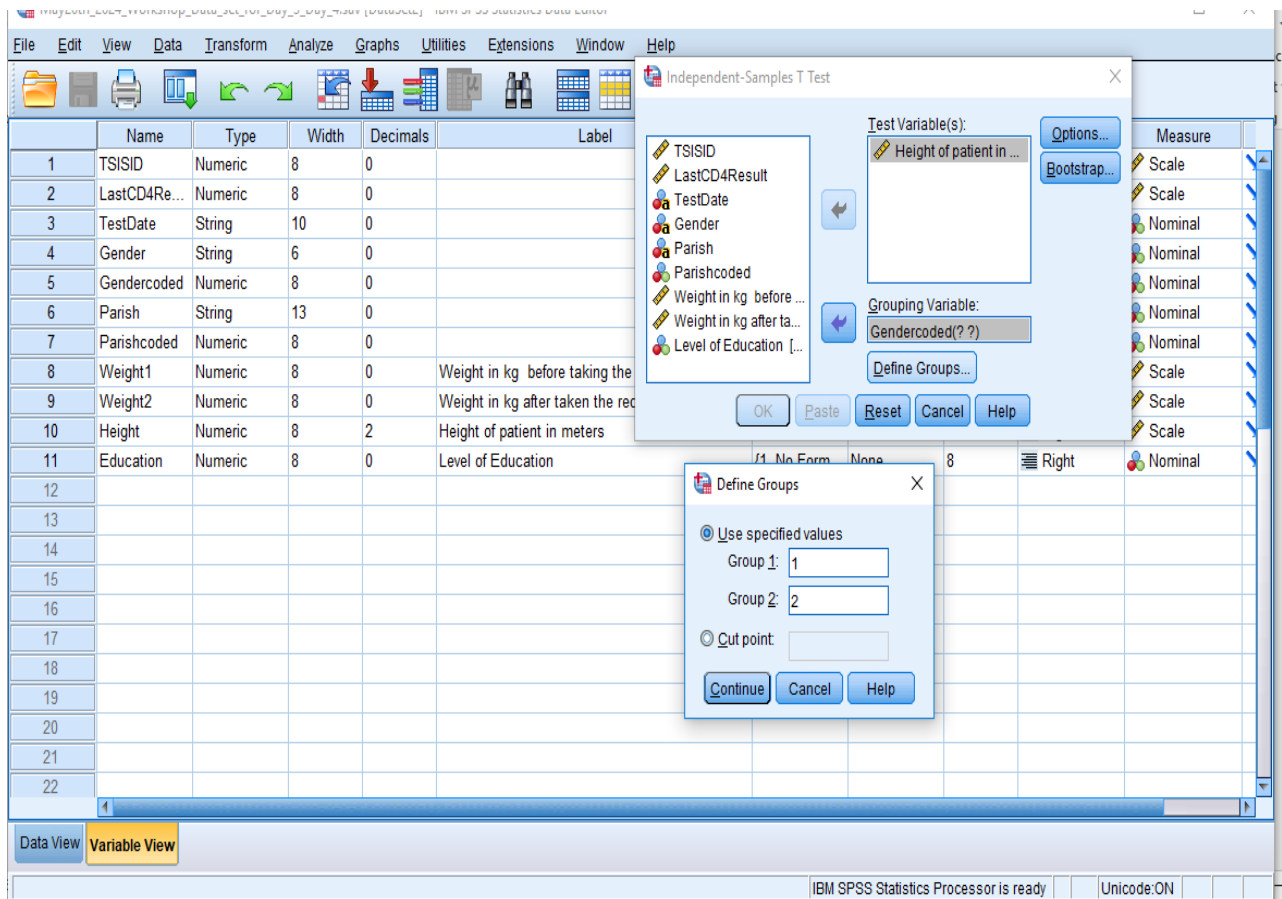


Figure 67.0

Group Statistics

	Gendercoded	N	Mean	Std. Deviation	Std. Error Mean
Height of patient in meters	Male	312	1.4147	.15722	.00890
	Female	312	1.4070	.15557	.00881

Independent Samples Test

		Height of patient in meters	
		Equal variances assumed	Equal variances not assumed
Levene's Test for Equality of Variances	F	.093	
	Sig.	.760	
t-test for Equality of Means	t	.617	.617
	df	622	621.931
	Sig. (2-tailed)	.538	.538
	Mean Difference	.00772	.00772
	Std. Error Difference	.01252	.01252
	95% Confidence Interval of the Difference	Lower -0.01687	Lower -0.01687
		Upper .03231	Upper .03231

Interpretation: There is no statistical significant difference in the height of males and females patient, $p=0.538 > 0.05$.

Parametric Test - One-Way Analysis of Variance

The One Way Analysis of Variance is similar to the Kruskal-Wallis Test. **Figure 68.0 to 70.0** illustrate the use of One-Way Anova to compare the height of the patients across the fourteen parishes.

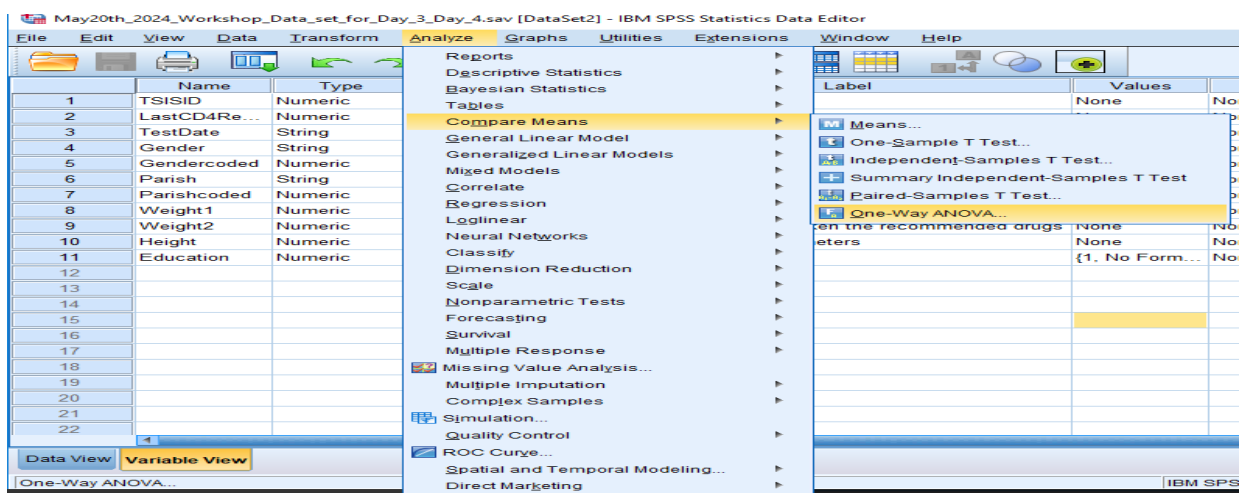


Figure 68.0

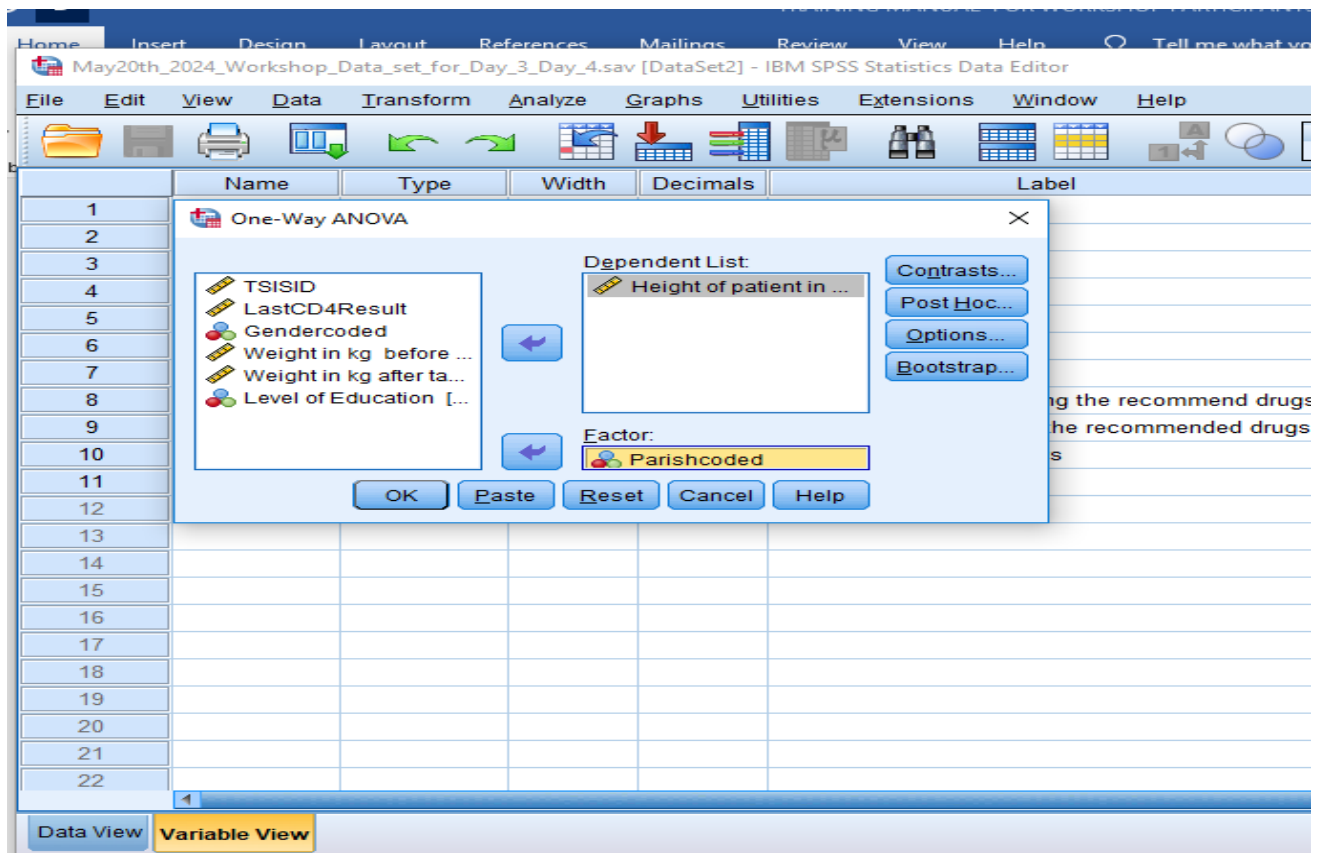


Figure 69.0

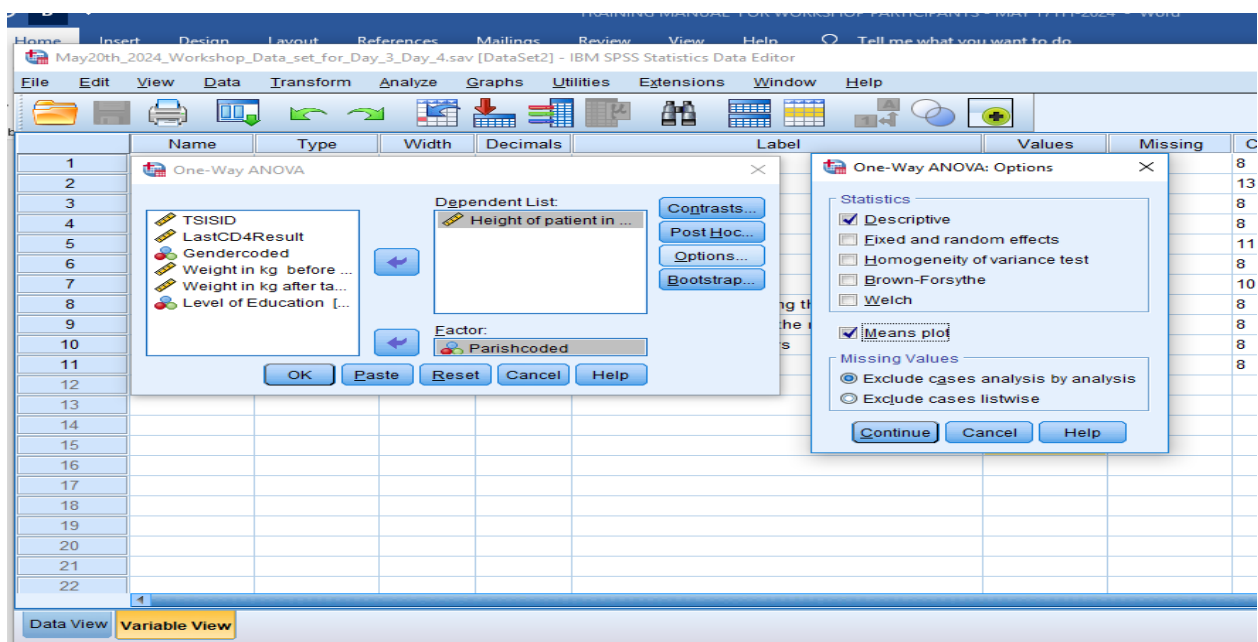


Figure 70.0

SPSS Outputs

Descriptives

Height of patient in meters

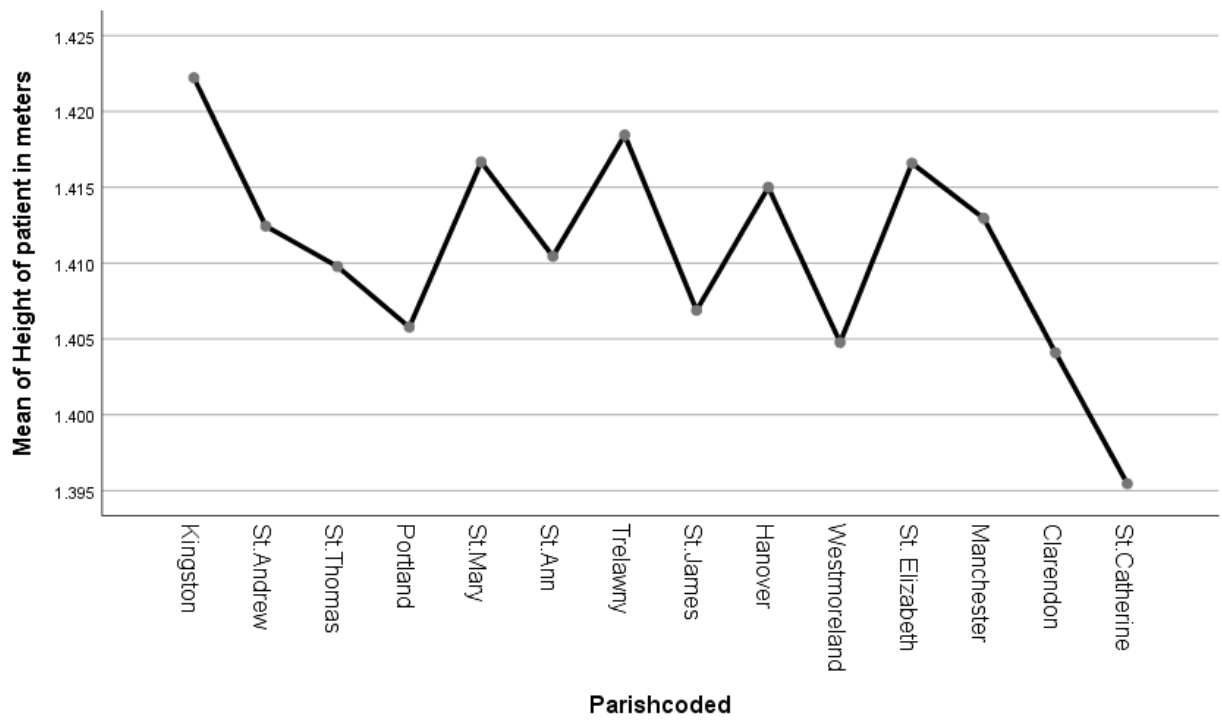
	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Kingston	45	1.4222	.15805	.02356	1.3747	1.4697	1.20	1.62
St.Andrew	45	1.4124	.15467	.02306	1.3660	1.4589	1.20	1.62
St.Thomas	45	1.4098	.16036	.02391	1.3616	1.4580	1.20	1.62
Portland	45	1.4058	.15286	.02279	1.3599	1.4517	1.20	1.62
St.Mary	45	1.4167	.15821	.02359	1.3691	1.4642	1.20	1.62
St.Ann	45	1.4104	.15703	.02341	1.3633	1.4576	1.20	1.62
Trelawny	45	1.4184	.15884	.02368	1.3707	1.4662	1.20	1.62
St.James	45	1.4069	.15448	.02303	1.3605	1.4533	1.20	1.62
Hanover	44	1.4150	.15924	.02401	1.3666	1.4634	1.20	1.62
Westmoreland	44	1.4048	.15893	.02396	1.3565	1.4531	1.20	1.62
St. Elizabeth	44	1.4166	.15352	.02314	1.3699	1.4633	1.20	1.62
Manchester	44	1.4130	.16229	.02447	1.3636	1.4623	1.20	1.62
Clarendon	44	1.4041	.16227	.02446	1.3548	1.4534	1.20	1.62
St.Catherine	44	1.3955	.15871	.02393	1.3472	1.4437	1.20	1.62
Total	624	1.4108	.15632	.00626	1.3986	1.4231	1.20	1.62

ANOVA

Height of patient in meters

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	.028	13	.002	.088	1.000
Within Groups	15.196	610	.025		
Total	15.224	623			

$p=1.00 > 0.05$, hence, there is no significant difference in the height of the patients across the fourteen parishes



Day 4

Learning Objectives for Day 4

At the end of the session, participants should be able to:

- Apply Correlation – Parametric: -Product Moment & Non-Parametric: -Spearman Rho to health-related dataset
- Use Simple Linear Regression to make prediction when only one independent variable is involved
- Use Multiple Linear Regression to make prediction when two or more independent variables are involved
- Apply Logistic Regression to the prediction of a binary outcome

Correlation – Parametric: -Product Moment & Non-Parametric: -Spearman Rho

Correlation is the degree of association between two variables. The Product Moment Correlation is used to measure the degree of association between two continuous variables. The non-parametric equivalent is the Spearman-Rho. In SPSS, you will go to Analyze, then Correlate, then Bivariate . Figure 71 to 72 illustrate the steps to identify any association in the three variables: LastCD4Results, Weight and Height of the patients.

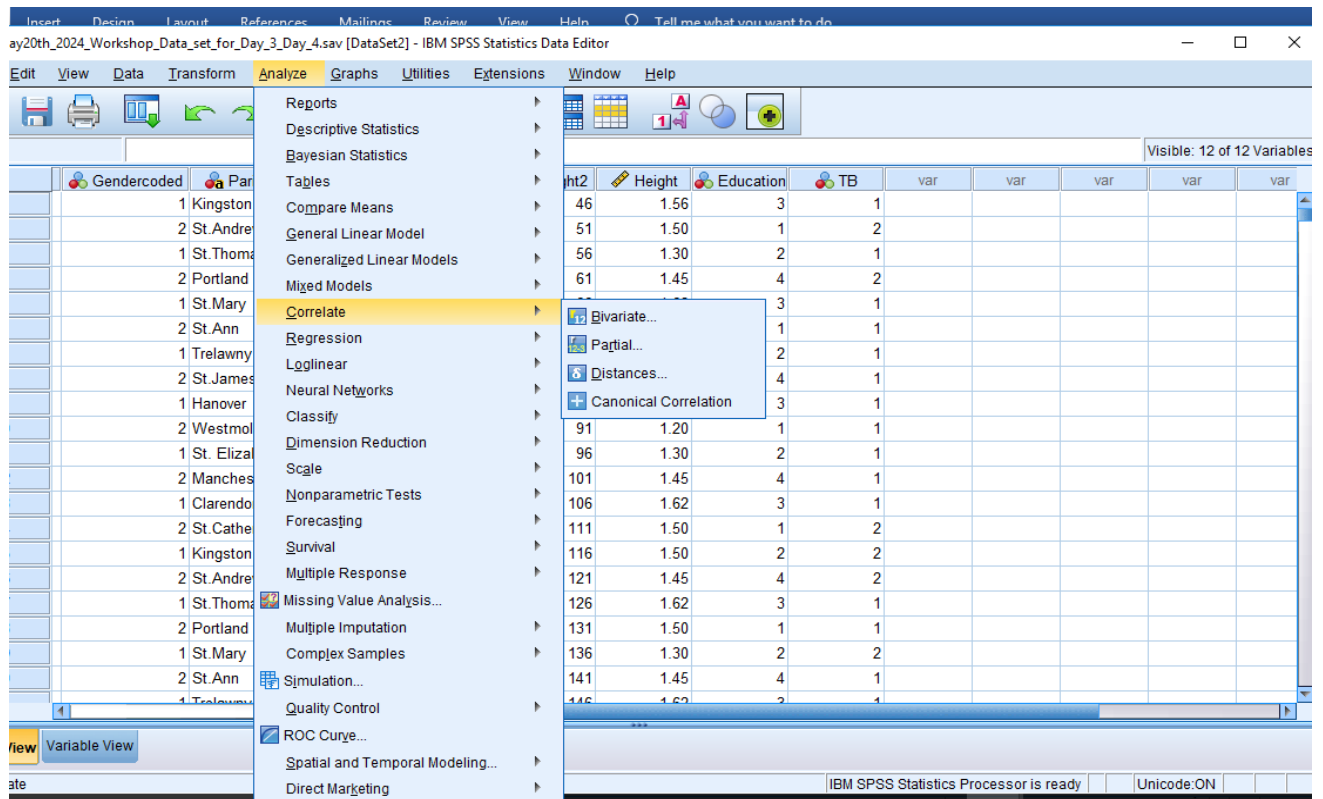


Figure 71.0

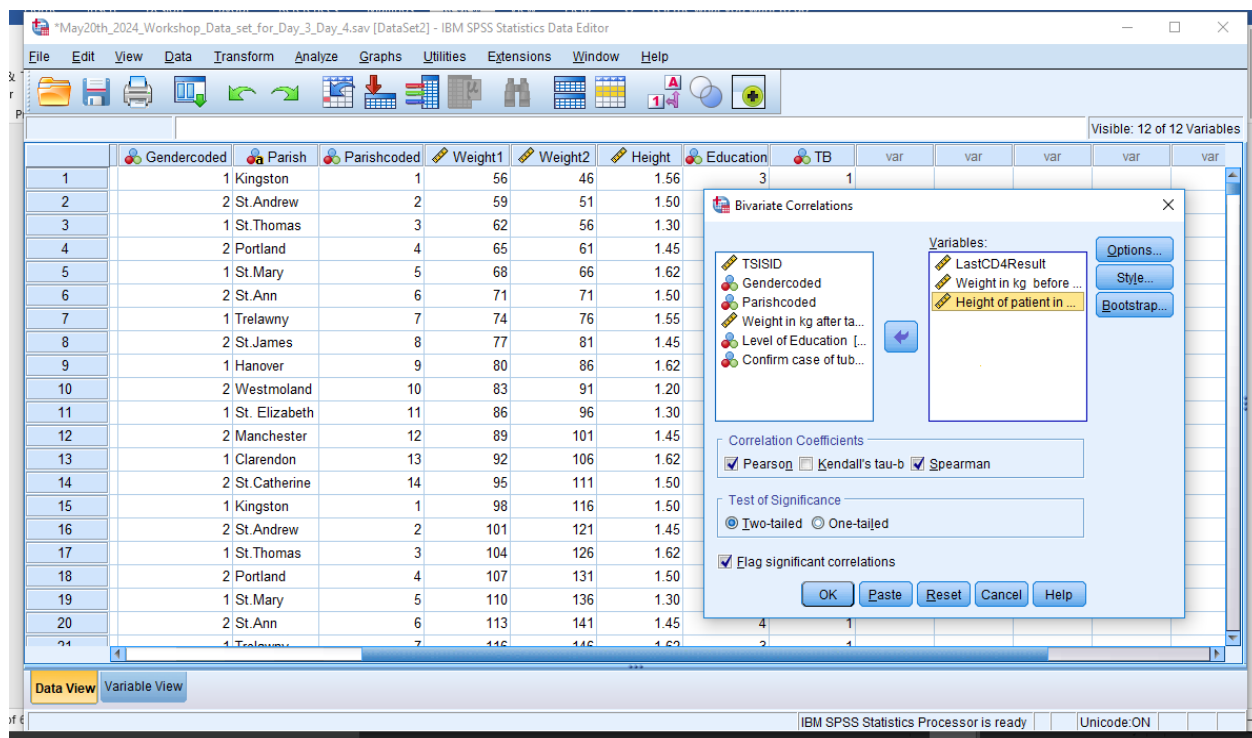


Figure72.0

Correlations

			Weight in kg before taking the recommend drugs	Height of patient in meters
LastCD4Result	Pearson Correlation	1	.002	.032
	Sig. (2-tailed)		.964	.419
	N	624	624	624
Weight in kg before taking the recommend drugs	Pearson Correlation	.002	1	-.052
	Sig. (2-tailed)	.964		.193
	N	624	624	624
Height of patient in meters	Pearson Correlation	.032	-.052	1
	Sig. (2-tailed)	.419	.193	
	N	624	624	624

The highlighted revealed no significant correlation among the variables.

Correlations

		LastCD4Result	Weight in kg before taking the recommend drugs	Height of patient in meters
Spearman's rho	LastCD4Result	Correlation Coefficient	1.000	.056
		Sig. (2-tailed)	.	.161
		N	624	624
	Weight in kg before taking the recommend drugs	Correlation Coefficient	-.011	1.000
		Sig. (2-tailed)	.784	.
		N	624	624
	Height of patient in meters	Correlation Coefficient	.056	-.052
		Sig. (2-tailed)	.161	.199
		N	624	624

Simple Linear Regression

This is used to make prediction for a continuous outcome using one independent variable.

Figure 73 illustrate the prediction of the weight of patient by the LastCD4Results

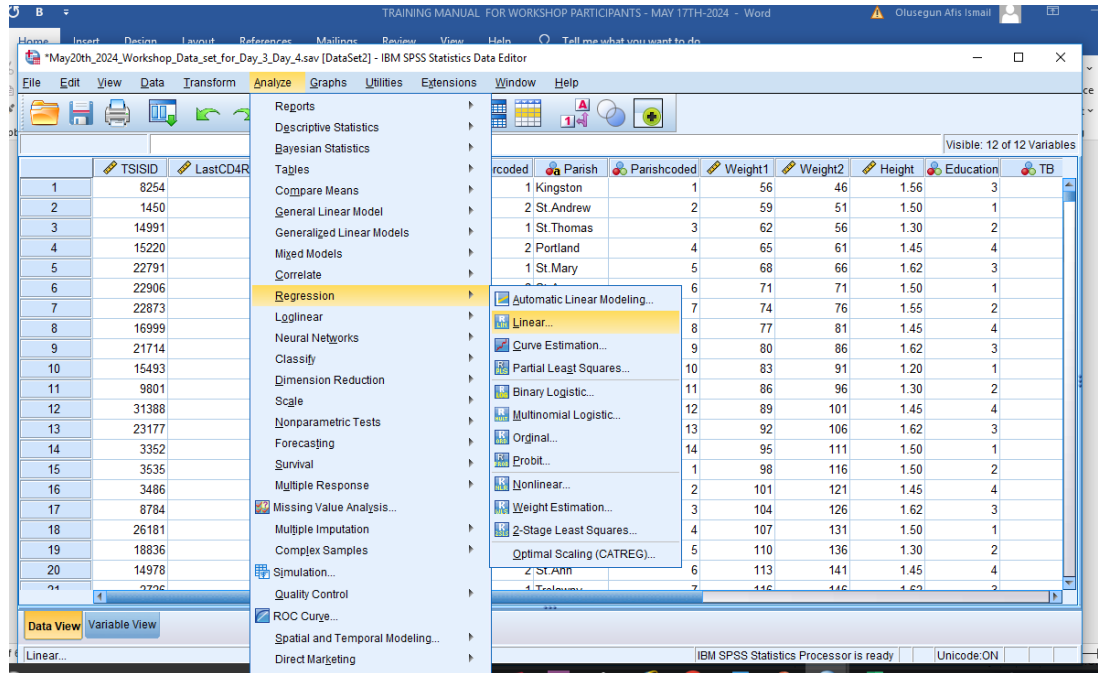


Figure 73.0

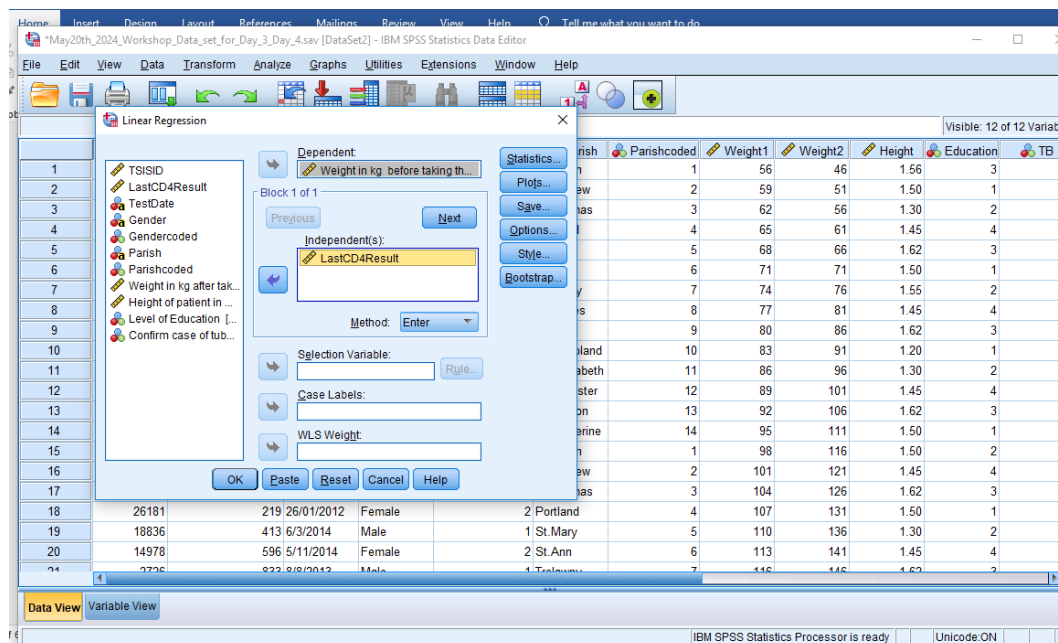


Figure 74.0

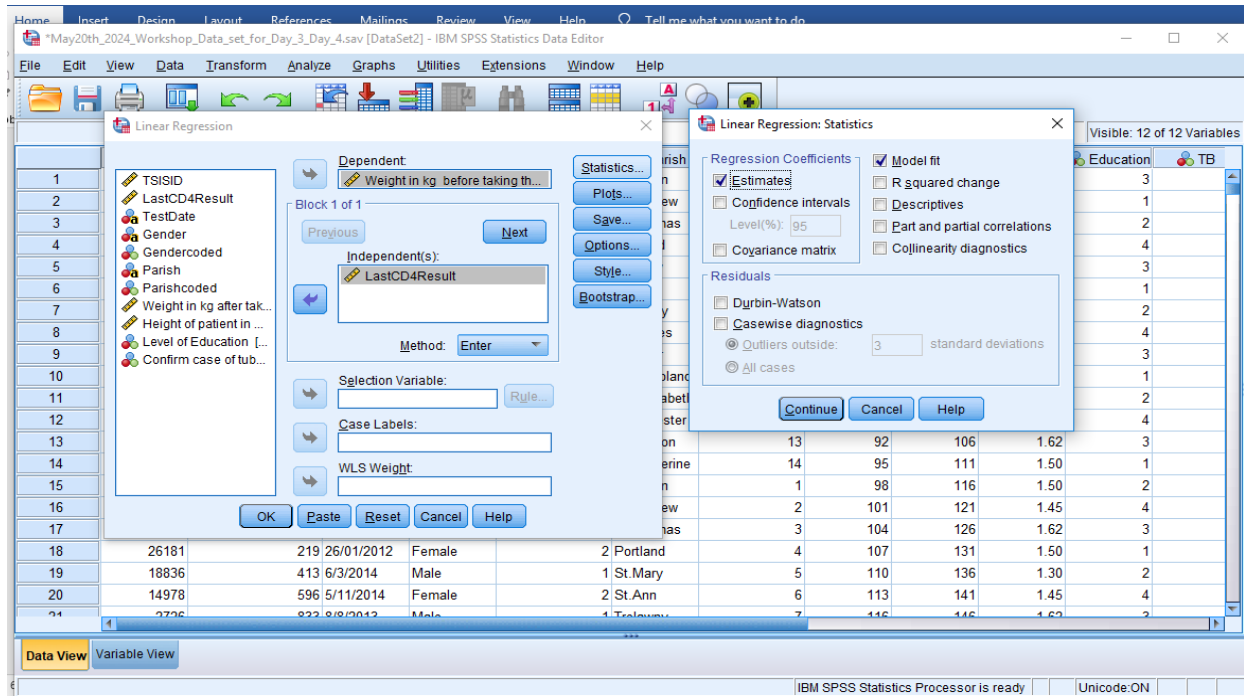


Figure 75.0

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	LastCD4Result ^b	.	Enter

a. Dependent Variable: Weight in kg before taking the recommend drugs

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.002 ^a	.000	-.002	40.248

a. Predictors: (Constant), LastCD4Result

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	3.308	1	3.308	.002	.964 ^b
	Residual	1007572.635	622	1619.892		
	Total	1007575.942	623			

a. Dependent Variable: Weight in kg before taking the recommend drugs

b. Predictors: (Constant), LastCD4Result

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	149.157	2.784		53.580	.000
	LastCD4Result	.000	.005	.002	.045	.964

a. Dependent Variable: Weight in kg before taking the recommend drugs

Multiple Linear Regression to make prediction when two or more independent variables are involved

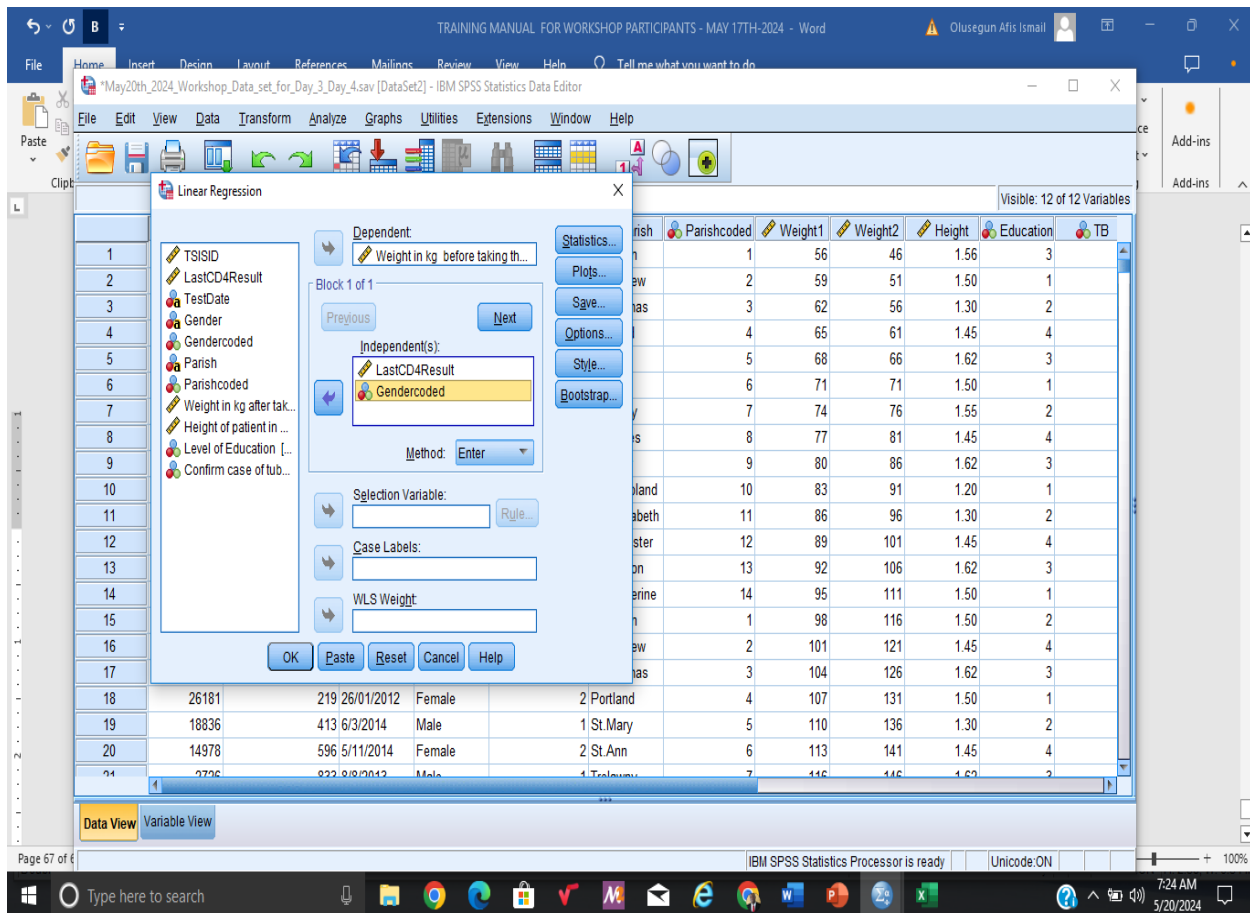


Figure 76.0

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.002 ^a	.000	-.003	40.280

a. Predictors: (Constant), Gendercoded, LastCD4Result

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	3.561	2	1.781	.001	.999 ^b
	Residual	1007572.38	621	1622.500		
	Total	1007575.94	623			

a. Dependent Variable: Weight in kg before taking the recommend drugs

b. Predictors: (Constant), Gendercoded, LastCD4Result

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	149.096	5.608		26.587	.000
	LastCD4Result	.000	.005	.002	.045	.964
	Gendercoded	.040	3.225	.001	.013	.990

a. Dependent Variable: Weight in kg before taking the recommend drugs

Logistic Regression to the prediction is used to predict a binary outcome

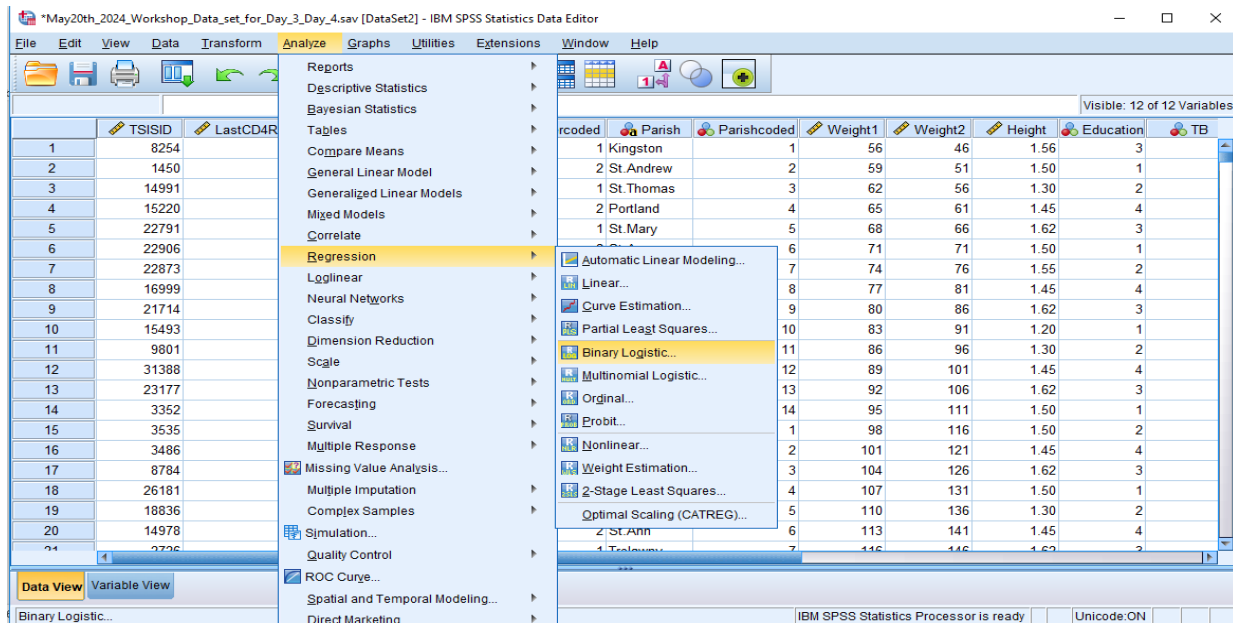


Figure 77.0

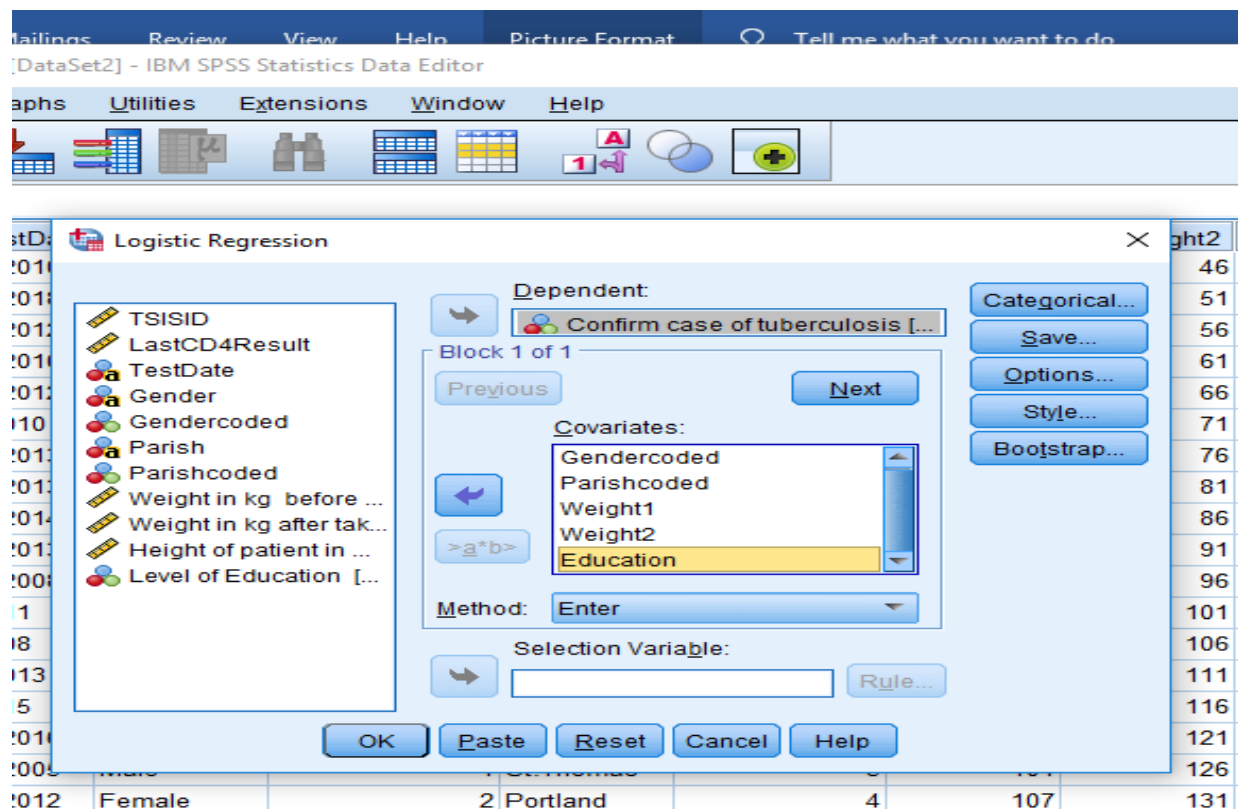


Figure 78.0

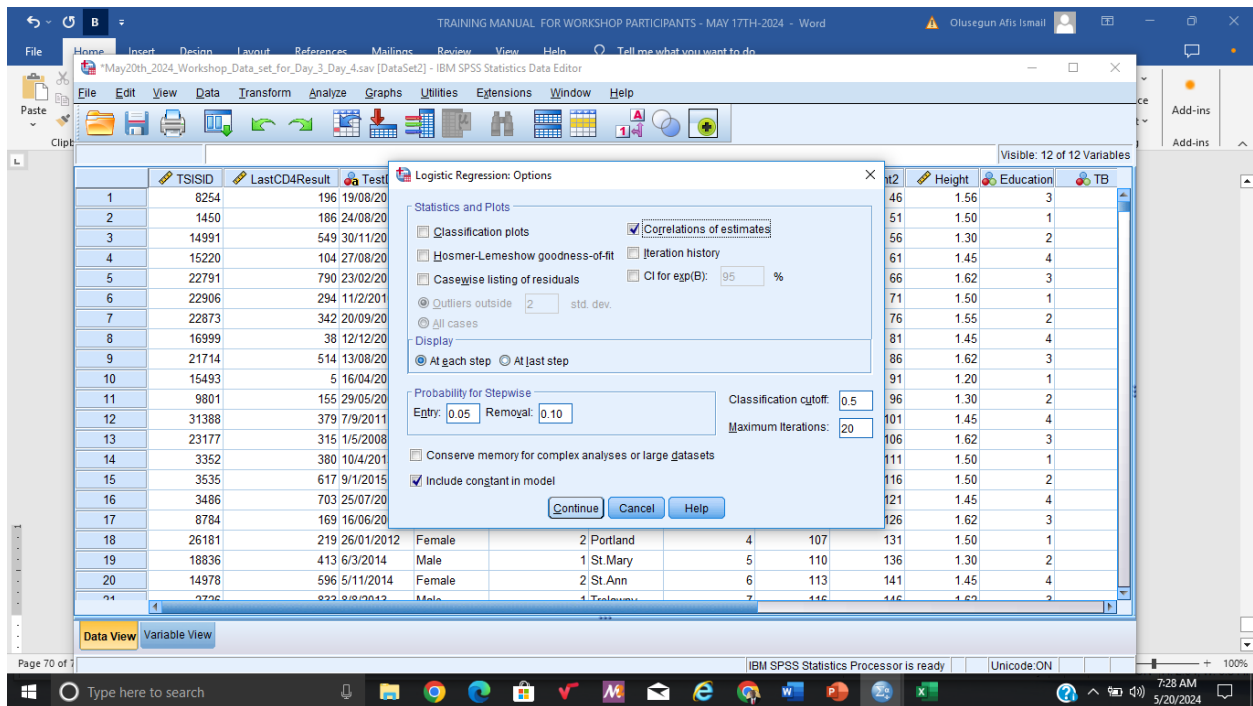


Figure 79.0

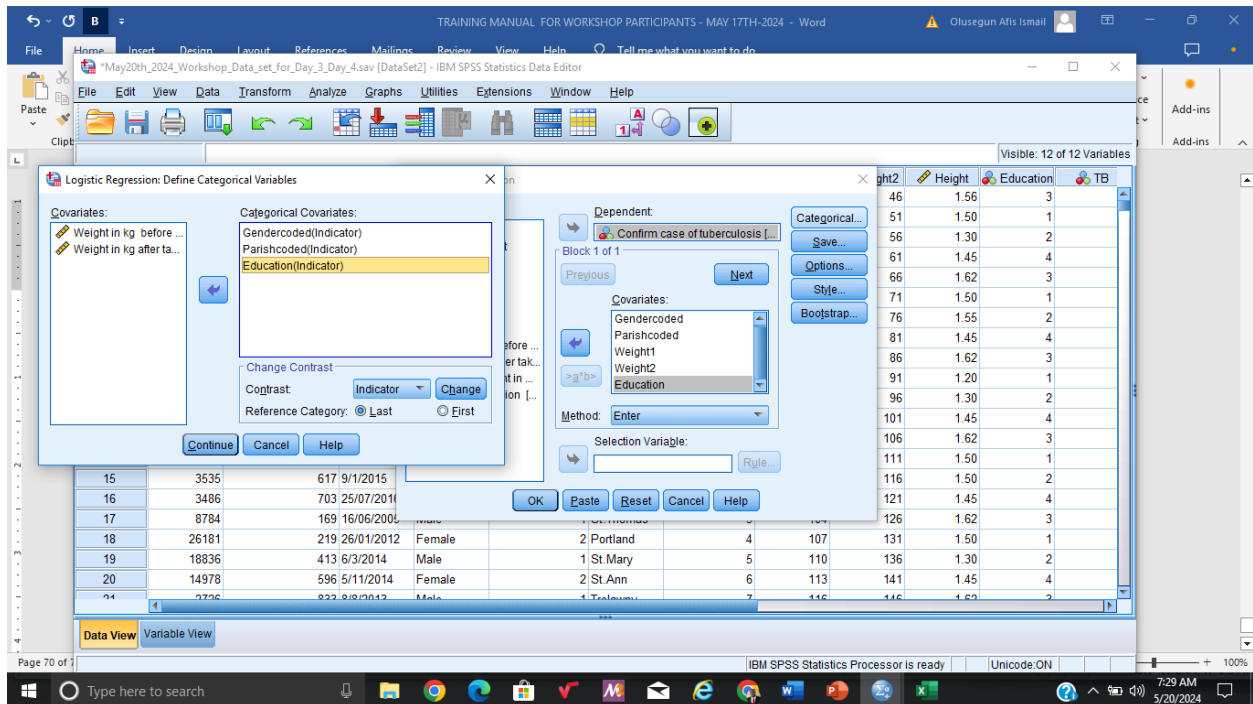


Figure 80.0

*Dependent Variable**Encoding*

Original Value	Internal Value
Yes	0
No	1

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	17.744	18	.473
	Block	17.744	18	.473
	Model	17.744	18	.473

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	696.929 ^a	.028	.041

a. Estimation terminated at iteration number 4 because parameter estimates changed by less than .001.

Classification Table^a

		Predicted		
		Confirm case of tuberculosis		Percentage Correct
	Observed	Yes	No	
Step 1	Confirm case of tuberculosis	Yes	462	0
		No	162	0
Overall Percentage				74.0

a. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1 ^a	Gendercoded(1)	-.017	1.096	.000	1	.988	.983
	Parishcoded			15.239	12	.229	
	Parishcoded(1)	-.438	.453	.932	1	.334	.646
	Parishcoded(2)	-.441	.453	.948	1	.330	.643
	Parishcoded(3)	-.928	.486	3.641	1	.056	.395
	Parishcoded(4)	-.794	.475	2.792	1	.095	.452
	Parishcoded(5)	-.791	.475	2.771	1	.096	.453
	Parishcoded(6)	-.922	.486	3.610	1	.057	.398
	Parishcoded(7)	-.922	.486	3.598	1	.058	.398
	Parishcoded(8)	-.925	.485	3.636	1	.057	.397
	Parishcoded(9)	-1.211	.517	5.490	1	.019	.298
	Parishcoded(10)	-.632	.467	1.836	1	.175	.531
	Parishcoded(11)	-.410	.454	.816	1	.366	.664
	Parishcoded(12)	-.190	.444	.182	1	.669	.827
	Weight in kg before taking the recommend drugs	-.003	.002	1.186	1	.276	.997
	Weight in kg after taken the recommended drugs	.001	.002	.211	1	.646	1.001
	Level of Education			.696	3	.874	
	Level of Education (1)	.002	.259	.000	1	.992	1.002
	Level of Education (2)	.134	1.020	.017	1	.895	1.144
	Level of Education (3)	-.088	1.016	.007	1	.931	.916
	Constant	-.223	.551	.164	1	.686	.800

a. Variable(s) entered on step 1: Gendercoded, Parishcoded, Weight in kg before taking the recommend drugs, Weight in kg after taken the recommended drugs, Level of Education .

References

**Darren George and Paul Mallery (2001). SPSS FOR WINDOWS .STEP BY STEP.
A SIMPLE GUIDE AND REFERENCE. PEARSON PUBLISHERS.**